EMV Support Manual

BASE24-pos®

Sep-2012, Release 6.0 Version 10 Publication Number: PS-CC986-01

Contents

Transaction Processing *continued*

Authorization Processing *continued*

ACI Worldwide, Inc.

What's New

This section highlights the major changes that have been made in updates to the ***BASE24-pos EMV Support Manual.*** The first column of the tables list the sections and appendixes in which major changes have been made. The second column of the tables describe the major changes for each section or appendix.

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Section/ Appendix	Major Changes
2	Adds the MANUAL FALLBACK ACTION field to CPF screen 13. This field allows card issuers to specify the authorization action required (continue processing, decline the transaction and return the card, or refer the transaction), when a manually-entered EMV fallback transaction is processed.

ACI Worldwide, Inc.

Preface

The EMV program is an add-on to the BASE24-atm and BASE24-pos products. The ***BASE24-pos EMV Support Manual*** provides information about configuring EMV to operate in conjunction with the BASE24-pos product and describes the processing of BASE24-pos EMV transactions. For information about the BASE24-atm EMV program, see the ***BASE24-atm EMV Support Manual.***

Audience

This manual is intended for personnel who must understand the implementation of the BASE24-pos EMV product.

Prerequisites

This manual assumes the reader is very familiar with the BASE24-pos product. Only the additions and changes that have been made for the EMV module are described in the manual. Where appropriate, references for more information are made to BASE24-pos manuals.

Additional Documentation

The BASE24 documentation set is arranged so that each BASE24 manual presents a topic or group of related topics in detail. When one BASE24 manual presents a topic that has already been covered in detail in another BASE24 manual, the topic is summarized and the reader is directed to the other manual for additional information. Information has been arranged in this manner to be more efficient for readers who do not need the additional detail and at the same time provide the source for readers who require the additional information.

This manual contains references to the following BASE24 publications:

- The ***BASE24 Base Files Maintenance Manual*** provides a comprehensive explanation of the BASE24 screens that access the base and switch files.
- The ***BASE24 Logical Network Configuration File Manual*** describes the standard BASE24-pos assigns and params.
- The ***BASE24 Tokens Manual*** describes the tokens that are used in standard BASE24 processing.
- The ***BASE24-atm EMV Support Manual*** describes how EMV operates.
- The ***BASE24-pos Multiple Currency Support Manual*** describes the support provided by the BASE24-pos multiple currency add-on product.
- The ***BASE24-pos Transaction Processing Manual*** describes how BASE24 processes standard BASE24-pos messages.

In addition, this manual references the following ISO standards:

- ISO 7816, ***Identification cards—Integrated circuit(s) cards with contacts***
- ISO 8583 (1987), ***Bank Card Originated Messages—Interchange Message Specifications—Content for Financial Transactions***
- ISO 3166, ***Codes for the Representation of Names of Countries***
- ISO 4217, ***Codes for the Representation of Currencies and Funds***

Software

This manual documents standard processing as of its publication date. Software that is not current and custom software modifications (that is, CSMs) may result in processing that differs from the material presented in this manual. The customer is responsible for identifying and noting these changes.

Manual Summary

The following is a summary of the contents of this manual.

“Conventions Used in This Manual” follows this preface and describes notation and documentation conventions necessary to understand the information in the manual.

Section 1, "Introduction," provides an overview of the BASE24 EMV add-on product.

Section 2, "Files Maintenance Screens and Logical Network Configuration File Assigns and Params," contains descriptions of the files maintenance fields and screens and the Logical Network Configuration File (LCONF) assigns and params that are specific to the EMV product.

Section 3, "Transaction Processing," describes the entities in the EMV transaction path and provides diagrams that illustrate EMV transaction flows.

Section 4, "Authorization Processing," describes additional authorization processing that is performed on EMV transactions.

Section 5, "CAP Token Validation," describes the processing that is performed to validate chip authentication program (CAP) tokens.

Section 6, "Device Handler Processing," describes the EMV processing that is performed by all Device Handler modules. Processing for specific Device Handler modules is described in device-specific appendixes.

Section 7, "Interchange Interface Processing," describes the EMV processing that is performed by Interchange Interface processes.

Appendix A, "SPDH Device Support," describes the functionality provided by the SPDH Device Handler EMV module.

Appendix B, "APACS 70 Device Support," describes the functionality provided by the BASE24-pos APACS 70 Device Handler EMV module.

Appendix C, "Hypercom Device Support," describes the functionality provided by the BASE24-pos Hypercom Device Handler EMV module.

Readers can use the index by field name to locate information about a particular screen field and the index by data name to locate information about a particular field from a file or record structure.

Publication Identification

Three entries appearing at the bottom of each page uniquely identify this BASE24 publication. The publication number (for example, PS-CC986-01 for the ***BASE24-pos EMV Support Manual***) appears on every page to assist readers in identifying the manual from which a page of information was printed. The publication date (for example, Sep-2009 for September, 2012) indicates the issue of the manual. The software release information (for example, R6.0v10 for release 6.0, version 10) specifies the software that the manual describes. This information matches the document information on the copyright page of the manual.

Conventions Used in This Manual

This section explains how field descriptions are documented in this manual.

Field Descriptions

Each field appearing on a files maintenance screen is listed by name and then described. Field descriptions briefly summarize the contents, purpose, and permissible values, as shown in the following example taken from the EMV-specific screen of the Card Prefix File (CPF).

Example

EMV CHECK — Determines whether the BASE24-pos Device Handler/Router/Authorization process should perform EMV authorization for this card prefix. Valid values are as follows:

Y = Yes, perform EMV authorization.
N = No, do not perform EMV authorization. Process transactions with this card prefix as a magnetic stripe transaction.

Field Length:	1 alphabetic character
Required Field:	Yes
Default Value:	Y
Data Name:	CPF.EMVCPF.EMV-CHK

Explanation

Each field description is completed by one or more of the following items of information:

Item	Description
Example	Illustrates a possible entry for the field to further clarify the value or values that can be entered.

Item	Description
Field Length	Specifies the size of the field and the type of characters that can be entered. This length refers to the field and valid values for the file maintenance screen, not the field in the DDLs. Possible values are alphabetic, alphanumeric, hexadecimal, and numeric. The term alphanumeric includes all alphabetic, numeric, and special characters that can be entered from a keyboard without using a control sequence. The term hexadecimal includes all numbers and the letters A through F. When a field value cannot be modified by the operator, the field length is System Protected.
Occurs	Indicates the number of times the field can be displayed on the screen. This information is provided only when the field can be displayed multiple times.
Required Field	Specifies whether a value has to be entered in the field. Possible values are Yes and No. Some fields are required only under certain conditions. In this case, the entry is Yes, followed by the conditions that determine when the field is required.
Default Value	Specifies the value that is automatically placed in the field when the screen is first displayed or when the **F8** key is pressed to clear the screen.
Data Name	Provides the DDL name associated with the field appearing on the screen. Data names are included in the documentation to assist in communicating screen and field issues to your technical staff. Note that screen data is not always stored in the BASE24 database as it appears on the screen or as it is described in the field description. If you need information on how screen data is actually stored, consult the DDLs.

1: Introduction

The BASE24-pos EMV product is an add-on to the BASE24-pos product. This section is an overview of the BASE24-pos EMV program and provides a brief introduction to the terminology used.

Europay, MasterCard, and Visa (EMV) have developed integrated circuit card (ICC) credit and debit specifications to facilitate interoperability across competing payment systems. That is, customers with EMV debit and credit cards could use any terminal that accepts EMV cards.

The BASE24-pos EMV product has been developed to support processing for both ICC and magnetic stripe transactions from a single terminal.

Integrated Circuit Card Technology

An integrated circuit card (ICC) is a plastic card, usually the size of a credit card, that contains an embedded microprocessor chip. This chip is capable of storing large amounts of cardholder information and can contain multiple applications. The terms chip card and smart card are sometimes used interchangeably with integrated circuit card. This manual uses both the terms integrated circuit card and ICC when describing the card that contains the embedded chip.

EMV cards have both a microprocessor chip and a magnetic stripe. Currently all debit and credit card schemes use the magnetic stripe with the microprocessor chip.

Card Standards

There are significant differences in the types of integrated circuit cards or EMV cards available. Almost all magnetic stripe cards are manufactured to a similar specification. However, each ICC manufacturer has offerings with a wide variety of chip memory sizes and processing capabilities.

Linked to the memory and processing potential is the operating system that resides on the chip itself. The operating system provides communications, file management, and security as well as application support. One of the goals of the EMV consortium is to establish a standard operating and application development system.

The following are technical standards that define the EMV card and its interface to the reading device.

- For bank payments, the card type conforms to ISO standards. The primary standard is the ISO 7816 standard, ***Identification cards—Integrated circuit(s) cards with contacts.***
- The EMV 1996 (v3.1.1) specification defines the business requirements for the use of the technology. By agreeing to use the ISO 7816 standard, EMV has established the principle of allowing a single terminal to support all ICC and non-ICC schemes.
- The EMV 2000 (v4.0), EMV 2004 (v4.1), and EMV 2008 (v4.2) specifications update the business requirements initially defined in the EMV 1996 specification.

Common Core Definitions

Common core definitions (CCD) are specified in EMV version 4.1, and establish a set of common data definitions and processes between the EMV chip card and the issuer host. Both Visa and MasterCard mutually recognize chip card applications conforming to these definitions for use on their respective branded cards. These definitions address the external data flowing between the chip card and the issuer and certain minimum card behavior.

Common core definitions should simplify chip implementation. With these definitions, issuers receive and return chip data in one common format from both Visa and MasterCard branded cards. With previous implementations of VIS and M/Chip, there has been considerable equivalent functionality. However, both Visa and MasterCard, in some instances, have chosen different implementations to accomplish the same functionality. These differences have resulted in issuers having to deal with each implementation separately. Common core definitions go a long way to defining a common implementation to be used by Visa, MasterCard, and other payment organizations choosing to adopt the CCD as their chip implementation specifications. These changes are focused from the external aspects of the card through delivery of the transaction (authorization and clearing) to the issuer.

Common Payment Application

The common payment application (CPA) is a CCD-compliant application specification developed by Visa and MasterCard. CPA takes CCD a step further, and defines the internals of the entire chip card application. While CCD focuses on the external interface of the card to issuer data and some processes, it does not provide a truly common payment application from a card functionality perspective. CPA is a complete CCD-compliant card application specification. CPA provides common functionality for the entire payment application. It specifies common data element definitions, card risk management options, common tags, card behaviors, common personalization, etc., that meet the requirements of EMV and CCD. CPA enables a single application implementation to be personalized with the same data elements and tags, including common risk management controls, for cards from all payment schemes supporting the CCD and CPA specifications. The CPA specification is owned, managed, and maintained by EMVCo, with endorsement and acceptance by both Visa and MasterCard.

CPA cards, by definition, are CCD-compliant. Issuers can also choose to support cards, which are CCD-compliant, but not necessarily CPA-compliant. It is important to note that use of CPA is optional on the part of the issuer.

EMV Implementations Supported by BASE24-pos

The BASE24-pos EMV product currently supports the following EMV implementations. The version number and date identify the EMV standard on which the implementation was originally based. Because versions of the EMV Standard are backwards-compatible, all implementations are still EMV-compliant.

	Specification Identifier	
Implementation	**Version Number**	**Date**
Common Core Definitions (CCD)	v4.1	2004
MasterCard M/Chip 2.1	v3.1.1	1996
MasterCard M/Chip 4.0	v4.0	2000
United Kingdom ICC Specification (UKIS)	v3.1.1	1996
Visa (VIS) version 1.3.2	v3.1.1	1996
Visa (VIS) version 1.4.0	v4.0	2000

Why Use an EMV Card?

The following are key reasons for banks to introduce integrated circuit cards for credit and debit cards:

- Reduce fraud losses
- Provide end-to-end security
- Save on processing costs
- Offer new value-added services

Magnetic stripe cards are relatively easy to replicate, while the complexities of the ICC, and the embedded security provided by the chip, make replication of an ICC card very difficult.

Traditionally, transaction security has been performed in security zones along the transaction path for magnetic-stripe card-based transactions. For example, a different MAC would be used for each message passed between systems, where each system is considered a separate security zone, and the PIN block would be translated every time a request message leaves a security zone. With EMV cards, the security is end-to-end (i.e., between the card and the issuer), rather than between various zones of the transaction path. Therefore, the function of an acquirer in an EMV transaction is basically to ensure that the data that comes in at one end is the same as the data that goes out at the other end.

The move to EMV technology enables banks to use the data storage and processing capabilities of the microchips to improve card and cardholder validation, as well as reduce communications costs by allowing increased offline authorization.

EMV Transaction Support

Integrated circuit cards have the usual attributes of a payment card: embossing, signature panel, hologram, and a magnetic stripe, allowing EMV-based payment methods to operate alongside existing traditional magnetic-stripe-based payment methods.

Four levels of transaction support provided by the BASE24-pos EMV product are based on the terminal capability, how the card details were read, and whether ICC data elements are included in the transaction message. Each level results in different processing requirements, as shown in the table below.

EMV-Capable Terminal?	Card Details Read from Chip or Magnetic Stripe?	Chip Data Elements Included?	Processing Requirements
No	Magnetic stripe	No	Standard BASE24 processing
Yes	Magnetic stripe	No	Add EMV Status token indicating EMV–capable terminal, but not a chip-read transaction.[1]
Yes	Microprocessor chip	No	Add EMV Status token indicating EMV–capable terminal and a chip-read transaction.[2]
Yes	Microprocessor chip	Yes, included in transaction message	Add EMV Status token indicating EMV–capable terminal and a chip-read transaction. Also add one or more tokens containing the chip data elements.

[1] In this situation, the card may be fraudulent and can result in fallback transaction processing.

[2] In this situation, the card may be fraudulent. However, this situation could occur because of Visa's Easy Entry program or MasterCard's Partial Grade program. The Easy Entry and Partial Grade options allow for the replication of only Track 2 magnetic stripe information on the chip (i.e., the chip does not contain any other chip data elements).

The type of transaction described in the last row of the table above is defined as an EMV transaction. The other three types of transactions are not EMV transactions.

Card Reader Interface

BASE24-pos supports three different methods to input EMV data: Contact EMV, contactless EMV, and contactless magnetic stripe. In all cases, card verification is performed by validating the Authorization Request Cryptogram (ARQC) included in the transaction data.

Contact EMV

Contact EMV provides card details to an authorization system that supports EMV processing. These transactions contain additional EMV data elements, including the ARQC, which are used in authorization processing by BASE24. The POS entry mode is set to 05x (integrated circuit card) in the transaction request to identify that EMV authorization processing should be performed.

Contactless EMV

Contactless EMV provides card details to an authorization system that supports EMV processing. These transactions contain additional EMV data elelments, and are treated like a contact EMV transaction in BASE24. The POS entry mode is set to 07x (contactless chip transaction) in the transaction request to identify that EMV authorization processing should be performed.

Contactless Magnetic Stripe

Generally, contactless magnetic stripe transactions do not contain EMV data. Instead, the card details emulate a contact (swiped) magnetic stripe transaction, and the authorization system uses dynamic card verification to validate the card.

However, Visa supports a contactless magnetic stripe implementation (MSD with CVN 17) that includes EMV data in the transaction. The POS entry mode is set to 91x (contactless magnetic stripe transaction) in the transaction request, which is the same value used to initiate dynamic card verification. However, the presence of EMV data in the transaction request allows BASE24 to identify that EMV authorization processing should be performed

Files Maintenance

The following files contain information about the BASE24 EMV product. Section 2 describes the EMV-specific fields in these files.

- The Institution Definition File (IDF) contains flags identifying the segments that can be included in the files belonging to the institution. For EMV authorization processing, a field on IDF screen 5 enables support of EMV transactions.
- The Card Prefix File (CPF) defines each card prefix that can be processed within a BASE24 logical network. For EMV authorization processing, screens 11 holds the CAM authentication control data for the card prefix, and screen 13 holds EMV information specific to BASE24-pos.
- The Cardholder Authorization File (CAF) contains one record for each cardholder whose card-issuing institution uses the Positive or Positive Balance Authorization method. Screen 13 is specific to EMV authorization processing, and holds the CAM authentication control data for the ICC.
- The ICC KEY File (KEYI) supports the key storage requirements for EMV. The KEYI holds multiple authentication keys (up to a maximum of 24) for an individual KEYI group. In addition, this file contains the key used for message integrity, verifying script commands using a message authentication code (MAC).
- The Logical Network Configuration File (LCONF) is the central location of file information and parameters for a BASE24 system. There is one LCONF for each logical network. The BASE24 EMV product modules use the standard LCONF assigns and params plus some EMV-specific assigns and params.
- The POS Terminal Data (PTD) File contains an EMV-specific screen. This screen enables you to specify the capabilities of each EMV terminal.

Maintaining the CAF EMV Segment

The CAF refresh record definition contains both mandatory and optional segments. Mandatory segments are those which must be present on tape for their entire length if the financial institution supports the segment. Optional segments must be present on the tape if the financial institution supports the segment, but the record can contain only the segment length field, set to the length of itself, instead of the entire segment. If only the length field of an optional segment is on the input file, that segment is not added to the disk record. If it is already present on disk, it will be deleted.

The Base and EMV segments of the CAF contain an application transaction counter (ATC) field. The ATC field you choose to update and check during transaction authorization affects your CAF maintenance procedures.

The EMV segment is processed as an optional segment, and therefore the segment on the CAF record is added or updated only if the entire segment is present in the CAF extract record. If the entire EMV segment is present in the refresh record, then the data fields in the CAF EMV segment (except the ATC field) are added or updated.

When a record is added (during either partial refresh or full refresh), the FILLER field in the EMV segment is initialized to ASCII zeros.

Note that a full CAF refresh resets the application transaction counter (ATC) value to zero in the Base segment and the EMV segment in records that contain that segment. Within the BASE24-pos Device Handler/Router/Authorization process, if the ATC limit is set to the value zero, ATC limit checking is not performed on EMV transactions. If the ATC value from the card is less than or equal to zero, the transaction is declined. After a full file refresh the ATC in the CAF is reset to zero. The ATC limit checking is bypassed for the next EMV transaction and the CAF ATC (Base or EMV segment, depending on which is checked and updated) is set to the value on the card. This logic avoids the requirement for impacting ATC values during a full refresh, which requires additional processing.

From Host Maintenance Process

The From Host Maintenance process ISO message format allows user fields to be sent in the message from the host to BASE24 for information defined in the EMV segment of the CAF or PBF.

Authorization Processing

This section provides an overview of authorization processing for EMV transactions. Section 4 describes this processing in more detail.

BASE24-pos EMV authorization provides support for the following authorization levels and methods:

- Level 1 (online only) as an optional prescreen check
- Level 2 (offline only):
  - Negative with Usage Accumulation Authorization method
  - Positive Authorization method
  - Positive with Balance Authorization method
  - Negative Authorization method
- Level 3 (online/offline):
  - As an optional prescreen check
  - As stand-in processing for all authorization methods listed above

For more detailed information about these authorization levels and methods, see the ***BASE24-pos Transaction Processing Manual.***

Card Authentication Method

EMV processing provides banks with the capability to capture and verify security-related data held on the microchip on the credit or debit card. This data is known as the card authentication method (CAM) data.

Part of the BASE24 processing of EMV cards includes calls to hardware security modules to verify card data. A separate set of security utilities, called CAM utilities, must be used in conjunction with the Base security utilities to support this verification.

On request messages, an authorization request cryptogram is generated by the card and the terminal and used by the issuer to authenticate the card. On response messages, an authorization response cryptogram is generated by the issuer and used by the card to authenticate the issuer. One call is made to the security module to verify the request cryptogram and generate the response cryptogram.

BASE24-pos allows issuers to validate the Card Verification Digits (CVD) during EMV authorization processing, where the CVD in the Track 2 equivalent data is different from the CVD on the magnetic stripe. Both CAM processing and card verification provide the business function of authenticating the card, but either or both authorization functions can be performed in a single transaction.

ATC Checking

The application transaction counter (ATC) is a value maintained by the microchip and updated for each transaction performed by the card application. A corresponding value is held in the CAF. The check is designed to detect fraudulent cards.

Dynamic card verification also uses an ATC. As a result, it is possible to maintain the ATC value on the Base segment or EMV segment of the CAF. However, this is typically done only if a card is likely to be used for both EMV and contactless magnetic stripe transactions. The dynamic card enhancement must be purchased to support this combination. If the card will be used only for EMV transactions, the ATC should be maintained only on the EMV segment.

Delayed Authorizations

Delayed authorizations are possible when an EMV card requests an online authorization but the terminal is unable to send the transaction online to the acquirer. This can result in the card declining the transaction, but the merchant has the option of accepting the transaction anyway. When the terminal is able to communicate with the acquirer again, the transaction can be sent online as a delayed authorization to determine whether the issuer is prepared to accept the transaction.

Issuers can configure whether delayed authorizations are supported at the card prefix level in the CPF. Acquirers can configure whether delayed authorizations are supported at the merchant level in the PRDF.

BASE24 uses the configured values when processing EMV authorization requests. A delayed authorization is identified by the presence of an AAC as the Application Cryptogram (i.e., a transaction declined by the card).

In issuer processing, ATC checking is inhibited when processing a delayed authorization because the transaction could have taken place before a subsequent transaction on the card that has already been processed by BASE24. ATC checking would fail in this situation because the ATC in the delayed authorization would be less than the ATC held on the CAF.

Status Field Checking

Status field checking is performed using preconfigured edit files (CVRTBL and TVRTBL) that are compiled into the Authorization module of the BASE24-pos Device Handler/Router/Authorization process. These files identify the field information to be checked and the required actions.

Reason Online Code Checking

Reason online code checking is performed using a preconfigured edit file (ROCTBL) that is compiled into the BASE24-pos Device Handler/Router/Authorization process. This file identifies the field values to be checked and the required actions.

Fallback Checking

If the microchip cannot be read on an ICC at an EMV-capable terminal, the system falls back or defaults to read the information from the magnetic stripe. The FALLBACK field in the CPF is used to configure whether to include fallback checking as a prescreen check.

Issuer Script Commands

Issuer script commands allow the issuer to alter the data or functionality of cards already in circulation. The terminal forwards a script command to the card, if returned from the issuer in a response to an online authorization request. The card uses the script data to reconfigure itself.

The BASE24 system supports the following script commands:

- CARD BLOCK (an irreversible disabling of the microchip)
- PUT DATA (updates the card data)

Script data passed between the issuer and the card is secured with a message authentication code (MAC) generated by a security module. Issuer Script Data is supported either in pass-through mode or as response scripts generated by the BASE24 system.

Additional Routing and Authorization Functions

The BASE24 system also supports the following additional routing and authorization functions.

- Prefix routing of EMV transactions using a prefix route field in the EMV segment of the CPF
- Enhanced routing and authorization of EMV transactions based on the issuer code field in the EMV segment of the CPF
- Processing EMV card transactions as standard Track 2 card transactions when card issuers start distributing integrated circuit cards before being able to support transactions initiated by users with those cards

Hardware Security Module Support

To be implemented as an issuer, the BASE24-pos EMV add-on product must have access to a hardware security module (HSM). The BASE24-pos EMV add-on product supports the use of Atalla or Thales e-Security (Racal) security modules. Whichever vendor is chosen, the security module must have a version of firmware that is capable of supporting the EMV add-on product.

Atalla

EMV 2000 processing without common core definitions (CCD) or common session key derivation is supported on the following Atalla HSMs: A*xx*100-V or A*xx*100 (AKB) running version 2.62 or later firmware. Support for common core definitions or common session key derivation requires Atalla A*xx*100-VAR or A*xx*100-AKB HSMs running version 2.80 or later firmware.

Thales e-Security

For Thales e-Security (Racal) HSMs, the minimum firmware version depends on hardware model and EMV key derivation method, as shown in the following table.

EMV Key Derivation Method	Hardware Model	Firmware Version
1996	7*x*00	5.05
2000	7*x*00	7.0
CCD or common session key derivation	Not available	
1996	7*x*10	5.05
2000	7*x*10	3.0
CCD or common session key derivation	Not available	
1996	8*xxx*	5.05
2000	8*xxx*	2.0
CCD or common session key derivation	8*xxx*	2.1, 2.2, 2.3

The following table shows which HSM commands must be supported if the BASE24-pos EMV product is configured to perform specific cryptographic processing.

Cryptographic Process	Atalla Command	Thales e-Security Command	
		EMV 1996	EMV 2000
ARQC verification/ARPC generation	350	KQ	KW
EMV MAC generation	352	KU	KY

Binary Data Support

EMV security processing requires the use of binary data. This requirement impacts certain HSMs and their interfaces with the BASE24-pos product.

The Thales e-Security Host Security Module (HSM) uses binary data for certain control commands. As a result, the interface between BASE24-pos and the Thales e-Security HSM must be configured with a protocol that supports the transmission of binary data (e.g., TCP/IP).

The Atalla Network Supervisor (NSP) uses binary-coded hexadecimal data instead of binary data for control commands. As a result, the interface between BASE24-pos and the Atalla NSP can use protocols that do not support the transmission of binary data (e.g., async, bisync) in addition to protocols that support the transmission of binary data.

Transaction Security Services Process Requirements

The BASE24-pos Device Handler/Router/Authorization process supports the following normal (EMV1996) cryptographic processing in an Atalla or Thales e-Security hardware module, with or without the use of the Transaction Security Services process.

- ARQC verification and ARPC generation
- EMV MAC generation

However, you must use the Transaction Security Services process with the BASE24-pos EMV product for the following functions:

- ARPC generation using Method 2 described in EMV version 4.1[1]
- Card key derivation using the Option B algorithm defined in EMV version 4.1[1]
- Issuer script processing with the Secure Messaging Format 1 defined in EMV version 4.1[1]
- Session key derivation defined in EMV 2000
- Session key derivation using the Common method defined in EMV version 4.1[1]

[1] Issuers requiring support for the CCD EMV scheme or common session key derivation must have at least version 06.4 of TSS.

For more information on the Transaction Security Services process and the functions it performs, refer to the ***BASE24 Transaction Security Services Processing Guide***.

Token Processing

The data required to authorize transactions is passed between the various processing modules in BASE24-pos Standard Internal Message (PSTM) format. These messages comprise a set of core (fixed) fields that are present in all internal messages, and a number of sets of function-specific fields known as tokens.

A token is a set of related fields added to an internal message as a method for passing data between modules in which new functionality has been implemented. EMV-related data is passed between the various processing modules within a BASE24 system by means of data tokens.

The following tokens contain information for the BASE24-pos EMV product.

- The EMV Status Token contains control information, not necessarily specific to EMV transactions.
- The EMV Request Data Token contains the 13 minimum data elements defined by EMV.
- The EMV Discretionary Data Token contains other data elements defined in more than one application message standard where EMV support has been implemented.
- The EMV Response Data Token contains data elements for response processing, and script command flags.
- The EMV Script Data Token contains issuer script commands.
- The EMV Supplementary Data token carries the supplementary data required by Visa in data element 55 in a contactless magnetic stripe transaction.

For more information about BASE24 tokens, including descriptions of tokens used by the BASE24-pos EMV product, see the ***BASE24 Tokens Manual.***

CAP Token Validation

Through the use of the Transaction Security Services process, BASE24 supports chip authentication program (CAP) token validation for MasterCard's OneSMART Authentication solution and Visa's Dynamic Passcode Authentication (DPA) solution. CAP token validation provides two factor authentication (2FA) in card-not-present point-of-sale (POS) environments. Because two factor authentication uses two security items—the chip on the card and a PIN—it is more secure than using only chip or PIN verification. The Transaction Security Services process supports CAP token validation in Thales e-Security host security modules (HSMs). For detailed information on the configuration requirements for CAP token validation, refer to section 2 and the ***BASE24 Transaction Security Services Processing Guide.***

CAP token validation can be used with contact and contactless EMV cards or on chip cards used only for CAP token validation. The Transaction Security Services process supports CAP token validation in Thales e-Security host security modules (HSMs).

BASE24-pos Device Handler EMV Support

The following BASE24-pos device handler modules support EMV processing:

- ACI Standard POS Device Handler
- APACS 70 POS Device Handler
- BASE24-pos Hypercom Device Handler

Refer to section 6 and the appendixes for more information about device support.

BASE24 Interchange Interface EMV Support

The following Interchange Interfaces support BASE24-pos EMV processing:

- AMEX Credit ISO Interchange Interface
- AMEX GNS Interchange Interface
- Banknet Interchange Interface
- BIC ISO Interchange Interface
- EPS-Net Interchange Interface
- JCB Interchange Interface
- MDS Maestro Interchange Interface
- Visa DPS Interface
- VisaNet Interchange Interface

Refer to section 7 for information about processing for the AMEX GNS, Banknet, and MDS Maestro interchange interfaces. Refer to interchange-specific documentation for the remaining interchange interfaces.

EMV Support in the BASE24 External Message

The BIC ISO Interchange Interface and BASE24 ISO Host Interface use data element P-63 in the BASE24 external message to carry EMV information. Both of these interfaces use existing tokens to carry EMV information internally, eliminating the need for mapping of data between the internal and external messages.

Both interface processes use the standard technique for configuring the EMV-specific tokens to be included in external messages. Refer to the ***BASE24 Tokens Manual*** for information on configuring the tokens used by these interface processes.

2: Files Maintenance Screens and Logical Network Configuration File Assigns and Params

This section provides details of BASE24 screens and Logical Network Configuration File (LCONF) assigns and params that support the BASE24-pos EMV extensions. For information about other BASE24-pos screens, see the ***BASE24 Base Files Maintenance Manual*** and the ***BASE24-pos Files Maintenance Manual.*** For additional information about LCONF assigns and params, see the ***BASE24 Logical Network Configuration File Manual.***

Institution Definition File (IDF)

The Institution Definition File (IDF) contains one record for each institution participating in the logical network and defines processing for each institution. The IDF contains routing tables for transaction routing within a BASE24 product and the parameters for cards, dates, processing control and sharing.

IDF Screen 5

IDF screen 5 contains flags identifying the segments that can be included in the files belonging to this institution. For EMV authorization processing, a field on the screen enables support of EMV transactions. This section describes the field on IDF screen 5 that has been modified to support EMV transaction processing.

```
BASE24-BASE  INSTITUTION FILE         LLLL        YY/MM/DD  HH:MM  05 OF 42

                  FIID:            FI-NAME:

                  FIID AUTH FILE SEGMENT INDICATORS (01 OF 02)

 Y BASE              N ATM              N POS              N EMV
 N PBF CR LINE       N PBF SHORT NM     N SSB BASE         N SSB CHECK
 N ADDR VERIF        N PREAUTH HOLD     N NON-CURR DSP     N ENHANCED PREAUTH











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FIID AUTH FILE SEGMENT INDICATORS

A series of flags indicate which segments the institution’s authorization files contain. These flags permit each institution to carry only the information needed for the BASE24 products supported by that institution. The following field is used to indicate EMV support.

EMV — Indicates whether EMV processing is supported. Valid values are as follows:

Y = Yes, EMV processing is supported.
N = No, EMV processing is not supported.

Field Length:	1 alphabetic character
Required Field:	Yes
Default Value:	N
Data Name:	IDF.IDFBASE.FIID-SEG-MAP

Card Prefix File (CPF)

The Card Prefix File (CPF) defines each card prefix that can be processed within a BASE24 logical network. One record must exist in the CPF for each card prefix to be processed. If one prefix is used with multiple primary account number (PAN) lengths, multiple records must exist in the CPF—one for each PAN length.

CPF records define the characteristics of each prefix and also contain prefix-specific parameters that allow institutions to define portions of their authorization processing that can be controlled at the prefix level. These parameters include expiration date checks, card and PIN verification controls, withdrawal limits, and credit account minimum standard increments.

CPF Screen 3

For EMV authorization processing, CPF screen 3 contains one field that can be used to control application transaction counter (ATC) checking. CPF screen 3 is shown below, followed by a description of the ATC CHECK field.

```
BASE24-BASE     CARD PREFIX            LLLL        YY/MM/DD  HH:MM  03 OF 22
 PREFIX:                     PAN LENGTH: 00                  FIID:

                            AUTHORIZATION INFORMATION

          EXPIRATION DATE PROCESSING FLAG:  0  (EXP DATE NOT REQUIRED)
               EXPIRATION DATE COMPARISON:  0  (COMPARISON NOT REQUIRED)
            CARD VALIDITY PERIOD (MONTHS):  000
      CHECK IF HOST ONLINE - SERVICE CODE:  N
       SERVICE CODE CHECKING ACTION INDEX:  1  (1-4)

                      DYNAMIC CARD VERIFICATION INFORMATION
     DYN CARD VERIF KEY LOCATOR:                      DCV CHECK TYPE:
     TRK1 DCVD OFST:     DCVD LEN:    ATC OFST:     ATC LEN:    UNPRED NUM OFST:
     TRK2 DCVD OFST:     DCVD LEN:    ATC OFST:     ATC LEN:    UNPRED NUM OFST:
       ATC CHECK:                             CHECK IF HOST ONLINE DCV:
                                                                   ATC:
       BAD ATC ACTION:                           BAD DCV ACTION:


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ATC CHECK — A flag indicating whether the application transaction counter (ATC) check is to be performed. Setting this field causes the ATC fields in the Base segment of the CAF to be updated. Valid values are as follows:

0 = Do not perform the ATC check.
1 = Perform the ATC check on EMV transactions. The ATC CHECK TYPE field on CPF screen 11 must be set to a value of 8 or 9 for this check to be performed.
2 = Perform the ATC check on contactless magnetic stripe transactions.
3 = Perform the ATC check on EMV and contactless magnetic stripe transactions. The ATC CHECK TYPE field on CPF screen 11 must be set to a value of 8 or 9 for this check to be performed on EMV transactions.

Field Length:	1 numeric character
Required Field:	Yes
Default Value:	0
Data Name:	ATC-CHK

CPF Screen 11

For EMV authorization processing, screen 11 holds the CAM authentication control data for the card prefix. This section describes the fields present on CPF screen 11.

```
BASE24-EMV     CARD PREFIX            LLLL        YY/MM/DD  HH:MM  11 OF 22
 PREFIX:                      PAN LENGTH: 00                 FIID:

                              EMV INFORMATION
       EMV CHECK: Y (Y/N)                         KEYI GROUP:
  CAM CHECK TYPE: 0 (CHECK DISABLED)      ATC CHECK TYPE: 0 (DISABLED)
        EMV IAD FORMAT: 0 (UNDEFINED)         STATUS CHECK ACTION INDEX: 1 (1-6)
  ALTERNATE IAD FORMAT: 0 (UNDEFINED)            ALTERNATE ACTION INDEX: 1 (1-6)
   SCRIPT TEMPLATE TAG: 0 (USE LCONF VALUE)           SCRIPT MAC LENGTH: 0
    CHECK IF HOST ONLINE -     CAM: N (Y/N)                    FALLBACK: N (Y/N)
                           TVR/CVR: N (Y/N)         REASON ONLINE CODE: N (Y/N)
    BAD CAM ACTION - DATA RELIABLE: 1 (DENY AND RETURN)
  BAD CAM ACTION - DATA UNRELIABLE: 1 (DENY AND RETURN)
              EMV PREFIX ROUTING:
        CARD VERIFY - CHECK TYPE: 0 (NONE)              EMV CV DATE: 0501
                    CHECK METHOD: 0 (STANDARD/STANDARD)    CV DATA: 000
  UPDATE COUNTERS -       DECLINED TRANSACTION: 0 (*******************)
              APPROVED FINANCIAL TRANSACTION: 0 (*******************)
          APPROVED NON-FINANCIAL TRANSACTION: 0 (*******************)

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EMV INFORMATION

EMV CHECK — Determines whether the BASE24-pos Device Handler/Router/Authorization process should perform EMV authorization for this card prefix. Valid values are as follows:

Y = Yes, perform EMV authorization.
N = No, do not perform EMV authorization. Process transactions with this card prefix as a magnetic stripe transaction.

Field Length:	1 alphabetic character
Required Field:	Yes
Default Value:	Y
Data Name:	CPF.EMVCPF.EMV-CHK

KEYI GROUP — A value that matches the KEYI GROUP field in the KEYI. This field is not required if the BASE24 system is acquiring EMV transactions and passing them through to a host or interchange for authorization. However, this field is required if the BASE24 system is authorizing EMV transactions for this prefix or doing any prescreening (e.g., verification of request cryptograms).

Field Length:	16 alphanumeric characters
Required Field:	Yes, if the BASE24 system is authorizing or prescreening EMV transactions for this prefix
Default Value:	No default value
Data Name:	CPF.EMVCPF.KEYI-GRP

CAM CHECK TYPE — The type of CAM verification to be performed. Valid values are as follows:

0 = Do not perform CAM check.
1 = Perform CAM check only if CAM data is reliable.
2 = Perform CAM check whether CAM data is reliable or unreliable.
9 = Process transaction as magnetic stripe instead of chip.

Field Length:	1 numeric character
Required Field:	Yes
Default Value:	0
Data Name:	CPF.EMVCPF.CAM-CHK-TYP

ATC CHECK TYPE — Determines whether the application transaction number (ATC) check is to be performed and the location of the ATC. Valid values are as follows:

0 = Do not perform ATC checking for EMV transactions.

1 = Perform ATC checking for EMV transactions. Use this value when the card does not support contactless magnetic stripe transactions, or when the card maintains a separate ATC value for EMV transactions and contactless magnetic stripe transactions.

8 = Perform ATC checking for EMV transactions and maintain one ATC value for both EMV transactions and contactless magnetic stripe transactions. Use this value only if the card does not support contact EMV transactions, but does support contactless transactions that include an EMV cryptogram (for which a CPF EMV segment is required, but a CAF EMV segment is not). The ATC CHECK field on CPF screen 3 must be set to a value of 1 or 3 for this check to be performed.

9 = Perform ATC checking for EMV transactions and maintain one ATC value for both EMV transactions and contactless magnetic stripe transactions. Use this value only if there is one ATC on the card that is used for both contactless magnetic stripe transactions and EMV transactions. The ATC CHECK field on CPF screen 3 must be set to a value of 1 or 3 for this check to be performed.

Field Length:	1 numeric character
Required Field:	Yes
Default Value:	Y
Data Name:	CPF.EMVCPF.ATC-CHK

EMV IAD FORMAT — The primary format of the issuer application data (IAD) used for this card type. Two IAD formats can be used for a card prefix, enabling cards with the same prefix to be migrated from one format to another as they are reissued.

A number of predefined formats for issuer application data exist, as specified by external entities (e.g., Visa, MasterCard, and American Express), but issuer-specific formats also can be defined. The data typically contains information that is used in the derivation of the authorization cryptogram, and the detailed layout is therefore required for issuer processing. Decoding procedures are included for recognized formats. Custom encoding and decoding procedures are required for proprietary formats.

The following data elements, which are included in the Visa, MasterCard, and American Express definitions, are used in EMV authorization processing and generation of the authorization cryptograms:

- Derivation key index
- Cryptogram version number
- Card verification results

Valid values are as follows:

0	=	Undefined. The format is unknown. The contents are transported as passthru data, but no checks requiring the IAD are performed by the BASE24 system.
1	=	Visa/UKIS. This is the IAD format defined by Visa in the Visa Integrated Circuit Card (ICC) Specification.
2	=	M/Chip 2.1. This is the IAD format recommended for MasterCard MCPA M/Chip 2.1 cards.
3	=	M/Chip 4. This is the IAD format recommended for MasterCard MCPA M/Chip 4 cards.
4	=	CCD-A. This is the IAD format mandated for CCD (IAD Format A) cards.
5	=	AMEX. This is the IAD format defined for American Express cards.
6 to 9	=	Reserved for other national or international scheme formats.
A to Z	=	Proprietary format defined for the issuer. (Custom formatting procedures are required.)

If this field is set to 1 (Visa/UKIS), the Visa format of issuer application data is used to extract the fields.

If this field is set to 2 (M/Chip 2.1), the MCPA M/Chip 2.1 format of issuer application data is used to extract the fields.

If this field is set to 3 (M/Chip 4), the MCPA M/Chip 4 format of issuer application data is used to extract the fields.

If this field is set to 4 (CCD-A), the CCD (IAD format A) format of issuer application data is used to extract the fields.

If this field is set to 5 (AMEX), the AMEX format of Issuer Application Data (IAD) is used to extract the fields from the IAD to populate the EMV tokens. The AMEX format of the Issuer Application Data matches the Visa format.

If this field is set to any other value, the data fields cannot be extracted because the format is currently unknown. If the fields are extracted, they are passed to the CAM utilities for request cryptogram verification and response cryptogram generation.

Field Length:	1 alphabetic or numeric character
Required Field:	Yes
Default Value:	0
Data Name:	CPF.EMVCPF.EMV-ISS-APPL-DATA-FRMT

STATUS CHECK ACTION INDEX — Identifies which of the preset rules for status field or Reason Online code processing are to be used for this card type with the primary IAD format. These preset rules are defined in the CVR and TVR tables or Reason Online code table. For more information on these tables, see the topics “Status Field Verification” and “Reason Online Code” in section 4. The range of valid values is 1–6.

Field Length:	1 numeric character
Required Field:	Yes
Default Value:	1
Data Name:	CPF.EMVCPF.ACTION-TABLE-INDEX

ALTERNATE IAD FORMAT — The alternate format of the issuer application data (IAD) used for this card type. Two IAD formats can be used for a card prefix, enabling cards with the same prefix to be migrated form one format to another as they are reissued.

A number of predefined formats for issuer application data exist, as specified by external entities (e.g., Visa, MasterCard, and American Express), but issuer-specific formats also can be defined. The data typically contains information that is used in the derivation of the authorization cryptogram, and the detailed layout is therefore required for issuer processing. Decoding procedures are included for recognized formats. Custom encoding and decoding procedures are required for proprietary formats.

The following data elements, which are included in the Visa, MasterCard, and American Express definitions, are used in EMV authorization processing and generation of the authorization cryptograms:

- Derivation key index
- Cryptogram version number
- Card verification results

Valid values are as follows:

0	=	Undefined. The format is unknown. The contents are transported as passthrough data, but no checks requiring the IAD are performed by the BASE24 system.
1	=	Visa/UKIS. This is the IAD format defined by Visa in the Visa Integrated Circuit Card (ICC) Specification.
2	=	M/Chip 2.1. This is the IAD format recommended for MasterCard MCPA M/Chip 2.1 cards.
3	=	M/Chip 4. This is the IAD format recommended for MasterCard MCPA M/Chip 4 cards.
4	=	CCD-A. This is the IAD format mandated for CCD (IAD Format A) cards.
5	=	AMEX. This is the IAD format defined for American Express cards.
6 to 9	=	Reserved for other national or international scheme formats.
A to Z	=	Proprietary format defined for the issuer. (Custom formatting procedures are required.)

If this field is set to 1 (Visa/UKIS), the Visa format of issuer application data is used to extract the fields.

If this field is set to 2 (M/Chip 2.1), the MCPA M/Chip 2.1 format of issuer application data is used to extract the fields.

If this field is set to 3 (M/Chip 4), the MCPA M/Chip 4 format of issuer application data is used to extract the fields.

If this field is set to 4 (CCD-A), the CCD (IAD format A) format of issuer application data is used to extract the fields.

If this field is set to 5 (AMEX), the AMEX format of Issuer Application Data (IAD) is used to extract the fields from the IAD to populate the EMV tokens. The AMEX format of the Issuer Application Data matches the Visa format.

If this field is set to any other value, the data fields cannot be extracted because the format is currently unknown. If the fields are extracted, they are passed to the CAM utilities for request cryptogram verification and response cryptogram generation.

Field Length:	1 alphabetic or numeric character
Required Field:	Yes
Default Value:	0
Data Name:	CPF.EMVCPF.EMV-ISS-APPL-DATA-FRMT-ALT

ALTERNATE ACTION INDEX — Identifies which of the preset rules for status field or Reason Online code processing are to be used for this card type with the alternate IAD format. These preset rules are defined in the CVR and TVR tables or Reason Online code table. For more information on these tables, see the topics “Status Field Verification” and “Reason Online Code” in section 4. The range of valid values is 1–6.

Field Length:	1 numeric character
Required Field:	Yes
Default Value:	1
Data Name:	CPF.EMVCPF.ACTION-TABLE-INDEX

SCRIPT TEMPLATE TAG — Specifies the template tag to be used when an issuer script is sent to the card. Valid values are as follows:

0 = Use the value from the LCONF parameter. See the topic “Issuer Script Commands” in section 4 for additional information about the available parameters.
1 = Use a value of 71.
2 = Use a value of 72.

Field Length:	1 numeric character
Required Field:	Yes
Default Value:	1
Data Name:	CPF.EMVCPF.SCRIPT-TPLT-TAG

SCRIPT MAC LENGTH — Specifies the MAC length to be used when an issuer script is sent to the card. Valid values are as follows:

0 = Use the value from the LCONF parameter. See the topic “Issuer Script Commands” in section 4 for additional information about the available parameters.
4 = Use a length of 4 bytes.
6 = Use a length of 6 bytes.
8 = Use a length of 8 bytes.

Field Length:	1 numeric character
Required Field:	Yes
Default Value:	0
Data Name:	CPF.EMVCPF.SCRIPT-MAC-LGTH

CHECK IF HOST ONLINE

The following four fields determine what prescreening checks the BASE24-pos Device Handler/Router/Authorization process should perform if the host is online:

CAM — Determines whether the BASE24-pos Device Handler/Router/Authorization process should perform an authorization request cryptogram prescreening check. Valid values are as follows:

Y = Yes, perform a prescreen check.
N = No, do not perform a prescreen check.

Field Length:	1 alphabetic character
Required Field:	Yes
Default Value:	N
Data Name:	CPF.EMVCPF.PRE-SCRN-CHK

FALLBACK — Determines whether the BASE24-pos Device Handler/Router/Authorization process should perform an EMV fallback prescreening check. Valid values are as follows:

Y = Yes, perform a prescreen check.
N = No, do not perform a prescreen check.

Field Length: 1 alphabetic character
Required Field: Yes
Default Value: N
Data Name: CPF.EMVCPF.PRE-SCRN-CHK-FALLBACK

TVR/CVR — Determines whether the BASE24-pos Device Handler/Router/Authorization process should perform a TVR and CVR prescreening check. Valid values are as follows:

Y = Yes, perform a prescreen check.
N = No, do not perform a prescreen check.

Field Length: 1 alphabetic character
Required Field: Yes
Default Value: N
Data Name: CPF.EMVCPF.PRE-SCRN-CHK-TVR-CVR

REASON ONLINE CODE — Determines whether the BASE24-pos Device Handler/Router/Authorization process should perform a Reason Online Code prescreening check. Valid values are as follows:

Y = Yes, perform a prescreen check.
N = No, do not perform a prescreen check.

Field Length: 1 alphabetic character
Required Field: Yes
Default Value: N
Data Name: CPF.EMVCPF.PRE-SCRN-CHK-RSN-ONL-CDE

BAD CAM ACTION—DATA RELIABLE — Indicates what actions the BASE24-pos Device Handler/Router/Authorization process is to take if the request cryptogram fails verification when the CAM data is reliable. Valid values are as follows:

0 = Continue processing the transaction
1 = Decline the transaction and return the card
2 = Decline the transaction and retain the card
3 = Refer the transaction

If the request cryptogram verification fails, an authorization response cryptogram is not returned (i.e., the EMV Response token is not included in the response message) regardless of the value of this flag.

Field Length:	1 numeric character
Required Field:	Yes
Default Value:	1
Data Name:	CPF.EMVCPF.BAD-CAM-ACT-RELIABLE

BAD CAM ACTION—DATA UNRELIABLE — Indicates what actions the BASE24-pos Device Handler/Router/Authorization process should take if the request cryptogram fails verification when the CAM data is unreliable. Valid values are as follows:

0 = Continue processing the transaction
1 = Decline the transaction and return the card
2 = Decline the transaction and retain the card
3 = Refer the transaction

If the request cryptogram verification fails, an authorization response cryptogram is not returned (i.e., the EMV Response token is not included in the response message) regardless of the value of this flag.

Field Length:	1 numeric character
Required Field:	Yes
Default Value:	1
Data Name:	CPF.EMVCPF.BAD-CAM-ACT-UNRELIABLE

EMV PREFIX ROUTING — A code indicating the routing group for this card prefix. Prefix routing groups are used to group prefixes for different types of authorization processing. The EMV prefix routing group number can be used to define a routing group to be used for EMV transactions, which can be different from the default routing group for this prefix, as specified in the base segment of the CPF. Valid values are as follows:

0–9	=	A prefix routing group number. Any number specified should have a corresponding group number entry in the IDF.
A	=	Any prefix not to be included in a special routing group.

Blank = Use the default value for this prefix, as set in the base segment of the CPF (i.e., EMV cards will use the same routing group as non-EMV cards).

Field Length:	1 alphabetic or numeric character
Required Field:	No
Default Value:	No default value
Data Name:	CPF.EMVCPF.EMV-PREFIX-RTE

CARD VERIFY – CHECK TYPE — Specifies the circumstances under which card verification is to be performed on EMV transactions. Valid values are as follows:

0 = Do not perform card verification.
1 = Perform card verification only when an ARQC is not present.
2 = Perform card verification when an ARQC is not present or when the ARQC has not been verified successfully.
3 = Always perform card verification.

Field Length:	1 numeric character
Required Field:	Yes
Default Value:	0
Data Name:	CPF.EMVCPF.EMV-CV-CHK-TYP

EMV CV DATE — Specifies the earliest card expiration date for which card verification is to be performed on EMV transactions. Valid values are dates in the format YYMM.

Field Length:	4 numeric characters
Required Field:	Yes
Default Value:	0501
Data Name:	CPF.EMVCPF.EMV-CV-EFF-DAT

CHECK METHOD — Specifies the algorithm and data to be used when card verification is performed on EMV transactions. Valid values are as follows:

0 = Use non-EMV card verification regardless of card expiration date.

1 = Replace the service code with the value in the CV DATA field on this screen for cards expiring on or after the date in the EMV CV DATE field, and use non-EMV card verification for cards expiring before the date in the EMV CV DATE field.

2 = Use non-EMV card verification for cards expiring on or after the date in the EMV CV DATE field, and do not perform card verification for cards expiring before the date in the EMV CV DATE field.

3 = Replace the service code with the value in the CV DATA field on this screen for cards expiring on or after the date in the EMV CV DATE field, and do not perform card verification on EMV transactions for cards expiring before the date in the EMV CV DATE field.

Field Length:	1 numeric character
Required Field:	Yes
Default Value:	0
Data Name:	CPF.EMVCPF.EMV-CV-CHK-MTHD

CV DATA — Contains any additional data required when card verification is performed on EMV transactions. Valid values are 000 through 999. For Visa issuers, the value is always 999.

Field Length:	3 numeric characters
Required Field:	Yes
Default Value:	000
Data Name:	CPF.EMVCPF.EMV-CV-DATA

UPDATE COUNTERS

The following three fields specify how a card should update the offline counters when processing a response message received from the issuer. These fields are used only for M/Chip 4.0 and CCD cards.

DECLINED TRANSACTION — Specifies how the card should update the offline counters when processing a response message for a declined transaction. Valid values are as follows:

0 = Do not update the offline counter.
1 = Set the counter to the upper offline limit.
2 = Reset the counter to zero.
3 = Add the transaction to the counter.

Field Length:	1 numeric character
Required Field:	Yes
Default Value:	0
Data Name:	CPF.EMVCPF.EMV-UPDATE-CNTRS-DCLN-TXN

APPROVED FINANCIAL TRANSACTION — Specifies how the card should update the offline counters when processing a response message for an approved financial transaction. Valid values are as follows:

0 = Do not update the offline counter.
1 = Set the counter to the upper offline limit.
2 = Reset the counter to zero.
3 = Add the transaction to the counter.

Field Length:	1 numeric character
Required Field:	Yes
Default Value:	0
Data Name:	CPF.EMVCPF.EMV-UPDATE-CNTRS-APPRV-FNCL-TXN

APPROVED NON-FINANCIAL TRANSACTION — Specifies how the card should update the offline counters when processing a response message for an approved non-financial transaction. Valid values are as follows:

0 = Do not update the offline counter.
1 = Set the counter to the upper offline limit.
2 = Reset the counter to zero.
3 = Add the transaction to the counter.

Field Length:	1 numeric character
Required Field:	Yes
Default Value:	0
Data Name:	CPF.EMVCPF.EMV-UPDATE-CNTRS-APPRV-NON-FNCL-TXN

CPF Screen 12

Screen 12 holds BASE24-atm EMV processing parameters for the card prefix. All fields on CPF screen 12 are used only by BASE24-atm. Refer to the ***BASE24-atm EMV Support Manual*** for the screen layout and field descriptions.

CPF Screen 13

Screen 13 holds BASE24-pos EMV processing parameters for the card prefix. This section describes the fields present on CPF screen 13.

```
BASE24-EMV     CARD PREFIX            LLLL        YY/MM/DD  HH:MM  13 OF 22
 PREFIX:                     PAN LENGTH: 00                 FIID:

                            POS EMV INFORMATION
                    FALLBACK ACTION: 0 (CONTINUE)
             MANUAL FALLBACK ACTION: 0 (CONTINUE)
        FORCE ONLINE FALLBACK CHECK: N (Y/N)
                         EMV ISSUER: 00 (**********)
               DELAYED AUTH SUPPORT: N (Y/N)
                          CAP GROUP:
                     CAP ATC UPDATE: 0 (CAP)
     BAD CAP TOKEN VALIDATION LIMIT:    0
   HOST CAP TOKEN VALIDATION OPTION: 0 (APPROVE, DON'T SEND TO HOST)







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POS EMV INFORMATION

The following fields contain BASE24-pos EMV processing parameters for the card prefix.

FALLBACK ACTION — Determines what action the BASE24-pos Device Handler/Router/Authorization process takes for POS fallback transactions. Valid values are as follows:

0 = Continue processing the transaction
1 = Decline the transaction and return the card
3 = Refer the transaction

Field Length:	1 numeric character
Required Field:	Yes
Default Value:	0
Data Name:	CPF.EMVCPF.FALLBACK-ACT-POS

MANUAL FALLBACK ACTION — Determines what action the BASE24-pos Device Handler/Router/Authorization process takes for manually-entered EMV fallback transactions. Valid values are as follows:

0 = Continue processing the transaction (Default)
1 = Decline the transaction and return the card
3 = Refer the transaction (POS only)

Field Length:	1 numeric character
Required Field:	Yes
Default Value:	0
Data Name:	CPF.EMVCPF.FALLBACK-ACT-MANUAL

FORCE ONLINE FALLBACK CHECK — Determines whether a POS fallback transaction must be forced online to a host or issuer system for authorization. This field is used only for POS transactions configured for level 3 authorization. Valid values are as follows:

Y = Yes, force online.
N = No, do not force online.

Field Length:	1 alphabetic character
Required Field:	Yes
Default Value:	N
Data Name:	CPF.EMVCPF.FORCE-ONL-FALLBACK-CHK

EMV ISSUER — A user-defined indicator identifying the issuer of the EMV card with this prefix. Valid values are as follows:

00 to 29 = On-us transaction (i.e., the issuer has an associated IDF record)
30 to 99 = Not-on-us transaction (i.e., the issuer does not have an associated IDF record)

This field is followed by a user-defined text description of the EMV card issuer.

Field Length:	2 numeric characters, followed by 10 alphanumeric characters
Required Field:	Yes
Default Value:	00, followed by **********
Data Name:	CPF.EMVCPF.EMV-ISS CPF.EMVCPF.EMV-ISS-DESCR

DELAYED AUTH SUPPORT — Indicates whether the issuer accepts delayed authorizations. Valid values are as follows:

Y = Yes, the issuer accepts delayed authorizations.
N = No, the issuer does not accept delayed authorizations.

Field Length:	1 alphabetic character
Required Field:	Yes
Default Value:	N
Data Name:	CPF.EMVCPF.DELAYED-AUTH-SPPT

CAP GROUP — The CAP profile that is used by BASE24 Transaction Security Services (TSS) to retrieve the static data required to validate the CAP token in an authorization message. A value of spaces indicates that CAP token verification is not supported for this prefix.

Field Length:	1–4 alphanumeric characters
Required Field:	No
Default Value:	No default value
Data Name:	CPF.EMVCPF.CAP-GRP

CAP ATC UPDATE — A code that indicates which ATC field on the CAF should be updated on a CAP token validation request. If the CAP application is separate to the EMV application on the card, the CAP ATC NUMBER field on CAF screen 13 should be updated. If the CAP application is part of the EMV application on the card, then the ATC NUMBER field on CAF screen 13 or the ATC NUMBER field on CAF screen 2 should be updated (depending on which ATC NUMBER field is used in EMV processing). Valid values are as follows:

0 = Update the CAP ATC field. This is the default value.
1 = Update the ATC field used in EMV processing.

Field Length:	1 numeric character
Required Field:	Yes
Default Value:	0
Data Name:	CPF.EMVCPF.CAP-ATC-UPDT

BAD CAP TOKEN VALIDATION LIMIT — The maximum number of consecutive failed CAP Token validation attempts permitted. When this limit is reached for a card, any subsequent requests for CAP token validation will be declined without the validation being performed. Valid values are 0 to 9999. A value of zero disables the limit check.

Field Length:	1–4 numeric characters
Required Field:	Yes
Default Value:	0
Data Name:	CPF.EMVCPF.BAD-CAP-TKN-CHK-LMT

HOST CAP TOKEN VALIDATION OPTION — A code identifying whether BASE24-pos can authorize a CAP token validation request, even if Level 3 (online/offline) authorization is configured for the card. If BASE24-pos validates the CAP token, there would be no requirement to forward the request to an issuer host system because there is no further authorization processing to be performed for the transaction. Valid values are as follows:

0 = Do not send to host. Authorize and log to the PTLF.
1 = Send to host. Authorize if host is down.
2 = Send to host. Deny if host is down.

Field Length:	1 numeric character
Required Field:	Yes
Default Value:	0
Data Name:	CPF.EMVCPF.HOST-CAP-TKN-OPT

Cardholder Authorization File (CAF)

The Cardholder Authorization File (CAF) contains one record for each cardholder whose card-issuing institution uses the Positive or Positive Balance Authorization method. CAF records contain authorization parameters and usage accumulation information for the card issuer's cardholders and are used in authorizing transaction requests.

CAF Screen 2

CAF screen 2 displays the application transaction counter (ATC) tracked in the Base segment of the CAF, which may or may not be used for EMV transactions, depending on how the CPF is configured. CAF screen 2 is shown below, followed by the description of the ATC NUMBER field.

```
BASE24-BASE  CARDHOLDER FILE          LLLL        YY/MM/DD  HH:MM  02 OF 21

        PAN:                                   MEMBER: 000    FIID:


                        CARD USAGE CONTROL

                        BAD PIN TRIES:    0

               EXPIRATION DATE (YYMM): 0000

                      DATE FIRST USED:

                      LAST RESET DATE:

                           ATC NUMBER:


*********************************** BASE24 ***********************************
 NEW PAGE:        FILE DESTINATION:       NEW LOGICAL NETWORK ID:
                        F12 - HELP
```

ATC NUMBER — Stores the last application transaction counter (ATC) received by BASE24 from the card. This field is used to record the number of transactions performed on the card when the ATC is contained in the Base segment instead of the EMV segment.

Field Length:	System protected
Data Name:	CAF.CAFBASE.ATC

CAF Screen 7

CAF Screen 7 enables institutions to set the expiration date and card status for the second card. This screen also displays the second card ATC for EMV transactions tracked in the Base segment of the CAF, which may or may not be used, depending on how the CPF is configured. Many institutions reissue cards, and a period exists when two physical cards with the same card number are in circulation. CAF screen 7 is shown below, followed by a description of the SECOND CARD ATC NUMBER field.

```
BASE24-BASE  CARDHOLDER FILE          LLLL        YY/MM/DD  HH:MM  07 OF 21

        PAN:                                 MEMBER: 000    FIID:


                    SECOND CARD USAGE CONTROL

             EXPIRATION DATE (YYMM): 0000

                        CARD STATUS: 6   (CARD NOT USED)

             SECOND CARD ATC NUMBER:








*********************************** BASE24 ***********************************
 NEW PAGE:        FILE DESTINATION:       NEW LOGICAL NETWORK ID:
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```

SECOND CARD ATC NUMBER — The last application transaction counter (ATC) received by BASE24 from the second card. This field is used to record the number of transactions performed on the second card when the ATC is contained in the Base segment instead of the EMV segment.

Field Length: System protected
Data Name: CAF.CAFBASE.ATC-SCND-CRD

CAF Screen 13

Screen 13 is specific to EMV authorization processing, and holds the CAM authentication control data for the ICC. CAF screen 13 is shown below, followed by descriptions of its fields.

```
BASE24-EMV   CARDHOLDER FILE          LLLL        YY/MM/DD  HH:MM  13 OF 21

         PAN:                                  MEMBER: 000    FIID:

                              EMV INFORMATION
            ATC LIMIT: 9999             BAD CAP TOKEN OVERRIDE FLAG: N
       EMV IAD FORMAT: 0 (USE CPF VALUE)  STATUS CHECK ACTION INDEX: 1 (1-4)
  SCRIPT TEMPLATE TAG: 0 (USE CPF VALUE)          SCRIPT MAC LENGTH: 0
          SEND CARD BLOCK: N                          SEND PUT DATA: N
                                   LOWER CONSECUTIVE OFFLINE LIMIT:  0
         SEND PIN UNBLOCK: 0 (NO ACTION)
          SEND PIN CHANGE: 0 (NO ACTION)
  ONLINE/OFFLINE PIN SYNC: 0 (NO ACTION)

                     ATC     CAP ATC      BAD CAP
                   NUMBER    NUMBER    TOKEN COUNTER   CAP APSN    CAP DKI
    PRIMARY CARD:     0          0            0                      00
  SECONDARY CARD:     0          0            0                      00


*********************************** BASE24 ***********************************
 NEW PAGE:        FILE DESTINATION:       NEW LOGICAL NETWORK ID:
                    F12 - HELP
```

EMV INFORMATION

The following fields contain information for EMV authorization.

ATC LIMIT — Specifies the acceptable number of transactions that can be performed on the card before issuer authorization is required. The range of valid values is 0–9999. A value of zero disables the limit check.

Field Length:	4 numeric characters
Required Field:	Yes
Default Value:	9999
Data Name:	CAF.EMVCAF.ATC-LMT

BAD CAP TOKEN OVERRIDE FLAG — A flag controlling whether additional CAP Token validation attempts can be performed if the number of consecutive failed CAP Token validation errors specified in the BAD CAP TOKEN VALIDATION LIMIT field on Card Prefix File (CPF) screen 13 has been reached. Valid values are as follows:

Y = Yes, additional validation attempts can be performed.
N = No, additional validation attempts cannot be performed.

If the validation is successful, the values in the BAD CAP TOKEN COUNTER fields on this CAF screen are reset to zero. This mechanism is the only way that the values in the BAD CAP TOKEN COUNTER fields can be reset once the limit has been reached without deleting and re-adding or replacing the CAF record.

Field Length:	1 alphabetic character
Required Field:	Yes
Default Value:	N
Data Name:	CAF.EMVCAF.BAD-CAP-TKN-OVRRD-FLG

EMV IAD FORMAT — Specifies the format of the EMV Issuer Application Data (IAD) that will be used for the card.

Note: The values in the EMV IAD FORMAT and ALTERNATE IAD FORMAT fields on CPF screen 11 identify which format is being used on a card. Prior to the ALTERNATE IAD FORMAT field on CPF screen 11 being available, the EMV IAD FORMAT field on CAF screen 13 enabled a card issuer to use the M/Chip 4 format for specific accounts within a card prefix that was still using the M/Chip 2.1 format.

Valid values are as follows:

0 = Use the format identified by the setting in the EMV IAD FORMAT field on CPF screen 11.

1 = The format as defined by Visa in the Visa Integrated Circuit Card (ICC) specification. This format is also used for UKIS cards.

The format is defined by the VISA-APPL-DATA data elements of the EMV Request Data token.

2 = The format recommended for MasterCard MCPA M/Chip 2.1 cards.

The format is defined by the MCPA-APPL-DATA data elements of the EMV Request Data token.

3 = Use the IAD format recommended for MasterCard MCPA M/Chip 4 cards.

The format is defined by the MCHIP4-APPL-DATA data elements of the EMV Request Data token.

Field Length:	1 numeric character
Required Field:	Yes
Default Value:	0
Data Name:	CAF.EMVCAF.ISS-APPL-DATA-FRMT

SCRIPT TEMPLATE TAG — Specifies the template tag to be used when an issuer script is sent to the card. Valid values are as follows:

0 = Use the value from the SCRIPT TEMPLATE TAG field on CPF screen 11.
1 = Use a value of 71.
2 = Use a value of 72.

Field Length:	1 numeric character
Required Field:	Yes
Default Value:	1
Data Name:	CAF.EMVCAF.SCRIPT-TPLT-TAG

SCRIPT MAC LENGTH — Specifies the MAC length to be used when an issuer script is sent to the card. Valid values are as follows:

0 = Use the value from the SCRIPT MAC LENGTH field on CPF screen 11.
4 = Use a length of 4 bytes.
6 = Use a length of 6 bytes.
8 = Use a length of 8 bytes.

Field Length:	1 numeric character
Required Field:	Yes
Default Value:	0
Data Name:	CAF.EMVCAF.SCRIPT-MAC-LGTH

STATUS CHECK ACTION INDEX — Identifies which of the rules for status field processing are to be used for the card. This field is used when M/Chip 2.1 and M/Chip 4 cards are issued with the same card prefix.

When the EMV IAD FORMAT field on CAF screen 13 is set to a value of 0 (use CPF value), the value in the STATUS CHECK ACTION INDEX field defined on CPF screen 11 (at the Prefix level) is used instead of the value in the STATUS CHECK ACTION INDEX field on CAF screen 13 (this field).

These rules are defined in the CVR and TVR tables. For more information on these tables, see the topic “Status Field Verification” in section 4. The range of valid values is 1–4.

Field Length:	1 numeric character
Required Field:	Yes
Default Value:	1
Data Name:	CAF.EMVCAF.ACTION-TABLE-INDEX

SEND CARD BLOCK — Determines whether a CARD BLOCK script command should be sent to the terminal to disable the microchip. Valid values are as follows:

Y = Yes, send CARD BLOCK script command.
N = No, do not send CARD BLOCK script command.

Note that this field is not automatically reset once an attempt to send the script has been made.

Field Length:	1 alphabetic character
Required Field:	Yes
Default Value:	N
Data Name:	CAF.EMVCAF.SEND-CARD-BLK

SEND PUT DATA — Determines whether a PUT DATA script command should be sent to the terminal to update the microchip data (e.g., new limits data). Valid values are as follows:

Y = Yes, send PUT DATA script command.
N = No, do not send PUT DATA script command.

This field is automatically reset once an attempt to send the script has been made.

Field Length:	1 alphabetic character
Required Field:	Yes
Default Value:	N
Data Name:	CAF.EMVCAF.SEND-PUT-DATA

LOWER CONSECUTIVE OFFLINE LIMIT — Specifies the maximum number of consecutive transactions that can be performed on the card before an attempt must be made to obtain authorization from the issuer. This field is used by BASE24-pos to generate the variable component of the PUT DATA script command so that a card can have a new limit downloaded. The range of valid values is 0–255.

Field Length:	4 numeric characters
Required Field:	Yes
Default Value:	0
Data Name:	CAF.EMVCAF.VEL-LMTS.LWR-CONSEC-LMT

SEND PIN UNBLOCK — Specifies the circumstances under which a PIN Unblock script command is generated and returned by the BASE24-atm system. Valid values are as follows:

0 = No action required (no EMV processing)

1 = Implicit (an EMV transaction indicates that the offline bad PIN try limit has been exceeded)

2 = Explicit (an EMV PIN Unblock transaction is received)

3 = Implicit and Explicit (an EMV transaction indicates that the offline bad PIN try limit has been exceeded or an EMV PIN Unblock transaction is received)

Field Length:	1 numeric character
Required Field:	Yes
Default Value:	0
Data Name:	CAF.EMVCAF.SEND-PIN-UNBLK

Note: This field is not currently supported by the BASE24-pos EMV product. For more information about this field, refer to the ***BASE24-atm EMV Support Manual.***

SEND PIN CHANGE — Specifies the circumstances under which a PIN Change script command is generated and returned by the BASE24-atm system. Valid values are as follows:

0 = Do not send script (no EMV processing)
1 = An EMV PIN Unblock transaction is received
2 = An EMV PIN Change transaction is received
3 = An EMV PIN Unblock transaction or and EMV PIN Change transaction is received

Field Length:	1 numeric character
Required Field:	Yes
Default Value:	0
Data Name:	CAF.EMVCAF.SEND-PIN-CHNG

Note: This field is not currently supported by the BASE24-pos EMV product. For more information about this field, refer to the ***BASE24-atm EMV Support Manual.***

ONLINE/OFFLINE PIN SYNC — Specifies whether synchronization of the online and offline PIN is required for the card. Valid values are as follows:

0 = PIN synchronization not required (no EMV processing)
1 = PIN synchronization required

Field Length:	1 numeric character
Required Field:	Yes
Default Value:	0
Data Name:	CAF.EMVCAF.PIN-CHNG-ACT

Note: This field is not currently supported by the BASE24-pos EMV product. For more information about this field, refer to the ***BASE24-atm EMV Support Manual.***

PRIMARY CARD DATA

The following five fields contain information about the primary card.

ATC NUMBER — The last application transaction counter (ATC) received by BASE24 from the primary card. This field is used to record the number of transactions performed on the primary card when the ATC is contained in the EMV segment instead of the Base segment.

Field Length:	System Protected
Data Name:	CAF.EMVCAF.ATC

CAP ATC NUMBER — The ATC received by BASE24 in the last approved chip authentication program (CAP) token validation transaction performed on the primary card.

Field Length:	System Protected
Data Name:	CAF.EMVCAF.CAP-DATA.ATC

BAD CAP TOKEN COUNTER — The number of consecutive failed CAP token validation attempts performed on the card. This value is reset to zero when a CAP token is validated successfully for the primary card.

Field Length:	System Protected
Data Name:	CAF.EMVCAF.CAP-DATA.BAD-CAP-TKN-CHK-ACCUM

CAP APSN — Specifies the CAP application PAN sequence number (APSN) for the card identified by the values in the PAN and MEMBER fields on each CAF screen. The APSN identifies and differentiates cards with the same PAN. In EMV transactions, this field is generally sent to BASE24 as part of the transaction data. However, for CAP transactions, it may not be present, which means the value has to be configured on the database.

Field Length:	1–2 numeric characters
Required Field:	No
Default Value:	No default value
Data Name:	CAF.EMVCAF.CAP-DATA.APSN

CAP DKI — Specifies the derivation key index (DKI) for the card identified by the values in the PAN and MEMBER fields on each CAF screen. The DKI identifies which of a set of master keys is to be used in the derivation of the authorization cryptogram for the card. In EMV transactions, this field is generally sent to BASE24 as part of the transaction data. However, for CAP transactions, it may not be present, which means the value has to be configured on the database.

Field Length:	2 numeric characters
Required Field:	Yes
Default Value:	00
Data Name:	CAF.EMVCAF.CAP-DATA.DKI

SECONDARY CARD DATA

The following five fields contain information about the secondary card.

ATC NUMBER — The last application transaction counter (ATC) received by BASE24 from the secondary card. This field is used to record the number of transactions performed on the secondary card when the ATC is contained in the EMV segment instead of the Base segment.

Field Length:	System Protected
Data Name:	CAF.EMVCAF.SCND-CRD-DATA.ATC-2

CAP ATC NUMBER — The ATC received by BASE24 in the last approved chip authentication program (CAP) token validation transaction performed on the secondary card.

Field Length:	System Protected
Data Name:	CAF.EMVCAF.SCND-CRD-DATA.CAP-DATA-2.ATC

BAD CAP TOKEN COUNTER — The number of consecutive failed CAP token validation attempts performed on the secondary card. This value is reset to zero when a CAP token is validated successfully for the card.

Field Length:	System Protected
Data Name:	CAF.EMVCAF.SCND-CRD-DATA.CAP-DATA-2.BAD-CAP-TKN-CHK-ACCUM

CAP APSN — Specifies the CAP application PAN sequence number (APSN) for the secondary card identified by the values in the PAN and MEMBER fields on each CAF screen. The APSN identifies and differentiates cards with the same

PAN. In EMV transactions, this field is generally sent to BASE24 as part of the transaction data. However, for CAP transactions, it may not be present, which means the value has to be configured on the database.

Field Length:	1–2 numeric characters
Required Field:	No
Default Value:	No default value
Data Name:	CAF.EMVCAF.SCND-CRD-DATA.CAP-DATA-2.APSN

CAP DKI — Specifies the derivation key index (DKI) used for the secondary card identified by the values in the PAN and MEMBER fields on each CAF screen. The DKI identifies which of a set of master keys is to be used in the derivation of the authorization cryptogram for the card. In EMV transactions, this field is generally sent to BASE24 as part of the transaction data. However, for CAP transactions, it may not be present, which means the value has to be configured on the database.

Field Length:	2 numeric characters
Required Field:	Yes
Default Value:	00
Data Name:	CAF.EMVCAF.SCND-CRD-DATA.CAP-DATA-2.DKI

ICC KEY File (KEYI)

The ICC KEY File (KEYI) supports the key storage requirements for the BASE24-pos EMV product. The KEYI is not used for customers who have implemented BASE24 Transaction Security Services (TSS). The key to the KEYI is the KEYI GROUP, BEGIN DATE, and END DATE fields.

The following keys are stored in the KEYI:

- A set of up to 24 double length DES keys used for authorization request cryptogram verification and authorization response cryptogram generation. These keys are indexed by the derivation key index obtained from the transaction request.
- A double length DES key used for secure messaging, the generation of the issuer script message authentication code (MAC).

The KEYI supports the key information required for ICC transactions. The KEYI holds multiple authentication keys (up to a maximum of 24) for an individual KEYI group.

A master key is held for use by the secure messaging facility for scripts. Secure messaging keys are different from those used by the card in request and response cryptogram processing.

KEYI Screen 1

KEYI screen 1 is shown below, followed by descriptions of its fields:

```
BASE24-EMV   EMV KEY FILE             LLLL        YY/MM/DD  HH:MM  01 OF 01
         KEYI GROUP:
 BEGIN DATE (YYYYMM): ******   END DATE (YYYYMM): ******          FIID:
                             EMV SCHEME: 0 (UKIS)
   CRYPTOGRAPHIC KEYS:
 1   0000000000000000 0000000000000000  2   0000000000000000 0000000000000000
 3   0000000000000000 0000000000000000  4   0000000000000000 0000000000000000
 5   0000000000000000 0000000000000000  6   0000000000000000 0000000000000000
 7   0000000000000000 0000000000000000  8   0000000000000000 0000000000000000
 9   0000000000000000 0000000000000000  10  0000000000000000 0000000000000000
 11  0000000000000000 0000000000000000  12  0000000000000000 0000000000000000
 13  0000000000000000 0000000000000000  14  0000000000000000 0000000000000000
 15  0000000000000000 0000000000000000  16  0000000000000000 0000000000000000
 17  0000000000000000 0000000000000000  18  0000000000000000 0000000000000000
 19  0000000000000000 0000000000000000  20  0000000000000000 0000000000000000
 21  0000000000000000 0000000000000000  22  0000000000000000 0000000000000000
 23  0000000000000000 0000000000000000  24  0000000000000000 0000000000000000
 SECURE MESSAGING KEYS       FOR INTEGRITY: 0000000000000000 0000000000000000
                       FOR CONFIDENTIALITY: 0000000000000000 0000000000000000

 *********************************** BASE24 ***********************************
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```

KEYI GROUP — The primary key to the KEYI for this prefix. The value in this field must be the same as the value in the KEYI GROUP field on CPF screen 11.

Field Length:	32 alphanumeric characters
Required Field:	Yes
Default Value:	No default value
Data Name:	KEYI.KEYICC.KEYI-GRP

BEGIN DATE — The first date in the range of expiration dates to which information in this KEYI record applies. If you do not want to use this functionality, place asterisks in the BEGIN DATE and END DATE fields. Asterisks are stored as WWWWWW and cause the same KEYI record to be used regardless of card expiration date. Valid values are as follows:

000000	=	Any card expiration date that is less than or equal to the specified value in the END DATE field uses this record.

yyyymm	=	Any card expiration date must be greater than or equal to the beginning date, but less than or equal to the end date.
******	=	Any card whose expiration date has not found a match on a date in a KEYI record uses this as a wild card match.

Field Length: 6 alphanumeric characters
Required Field: Yes
Default Value: ******
Data Name: KEYI.KEYICC.BEG-DAT

END DATE — The last date in the range of card expiration dates to which information in this KEYI record applies. If you do not want to use this functionality, place asterisks in the BEGIN DATE and END DATE fields. Asterisks are stored as WWWWWW and cause the same KEYI record to be used regardless of card expiration date. Valid values are as follows:

yyyymm	=	Any card expiration date must be greater than or equal to the beginning date, but less than or equal to the end date.
******	=	Any card whose expiration date has not found a match on a date in a KEYI record uses this as a wild card match.

Field Length: 6 alphanumeric characters
Required Field: Yes
Default Value: ******
Data Name: KEYI.KEYICC.END-DAT

FIID — The financial institution identifier of the institution to which this record applies.

Field Length: 4 alphanumeric characters
Required Field: No
Default Value: No default value
Data Name: KEYI.KEYICC.FIID

EMV SCHEME — The specific method used to perform the EMV security processing. EMV specifications define the high level EMV security requirements, but the individual payment schemes support different implementations. This field identifies the specific implementation associated with this KEYI record and is used by the security utilities to determine the specific security module calls and security functions to be performed. Valid values are as follows:

0 = UKIS security scheme
1 = Visa security scheme as defined in Visa ICC Specifications
2 = MCPA (M/Chip) security scheme as defined in MasterCard specifications

Field Length:	1 numeric character
Required Field:	No
Default Value:	0
Data Name:	KEYI.KEYICC.EMV-SEC-SCHEME

CRYPTOGRAPHIC KEYS — A set of up to 24 issuer master keys used to generate a unique EMV master key for generation of application cryptograms. The specific master key used for an individual chip card is indicated by the derivation key index that is personalized in the card and passed in the request message. Some issuer security schemes utilize a single master key. In this case a derivation key index of zero is supplied and the first entry in the array is used.

Field Length:	16 alphanumeric characters
Occurs	Up to 24 times
Required Field:	No
Default Value:	0000000000000000
Data Name:	KEYI.KEYICC.CRYPTO-KEYS

SECURE MESSAGING KEYS

The following two fields identify the secure messaging keys for integrity and confidentiality:

FOR INTEGRITY — The double length secure messaging key for message integrity. This is an issuer master key used to generate a unique EMV master key that is used in derivation of a session key used to authenticate script commands (using a MAC) for message integrity.

It is assumed that the secure messaging key retrieved from the KEYI and passed to the CAM utilities from the BASE24-pos Device Handler/Router/Authorization process is the appropriate key for the call to either the Thales or Atalla security module.

Field Length:	16 alphanumeric characters
Required Field:	No
Default Value:	0000000000000000
Data Name:	KEYI.KEYICC.SM-KEY

FOR CONFIDENTIALITY — The double length secure messaging key for message confidentiality.

Note: The EMV product supports the confidentiality aspect of secure messaging, (e.g., encrypting PIN data in issuer scripts) by communicating with the Transaction Security Services (TSS). Therefore, the Secure Messaging Key for Confidentiality is maintained on the TSS database, and this KEYI field is not used.

LCONF Assigns and Params

The BASE24 EMV product modules use several standard Logical Network Configuration File (LCONF) assigns and params, together with the EMV-specific assigns and params listed below.

One assign is specific to the EMV product: The KEYI assign identifies the location of the ICC Key file (KEYI).

The following params are specific to the EMV product:

- The EMV-CARD-BLOCK-SCRIPT param specifies the hexadecimal representation of binary data that is used to block an EMV card.
- The EMV-PUT-DATA-SCRIPT param specifies the hexadecimal representation of binary data that is used to reset the lower consecutive offline limit on an EMV card.

Refer to the “Issuer Script Commands” topic in section 4 for a description and examples of how these params are used.

Valid Values

The LCONF parameters EMV-CARD-BLOCK-SCRIPT and EMV-PUT-DATA-SCRIPT must contain the values specified in the table below.

The last two bytes of each of these params specifies the length of the MAC to be appended to the end of the script command. For example, a value of 04 in the last two bytes of the EMV-CARD-BLOCK-SCRIPT param indicates a 4-byte MAC, while a value of 07 in the last two bytes of the EMV-PUT-DATA-SCRIPT param indicates a 6-byte MAC (actually a 1-byte parameter value followed by the 6-byte MAC).

Format 1 secure messaging, which is used by the CCD EMV scheme, uses only a 4-byte MAC. Format 2 secure messaging, which is used by the other EMV schemes, uses 4-byte, 6-byte, and 8-byte MACs.

All lengths in the following table are specified in hexadecimal.

Byte Description	CARD BLOCK	PUT DATA
Template Tag	71 or 72	71 or 72
Template Length	0E or 12 10 or 14 12 or 16	0F or 13 11 or 15 13 or 17
Script ID Tag (2 bytes)	9F18	9F18
Script ID Length	00 or 04	00 or 04
Script ID Value (4 bytes) [1]	XXXXXXXX[2]	XXXXXXXX[2]
Script Tag	86	86
Script Length	09, 0B, or 0D	0A, 0C, or 0E
CLA	84	04
INS	16 or 1E	DA
P1	00	Data Element Tag (first byte)
P2	00	Data Element Tag (second byte)
Data Length	04, 06, or 08	05, 07, or 09
Command Data	Not present	FF
MAC Length	04, 06, or 08 bytes	04, 06, or 08 bytes

[1] The script ID value is optional. The script ID length identifies whether the script ID is present.

[2] The value X represents any hexadecimal digit in the script ID value.

Examples

The following are some examples for each of the params.

All lengths in the following table are specified in hexadecimal.

Script	Template		Script ID			Script Command[1, 2]				
	T	L	T	L	V	T	L	V		
Card Block	71	0E	9F18	00		86	09	84 16 00 00	04	M4
	71	12	9F18	04	12 34 56 78	86	09	84 16 00 00	04	M4
	71	10	9F18	00		86	0B	84 16 00 00	06	M6
	71	14	9F18	04	12 34 56 78	86	0B	84 16 00 00	06	M6
	71	12	9F18	00		86	0D	84 16 00 00	08	M8
	71	16	9F18	04	12 34 56 78	86	0D	84 16 00 00	08	M8
Put Data	72	0F	9F18	00		86	0A	04 DA 9F 58	05	M4
	72	13	9F18	04	12 34 56 78	86	0A	04 DA 9F 58	05	M4
	72	11	9F18	00		86	0C	04 DA 9F 58	07	M6
	72	15	9F18	04	12 34 56 78	86	0C	04 DA 9F 58	07	M6
	72	13	9F18	00		86	0E	04 DA 9F 58	09	M8
	72	17	9F18	04	12 34 56 78	86	0E	04 DA 9F 58	09	M8

1 To support an Application Block script instead of a Card Block script, the second binary byte of the V portion of the Script Command can be set to a value of 1E instead of 16.

2 To support an update of a 1-byte EMV data element other than the LCOL in the Put Data script, the last two binary bytes of the V portion of the Script Command can be set to the EMV tag of the data element to be updated instead of 9F58 (LCOL).

POS Terminal Data (PTD) File

Screen 17 of the POS Terminal Data (PTD) File is specific to the EMV product. This screen allows you to indicate the EMV capabilities of each terminal. PTD screen 17 is shown below, followed by descriptions of its fields.

```
BASE24-POS   TERMINAL DATA            LLLL        YY/MM/DD  HH:MM 17 OF 19

TERMINAL ID:                                          FIID:
                           EMV TERMINAL CAPABILITIES

                      KEY ENTRY:  N                       MAGNETIC STRIPE:  N
                    IC/CONTACTS:  N               RESERVED FOR FUTURE USE:  N
        RESERVED FOR FUTURE USE:  N               RESERVED FOR FUTURE USE:  N

 PLAINTEXT PIN/ICC VERIFICATION:  N    ENCIPHERED PIN/ONLINE VERIFICATION:  N
                      SIGNATURE:  N   ENCIPHERED PIN/OFFLINE VERIFICATION:  N
               NO CVM REQUIRED:  N               RESERVED FOR FUTURE USE:  N

     STATIC DATA AUTHENTICATION:  N           DYNAMIC DATA AUTHENTICATION:  N
                   CARD CAPTURE:  N               RESERVED FOR FUTURE USE:  N
   COMBINED DATA AUTHENTICATION:  N               RESERVED FOR FUTURE USE:  N

   ENTER 'Y' FOR EACH FACILITY WHICH IS SUPPORTED BY THIS TERMINAL

*********************************** BASE24 ***********************************
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EMV TERMINAL CAPABILITIES

The following three fields are used to indicate the methods supported by the terminal for entering the information from the card into the terminal:

KEY ENTRY
MAGNETIC STRIPE
IC/CONTACTS

Valid values for each of the three fields are as follows:

Y = Yes, the terminal has this capability.
N = No, the terminal does not have this capability.

The following five fields are used to indicate the methods supported by the terminal for verifying the identity of the cardholder at the terminal:

PLAINTEXT PIN/ICC VERIFICATION
ENCIPHERED PIN/ONLINE VERIFICATION
SIGNATURE
ENCIPHERED PIN/OFFLINE VERIFICATION
NO CVM REQUIRED

Valid values for each of the five fields are as follows:

Y = Yes, the terminal has this capability.
N = No, the terminal does not have this capability.

The following four fields are used to indicate the methods supported by the terminal for authenticating the card at the terminal and whether the terminal has the ability to capture a card.

STATIC DATA AUTHENTICATION
DYNAMIC DATA AUTHENTICATION
CARD CAPTURE
COMBINED DATA AUTHENTICATION

Valid values for each of the four fields are as follows:

Y = Yes, the terminal has this capability.
N = No, the terminal does not have this capability.

Note: Because of the way the TAL language handles bit operations on string variables, the bits set in the PTD record may not seem to match the EMV specifications. Refer to the EMV-TERM-CAP field in the DDLGDEFS file on the BA60DEFS subvolume for additional information.

3: Transaction Processing

When an integrated circuit card is presented at the terminal, the terminal determines which applications are supported by both the card and the terminal. The terminal displays the supported applications to the cardholder and the cardholder selects the appropriate application.

This section describes the transaction paths of the BASE24-pos EMV add-on product. Only the additional steps required by an EMV transaction are described. These diagrams do not include the standard BASE24-pos processing.

Token Usage

BASE24 uses six tokens to carry EMV-specific data during transaction processing.

The EMV Request Data token (token ID B2) contains the thirteen minimum request data elements required for inclusion in request messages, as defined by EMV. The Device Handler module of the BASE24-pos Device Handler/Router/Authorization process or the Interchange Interface process creates this token and adds it to the transaction message before sending it to the Router and Authorization modules of the BASE24-pos Device Handler/Router/Authorization process.

The EMV Discretionary Request Data token (token ID B3) consists of EMV-related data that is not required for authorization. However, each data element is supported by more than one EMV-compliant interface and, therefore, can be mapped between interfaces by BASE24.

The EMV Status token (token ID B4) holds data identifying the status of a transaction. The Device Handler modules and Interchange Interface processes create this token and add it to the PSTM before sending it to the Router and Authorization modules of the BASE24-pos Device Handler/Router/Authorization process. The acquiring endpoint adds the token when the transaction originates from an EMV-capable terminal, regardless of whether or not the data relates to an EMV transaction.

The EMV Response Data token (token ID B5) contains the response cryptogram, data required to generate the response cryptogram, and flags used to identify the scripts to be returned to the acquirer. If authorization is performed on BASE24, the BASE24-pos Device Handler/Router/Authorization process creates this token. If the transaction is routed to an interchange or a host for authorization, the BASE24 Interchange Interface process or the BASE24 ISO Host Interface process creates the token.

The EMV Script Data token (token ID B6) holds EMV script data. The issuer process creates this token. In the context of EMV transactions, the issuer process can be an Interchange Interface process if the issuer is external to BASE24, or the Router and Authorization modules of the BASE24-pos Device Handler/Router/Authorization process if BASE24 is configured for offline or online/offline authorization. The token is added to the PSTM before returning the message to the acquiring process. This token is present only if the transaction response contains script data.

The Authentication Data token (token ID CE) holds the request message data needed for the BASE24-pos Device Handler/Router/Authorization process to perform a chip authentication program (CAP) token validation request, the application transaction counter (ATC) returned from the host security module (HSM), and a code indicating the results of the CAP token validation request.

Refer to the ***BASE24 Tokens Manual*** for token formats, including the corresponding EMV tags, for each field in these tokens. All EMV tokens are considered Base message tokens, and are presented in the BASE24 Base Tokens section of the manual.

For more information about the EMV data elements, refer to the MasterCard M/Chip or Visa Smart Debit Credit (VSDC) documentation sets, or the EMVCo specification.

Offline PIN Support

Information about the offline PIN processing performed at the POS device must be passed through the BASE24 system in authorization request messages. In particular, the following requirements must be met. The EMV data elements that must be present are listed in parentheses.

- The POS terminal supports offline PINs (Terminal Capabilities, byte 2, bits 8 and 5).
- The POS terminal supports Dynamic Data Authentication (Terminal Capabilities, byte 3, bit 7).
- Cardholder verification was successful (Terminal Verification Results, byte 3, bit 8).
- The offline PIN try limit has been exceeded (Terminal Verification Results, byte 3, bit 6).
- A PIN was entered, given that PIN entry was required and a PIN pad was available (Terminal Verification Results, byte 3, bit 4).
- Offline PIN processing was performed (Card Verification Results). The location of the offline PIN processing indicator depends on the card format, as shown in the table below.
- Offline PIN processing was successful (Card Verification Results). The location of the offline PIN verification indicator depends on the card format, as shown in the table below.

Card Format	Offline PIN Processing Indicator Location	
	Performed	**Successful**
M/Chip 2.1,Visa, and AMEX	Byte 2, bit 3	Byte 2, bit 2
M/Chip 4	Byte 4, bit 6	Byte 4, bit 5
CCD	Byte 2, bit 4	Byte 2, bit 3

Note that the Terminal Capabilities field is an optional EMV field, and cannot be provided by some systems, for example VisaNet. However, for POS devices directly-connected to BASE24, for example the APACS 70 terminals, the EMV Terminal Capabilities can be configured on the BASE24 database and are always available.

Processing provides support for passing this data through the BASE24-pos system, provided that the external interfaces at both ends also support the data elements.

Required PIN Processing

BASE24-pos checks the value of the PIN PROCESSING FLAG field on CPF screen 7 to determine whether a PIN is required. Valid values are as follows:

0 = PIN entry is not required to complete any transaction.
1 = PIN entry is required for every transaction; transactions from terminals without PIN capabilities are declined.
2 = PIN entry is required with transactions from terminals with PIN capabilities; however, transactions from attended terminals without PIN capabilities are processed without the PIN. Transactions from unattended terminals without PIN capabilities are declined.

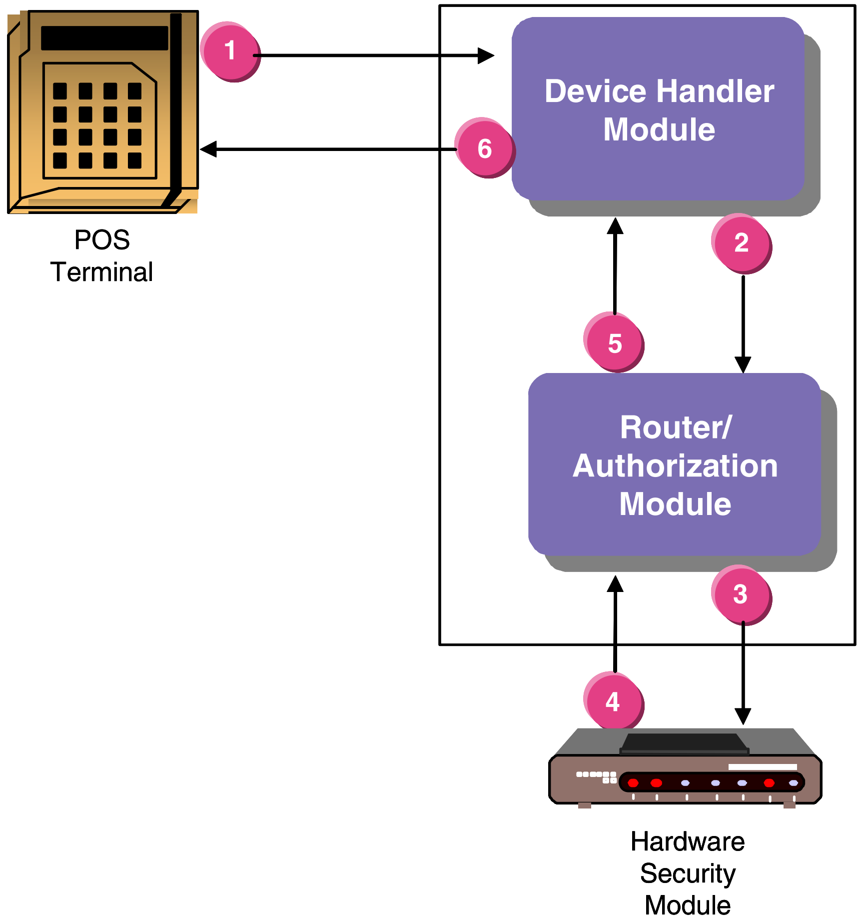
Regardless of the setting in the PIN PROCESSING FLAG field, a PIN that is entered incorrectly must be reentered, and the transaction is denied if the maximum number of PIN retries has been met, subject to the setting in the PIN TRIES RESET OPTION field. The PIN TRIES RESET OPTION field can be set at the institution level on IDF screen 2 or the card prefix level on CPF screen 2.

If a PIN is required, BASE24-pos looks for a PIN that it can verify or checks the Card Verification Results to determine whether offline PIN verification has been performed (see the “Offline PIN Support” topic in this section).

BASE24-pos declines the message with response code 096 (PIN required) in the TRAN.RESP-CDE field in the POS Standard Internal Message (PSTM) if offline PIN verification is required, but has not been performed. BASE24-pos sets the PIN-TRIES field in the PSTM to a value of Z if offline PIN verification has been performed.

BASE24 is the Issuer and the Acquirer

The following diagram illustrates the flow of an EMV transaction in which BASE24 is both the acquirer and the issuer.

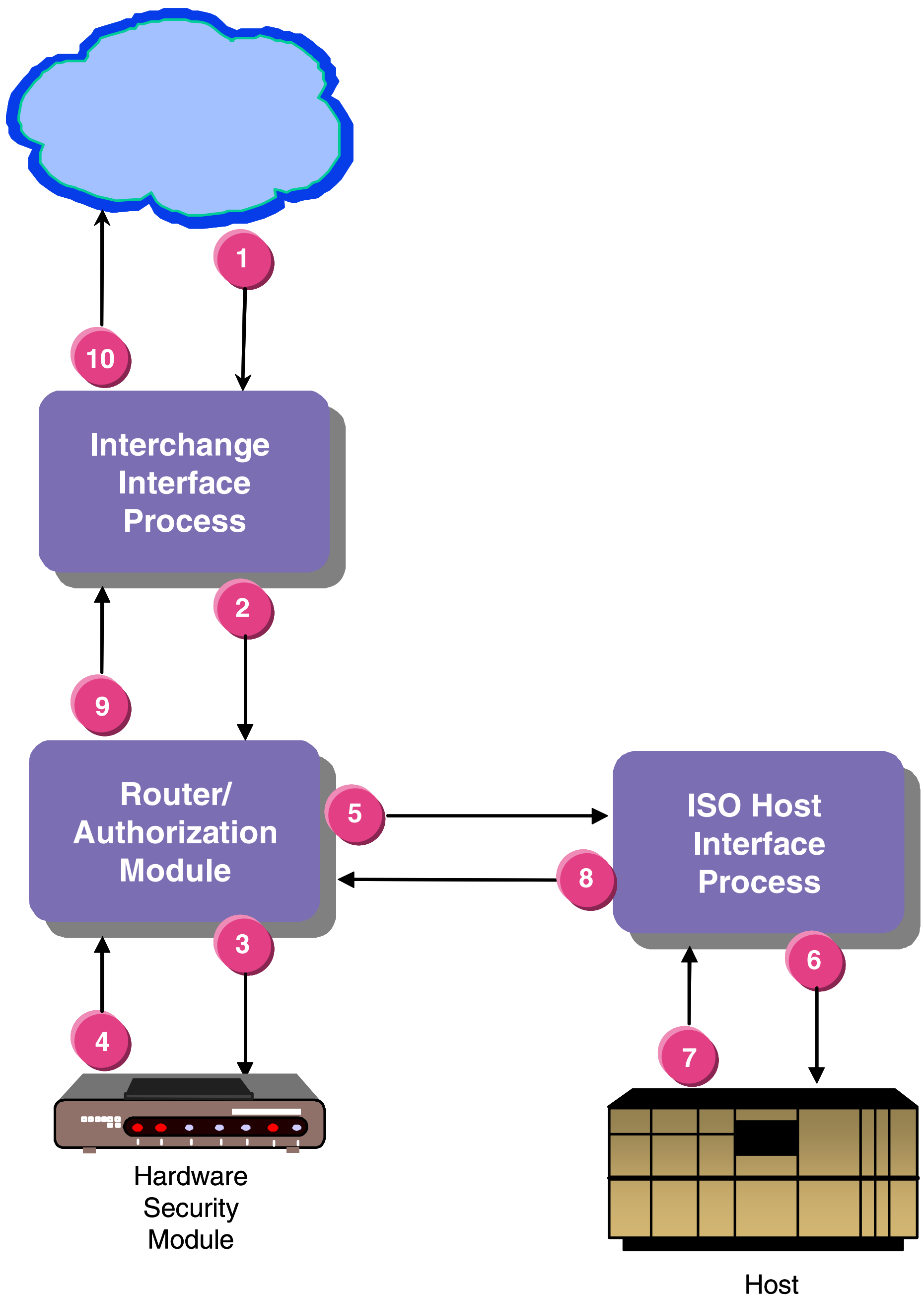


Steps	Processing
1	The card generates the authorization request cryptogram. The terminal transmits the request cryptogram and the data used to generate it in the request message and sends the message to the Device Handler module.

Steps	Processing
2	The Device Handler module checks the terminal capabilities indicators in the device-dependent data area of the TDF to determine which EMV tokens to send in the authorization message. For an EMV transaction, the Device Handler module formats the EMV Request Data token, the EMV Status token, and the EMV Discretionary Data token and adds them to the BASE24-pos Standard Internal Message (PSTM). The Device Handler module then sends the PSTM with the appended EMV tokens to the Router/ Authorization module.
3, 4	The presence of the EMV Request Data token and the appropriate configuration data in the IDF, CPF, CAF, and KEYI files initiates EMV authorization processing. This processing includes card and issuer authentication, using the request cryptogram and the KEYI check, generation of the authorization response cryptogram, ATC limit checking, EMV Status field checking (CVR and TVR), and issuer script handling. The Router/Authorization module performs CAM authentication using the hardware security module.
5	The Router/Authorization module formats an EMV Response Data token and, if applicable, an EMV Script Data token for the response message to the Device Handler module. If the transaction is not approved, a second call is made to the hardware security module using the response code to generate a new response cryptogram to be sent to the terminal.
6	The Device Handler module reformats the response message to the native format of the device and sends the message to the terminal. The card uses the information in the EMV Response Data token to authenticate the transaction authorizer by verifying the response cryptogram.

BASE24 Performs Prescreen Checks

The following diagram illustrates the flow of an EMV transaction in which an interchange acquires the transaction, BASE24 performs prescreen checks, and sends it to the host for authorization.



Steps	Processing
1	The interchange sends a request message with EMV-specific data in the ISO data elements to the BASE24 Interchange Interface process.
2	The Interchange Interface process formats the EMV Request Data token, the EMV Status token, and the EMV Discretionary Data token (according to the interchange EMV specifications) and adds them to the Standard Internal Message (STM). The Interchange Interface process sends the STM with the appended EMV tokens to the Router/Authorization module.
3, 4	The presence of the EMV Request Data token and the appropriate configuration data in the IDF, CPF, CAF, and KEYI files initiates EMV authorization processing. Prescreen processing can be configured to include card and issuer authentication, using the request cryptogram and the KEYI check, generation of the authorization response cryptogram, and EMV Status field checking (CVR and TVR). Issuer script handling can be performed after the transaction approval is received. The Router/Authorization module performs CAM authentication using the hardware security module.
5	The Router/Authorization module sends the transaction request to the ISO Host Interface process.
6	The ISO Host Interface process reformats the message into the external message (SEM) format and sends it to the host.
7	The host authorizes the transaction and returns a response to the ISO Host Interface process. The host can return a script in the Script Data token or indicate that BASE24 should generate a script by setting a script flag in the EMV Response Data token.
8	The ISO Host Interface process reformats the message into STM format and returns it to the Router/Authorization module.

Steps	Processing
9	The Router/Authorization module formats the EMV Response Data token and, if applicable, the EMV Script Data token. If the authorization level is level 3 (online/offline) and the authorization method is Positive or Positive Balance, the Router/Authorization module updates the ATC value in the CAF. If the transaction is not approved, a second call is made to the hardware security module using the response code to generate a new response cryptogram to be sent to the interchange. The Router/Authorization module sends the response message to the Interchange Interface process.
10	The Interchange Interface process reformats the response message into the external message format and sends the message to the interchange.

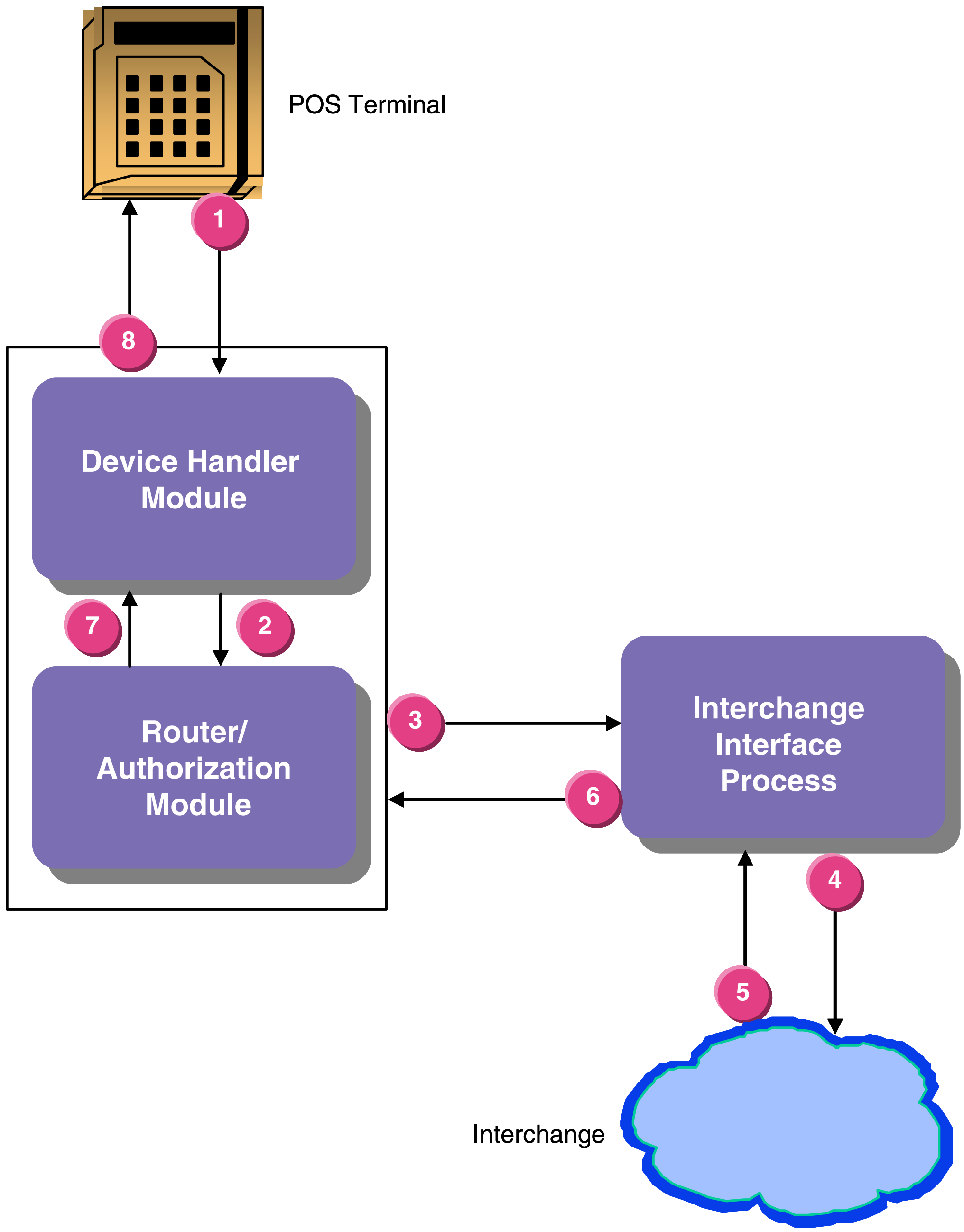
BASE24 is the Acquirer Only

In the following diagram, BASE24 is the acquirer only, sending the transaction to an Interchange for authorization.

Both the terminal and the card can perform offline risk management (e.g., transaction velocity checking, floor limit checking) to determine whether the transaction should be approved, declined, or transmitted online for authorization.

Traditionally, transaction security has been performed in security zones along the transaction path for magnetic-stripe card-based transactions. For example, a different MAC would be used for each message passed between systems, where each system is considered a separate security zone, and the PIN block would be translated every time a request message leaves a security zone. With EMV cards, the security is end-to-end (i.e., between the card and the issuer), rather than between various zones of the transaction path. Therefore, the function of an acquirer in an EMV transaction is basically to ensure that the data that comes in at one end is the same as the data that goes out at the other end.

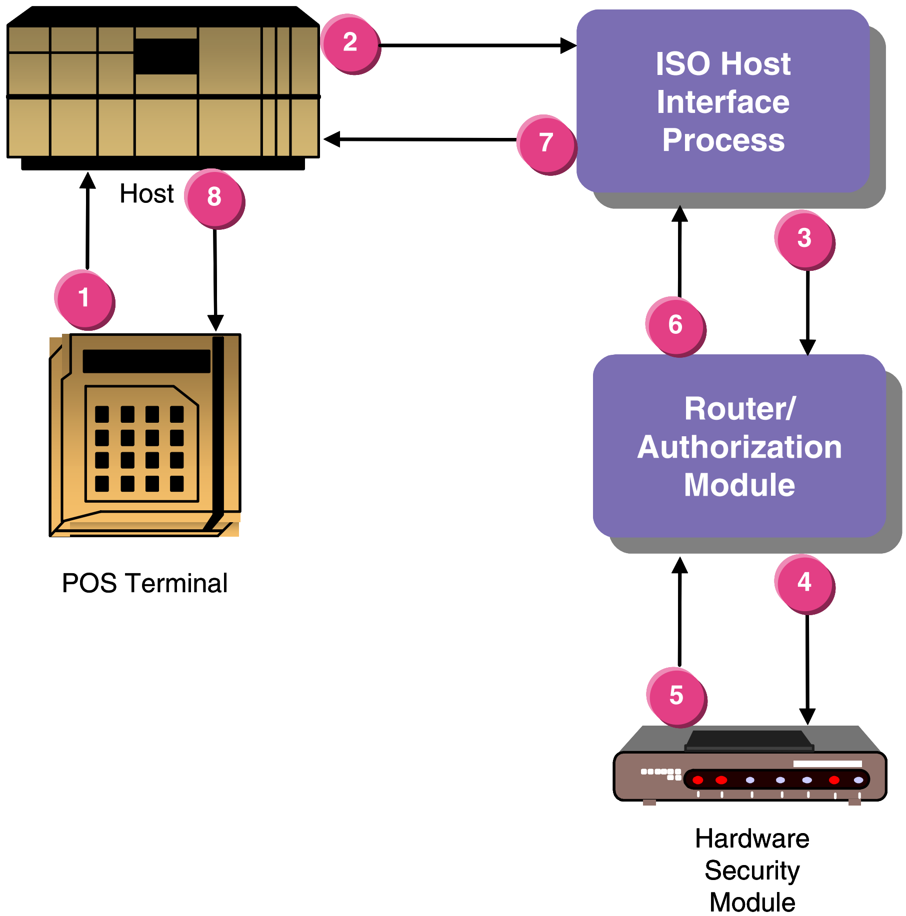
If both the card and the terminal indicate that the transaction satisfied the criteria for offline authorization, the transaction is approved offline, and the card generates a cryptogram, called the transaction certificate (TC), based on data related to the card, to the terminal, and to the transaction. This certificate and the data used to generate it are transmitted in the clearing message for future cardholder disputes and chargeback purposes. BASE24 does not verify the transaction certificate.


POS Terminal
1
8
Device Handler Module
7
2
Router/ Authorization Module
3
6
Interchange Interface Process
4
5
Interchange

Steps	Processing
1	If the criteria for offline authorization are not satisfied, the terminal generates an online request message indicating why the transaction was transmitted online, and the card generates an authorization request cryptogram. The terminal transmits the cryptogram and the data used to generate it in the request message.
2	If the terminal capabilities are not sent in the request message, the Device Handler module checks the terminal capabilities in the PTDF to determine which EMV tokens to send. The Device Handler module formats the EMV Request Data token, the EMV Status token, and the EMV Discretionary Data token and generates the BASE24-pos Standard Internal Message (PSTM). The Device Handler module sends the PSTM with the appended EMV tokens to the Router/Authorization module.
3	The Router/Authorization module sends the request message to the Interchange Interface process.
4	The Interchange Interface process reformats the message into the external message (SEM) format and sends it to the interchange for authorization. Message reformatting includes moving data from the EMV tokens to the external message.
5	The interchange authorizes the transaction and returns a response to the Interchange Interface process.
6	The Interchange Interface process adds the EMV Response Data token and, if applicable, the EMV Script Data token to the internal message, reformats the response and script from the Interchange into the standard internal message format, and sends it to the Router/Authorization module.
7	The Router/Authorization module sends the response message with the appended tokens to the Device Handler module.
8	The Device Handler module reformats the response into the native mode format of the device and sends the message to the terminal. The terminal uses the information in the EMV Response Data token to authenticate the issuer by verifying the authorization response cryptogram, and uses the information in the EMV Script Data token to perform issuer scripts.

BASE24 is the Issuer; the Host is the Acquirer

The following steps identify the flow of an EMV transaction in which the host is the acquirer and BASE24 is the issuer.



Steps	Processing
1	The card generates the authorization request cryptogram. The terminal transmits the request cryptogram and the data used to generate it in the request message and sends the message to the host.
2	The host reformats the message to the external message (SEM) format, adds EMV-specific data in the ISO data elements, and passes the message to the ISO Host Interface process.

Steps	Processing
3	The ISO Host Interface process formats the EMV Request Data token, the EMV Status token, and the EMV Discretionary Data token and adds them to the BASE24-pos Standard Internal Message (PSTM). The ISO Host Interface process sends the PSTM with the appended EMV tokens to the Router/Authorization module.
4, 5	The presence of the EMV Request Data token and the appropriate configuration data in the IDF, CPF, CAF, and KEYI files initiates EMV authorization processing. This processing includes card and issuer authentication, using the request cryptogram and the KEYI check, generation of the authorization response cryptogram, ATC limit checking, EMV Status field checking (CVR and TVR), and issuer script handling. The Router/Authorization module performs CAM authentication using the hardware security module.
6	The Router/Authorization module formats an EMV Response Data token and, if applicable, an EMV Script Data token for the response message to the ISO Host Interface process. If the transaction is not approved, a second call is made to the hardware security module using the response code to generate a new response cryptogram to be sent to the terminal.
7	The ISO Host Interface process reformats the response message into the external message (SEM) format and sends the message to the host.
8	The host reformats the response message to the native format of the device and sends the message to the terminal. The card uses the information in the message to authenticate the transaction authorizer by verifying the response cryptogram.

4: Authorization Processing

This section describes additional authorization processing that is performed on EMV transactions. This discussion does not include BASE24-pos standard processing. The following topics are described:

- Offline authorization
- Prescreen checking
- Issuer application data
- Application transaction counter checking
- Status field verification
- Reason online code
- Fallback checking
- Issuer script commands
- Prefix routing
- Processing EMV card transactions as standard Track 2 card transactions
- Authorization request and response cryptograms
- Cryptographic support

Authorization Summary

The main areas of POS authorization impacted by EMV processing are summarized in the table below. The processing functions are described in this section.

Authorization Level	Description	Processing Functions					
		CAM	ATC	CVR/ TVR	ROL	FB	IS
1	Prescreen processing	✓		✓	✓	✓	
1	Response processing						✓
2	Level 2 processing (Negative Authorization method)	✓		✓	✓	✓	
2	Level 2 processing (Positive Authorization method)	✓	✓	✓	✓	✓	✓
3	Prescreen processing	✓		✓	✓	✓	
3	Response processing						✓
3	Level 3 delegated authority processing (Negative Authorization method)	✓		✓	✓	✓	
3	Level 3 delegated authority processing (Positive Authorization method)	✓	✓	✓	✓	✓	✓
3	Stand-in processing (Negative Authorization method)	✓		✓	✓	✓	
3	Stand-in processing (Positive Authorization method)	✓	✓	✓	✓	✓	✓
	Default authorization processing						

Key	
CAM	Card and issuer authentication (CAM check)
ATC	ATC limit checking
CVR/TVR	EMV status field checking (CVR/TVR)

Key	
ROL	Reason online checking
FB	EMV fallback checking
IS	Issuer script handling

Response Codes

If a transaction results in a denial, a response message is generated using the following EMV-specific response codes.

BASE24-pos Response Code	Description
130	Request cryptogram failure (referral)
131	CVR referral
132	TVR referral
133	Reason online code referral
134	Fallback referral
400	Request cryptogram failure
401	Security module parameter error
402	Security module failure
403	KEYI record not found
404	ATC check failure
405	CVR decline
406	TVR decline
407	Reason online code decline
408	Fallback decline
910	Request cryptogram failure
911	CVR decline
912	TVR decline

Offline Authorization

Offline authorization is defined as the EMV-specific processing that is performed by the BASE24-pos Device Handler/Router/Authorization process in an offline situation.

For both level 2 authorization (offline only) and level 3 stand-in authorization, EMV-specific authorization processing provides the following:

- Authorization request cryptogram verification
- Application transaction counter checking, for Positive Authorization methods (CAF and CAF/PBF)
- Status field (CVR and TVR) checking (if the EMV Request Data token is present)
- Reason online code checking
- Fallback transaction checking
- Script formatting, for Positive Authorization methods (CAF and CAF/PBF)

Each of these is described in more detail later in this section.

Prescreen Checking

Prescreen checks are checks that are performed in situations where BASE24 shares authorization responsibilities with a host. In such situations, the BASE24-pos Device Handler/Router/Authorization process can make preliminary standard checks on the transactions and decline any that do not meet preliminary requirements.

EMV prescreen checking is currently invoked from level 1 online authorization and level 3 online/offline authorization. The following prescreen checks are available.

- Authorization request cryptogram verification, based on the value of the CAM field on Card Prefix File (CPF) screen 11.
- Fallback transaction checking, based on the value of the FALLBACK field on CPF screen 11.
- Reason online code checking, based on the value of the REASON ONLINE CODE field on CPF screen 11.
- Status field checking, based on the value of the TVR/CVR field on CPF screen 11.

Issuer Application Data

Issuer application data contains proprietary issuer data for transmission to the issuer in an online transaction. Typically this data contains information that is used in the derivation of the authorization cryptogram. The following data elements that are included in both the Visa and MasterCard definitions are used in EMV authorization processing and generation of the authorization cryptograms:

- Derivation key index
- Cryptogram version number
- Card verification results

The following fields are extracted from the issuer application data in the EMV Request Data token:

CRD-VERIF-RSLTS	Used as input when calculating the authorization cryptogram. This field is extracted for CVR status checks.
DERIV-KEY-INDEX	Used to identify which of a set of master keys is to be used in the derivation of the authorization cryptogram for the card.
CRYPTO-VER-NUM	Indicates the version of the method or algorithm used to generate the authorization cryptogram. The version defines the set of data fields to be used in generation of the cryptogram.

The BASE24-pos EMV product supports different layouts for issuer application data, depending on the card type. MasterCard and Europay specify different definitions of issuer application data than Visa does. The layout of the issuer application data for each of the currently supported formats is specified by redefinitions of the issuer application data field in the EMV Request Data token.

Valid values for the EMV IAD FORMAT and ALTERNATE IAD FORMAT fields on CPF screen 11 are as follows:

0 = Unspecified. The format is unknown. The contents are sent as passthrough data, but no cryptographic checks are performed on BASE24.

1 = The format as defined by Visa in the Visa Integrated Circuit Card (ICC) specification. This format is also used for UKIS cards.

The format is defined by the VISA-APPL-DATA data elements of the EMV Request Data token.

2 = The format recommended for MasterCard MCPA (M/Chip 2.1) cards.

The format is defined by the MCPA-APPL-DATA data elements of the EMV Request Data token.

3 = The format recommended for MasterCard MCPA (M/Chip 4) cards.

The format is defined by the MCHIP4-APPL-DATA data elements of the EMV Request Data token.

4 = The format mandated for CCD (IAD Format A) cards.

The format is defined by the CCD-A-APPL-DATA data elements of the EMV Request Data token.

Configuring Alternate Formats

The BASE24-atm product supports an alternate layout so cards with the same prefix can be migrated from one layout to another as they are reissued. The primary format of the data on a card for a given prefix is configurable using the EMV IAD FORMAT field on CPF screen 11. The alternate format of the data on a card for a given prefix is configurable using the ALTERNATE IAD FORMAT field on CPF screen 11.

Prior to the ALTERNATE IAD FORMAT field being available, the EMV IAD FORMAT field on CAF screen 13 enabled a card issuer to use the M/Chip 4 format for specific accounts within a card prefix that was still using the M/Chip 2.1 format.

Valid values for the EMV IAD FORMAT field on CAF screen 13 are as follows:

0 = Use the format specified in the EMV IAD FORMAT field on CPF screen 11. This is the default value.

1 = The format as defined by Visa in the Visa Integrated Circuit Card (ICC) specification. This format is also used for UKIS cards.

The format is defined by the VISA-APPL-DATA data elements of the EMV Request Data token.

2 = The format recommended for MasterCard MCPA (M/Chip 2.1) cards.

The format is defined by the MCPA-APPL-DATA data elements of the EMV Request Data token.

3 = The format recommended for MasterCard MCPA (M/Chip 4) cards.

The format is defined by the MCHIP4-APPL-DATA data elements of the EMV Request Data token.

Note: Once the EMV IAD FORMAT field on CPF screen 11 contains a value of 3 (M/Chip 4 format), both MasterCard values in the EMV IAD FORMAT field on CAF screen 13 have the same effect.

Determining the Issuer Application Data Format

During a migration period, cards with the same prefix can have more than one application (e.g., cards with a VIS or M/Chip format migrating to a Common Core Definitions (CCD) format).

BASE24-atm uses three components to determine the issuer application data (IAD) format.

- Transaction data
- EMV IAD FORMAT field on CPF screen 11
- ALTERNATE IAD FORMAT field on CPF screen 11

BASE24-atm determines the IAD format from the transaction data, then uses that IAD format and the two IAD formats defined on CPF screen 11 to determine the IAD format for the transaction.

Transaction Data

The following table shows how the IAD format on cards that comply with specific applications can be identified from the transaction data.

IAD Length	IAD Byte 1	IAD Byte 2	IAD Format from Transaction Data
Greater than 6	06	Not applicable	VIS
Greater than 7	Not applicable	0*x*[1]	M/Chip 2.1
Greater than 17	Not applicable	1*y*[1]	M/Chip 4.0
32	0F	A*z*[2]	CCD

[1] *x* and *y* can be any hexadecimal digit

[2] *z* can be a hexadecimal digit in the range of 5 through F

Issuer Application Data Format for a Specific Transaction

The following table shows the checks performed to derive the Issuer Application Data format for a specific transaction, using the two IAD formats defined on CPF screen 11 and the IAD format determined from the transaction data.

The first check must be on the two IAD format fields in the CPF record, as shown in the first row of the table. If these values match, the specified format is assumed, regardless of the format of the Issuer Application Data format that is determined from the transaction data.

It is possible for the IAD in a transaction to match more than one defined format. For example, a 10-byte IAD beginning %h06 %h00 in the first two bytes could indicate either a VIS or an M/Chip 2.1 format. If both IAD formats are configured on CPF screen 11, BASE24 uses the value in the EMV IAD FORMAT field rather than the value in the ALTERNATE IAD FORMAT field.

CPF Screen 11 Fields		IAD Format from Transaction Data	IAD Format Used for Transaction
EMV IAD FORMAT	**ALTERNATE IAD FORMAT**		
Any value	Matches value in EMV IAD FORMAT field	Not applicable	Value in EMV IAD FORMAT field
Non-zero	0	Matches value in EMV IAD FORMAT field	Value in EMV IAD FORMAT field
Non-zero	0	Does not match value in EMV IAD FORMAT field	0
0	Non-zero	Matches value in ALTERNATE IAD FORMAT field	Value in ALTERNATE IAD FORMAT field
0	Non-zero	Does not match value in ALTERNATE IAD FORMAT field	0
Non-zero	Non-zero	Matches value in EMV IAD FORMAT field	Value in EMV IAD FORMAT field
Non-zero	Non-zero	Matches value in ALTERNATE IAD FORMAT field	Value in ALTERNATE IAD FORMAT field
Non-zero	Non-zero	Does not match value in EMV IAD FORMAT field or ALTERNATE IAD FORMAT field	0

Application Transaction Counter Checking

An integrated circuit card (ICC) is a plastic card, usually the size of a credit card, that contains an embedded microprocessor chip. This chip is capable of storing large amounts of cardholder information and can contain multiple applications. The terms chip card and smart card are sometimes used interchangeably with integrated circuit card.

The application transaction counter (ATC) is a value maintained by the microchip and updated for each transaction performed by the card application. A single card can hold multiple ATCs, depending on the number of applications on the card. The ATC is also maintained in the BASE24 Cardholder Authorization File (CAF), as the value may be checked during transaction processing.

Application transaction counter checking is used only with the Positive and Positive Balance authorization methods.

To reflect the possibility of multiple ATCs being maintained on the card, there are multiple CAF fields that can be used to hold the ATC. The CAF field used in a particular transaction is determined by the setting of various fields on the CPF:

- ATC CHECK field on CPF screen 3
- ATC CHECK TYPE field on CPF screen 11
- CAP ATC UPDATE field on CPF screen 13

You should set these fields based on which of the following you want to maintain:

- Separate ATCs for contactless magnetic stripe transactions, EMV transactions, and CAP token validation transactions.
- A single ATC for all three types of transactions.
- A single ATC for both contactless magnetic stripe and EMV transactions, and a separate ATC for CAP token validation transactions (or where CAP token validation transactions are not supported).
- A single ATC for both EMV and CAP token validation transactions, and a separate ATC for contactless magnetic stripe transactions (or where contactless magnetic stripe transactions are not supported).
- Separate ATCs for contactless magnetic stripe transactions and EMV transactions (where CAP token validation transactions are not supported).

- Separate ATCs for contactless magnetic stripe transactions and EMV transactions (where contactless magnetic stripe transactions are not supported).
- A single ATC for just contactless magnetic stripe transactions (where EMV transactions and CAP token validation transactions are not supported).
- A single ATC for just EMV transactions (where contactless magnetic stripe transactions and CAP token validation transactions are not supported).
- A single ATC for just CAP token validation transactions (where contactless magnetic stripe transactions and EMV transactions are not supported).
- No ATCs at all.

The relationship between these fields is described in more detail below.

CAP ATC UPDATE — code indicating whether the application transaction counter (ATC) for CAP token validation transactions is maintained separately or is combined with the ATC used for EMV transactions. Essentially, this depends on whether the CAP application on the card is separate from the EMV application or part of the EMV application.

Set this field to 0 if a separate ATC value is to be maintained for CAP token validation transactions. Use this value if there is a separate ATC on the card for CAP token validation transactions and EMV transactions, or where EMV transactions are not supported by the card.

Set this field to 1 if a single ATC value is to be maintained for both CAP token validation transactions and EMV transactions. Use this value only if there is one ATC on the card that is used for both CAP token validation transactions and EMV transactions.

ATC CHECK TYPE — A code indicating whether the application transaction counter (ATC) for EMV transactions is maintained separately or is combined with the ATC used for contactless magnetic stripe transactions. Essentially, this depends on whether the contactless magnetic stripe application on the card is separate from the EMV application or part of the EMV application. This code also indicates the type of application transaction counter (ATC) checking to be performed for EMV cards (in contact or contactless EMV transactions).

Set this field to 0 if you do not want to check the ATC for EMV transactions.

Set this field to 1 if you want to perform ATC checking for EMV transactions and maintain a separate ATC value for EMV transactions and contactless magnetic stripe transactions, or where contactless magnetic stripe transactions are not supported by the card.

Set this field to 8 if you want to perform ATC checking for EMV transactions and maintain one ATC value for both EMV transactions and contactless magnetic stripe transactions. Use this value only if the card does not support contact EMV transactions, but does support contactless transactions that include an EMV cryptogram (for which a CPF EMV segment is required, but a CAF EMV segment is not).

Set this field to 9 if you want to perform ATC checking for EMV transactions and maintain one ATC value for both EMV transactions and contactless magnetic stripe transactions. Use this value only if there is one ATC on the card that is used for both contactless magnetic stripe transactions and EMV transactions.

ATC CHECK — A code indicating the type of application transaction counter (ATC) checking to be performed for contactless magnetic stripe transactions.

Set this field to 0 if you do not want to check the ATC for contactless magnetic stripe transactions.

Set this field to 1 if you do not want to check the ATC for contactless magnetic stripe transactions, but you do want to perform ATC checking for EMV transactions. Use this value only if there is one ATC on the card that is used for both contactless magnetic stripe transactions and EMV transactions.

Set this field to 2 if you want to perform ATC checking for contactless magnetic stripe transactions. Use this value if there is a separate ATC on the card for EMV transactions and contactless magnetic stripe transactions, or where EMV transactions are not supported by the card.

Set this field to 3 if you want to perform ATC checking for EMV transactions and contactless magnetic stripe transactions. Use this value only if there is one ATC on the card that is used for both contactless magnetic stripe transactions and EMV transactions.

Second Card Processing

Second card processing is an optional feature that is enabled by LCONF parameter EXP-DAT-CMP and different expiration dates for the primary (original) and second (reissued) cards in the EXPIRATION DATE fields on CAF screens 2 and 7. This feature enables an institution to issue two cards with the same primary account number (PAN) and member number. The two cards are differentiated by comparing the expiration date on the card with two expiration dates held in the CAF, one for the primary card and another for the second card.

The system can maintain one or two additional ATC in the CAF for the second card. The SECOND ATC NUMBER fields are located on CAF screens 7 and 13. The values in the ATC CHECK field on CPF screen 3 and ATC CHECK TYPE field on CPF screen 11 control which SECOND ATC NUMBER fields are used. When a transaction is received from the second card and ATC checking is enabled, the ATC from the card's microchip is checked against the value in the appropriate SECOND ATC NUMBER field. Additionally, if the system is configured by LCONF parameter CAF-SCND-CRD-PROMOTE to automatically make the second card the new primary card, the second card's ATC is included in the data that is moved to the primary card's data.

Updating the ATC

The following fields in the CAF are updated by BASE24 during transaction processing:

- ATC NUMBER on CAF screen 2 (BASE)
- SECOND CARD ATC NUMBER on CAF screen 7 (BASE)
- PRIMARY CARD ATC NUMBER on CAF screen 13 (EMV)
- SECONDARY CARD ATC NUMBER on CAF screen 13 (EMV)
- PRIMARY CARD CAP ATC NUMBER on CAF screen 13 (CAP)
- SECONDARY CARD CAP ATC NUMBER on CAF screen 13 (CAP)

These fields are also maintained on the Dynamic Cardholder Authorization File (CAFD), if that file is configured for the institution.

The expiration date in the transaction message determines whether the primary ATC or the corresponding secondary ATC is updated. The values in the CPF fields mentioned above specify whether the BASE ATC, the EMV ATC or the CAP ATC is updated for each type of transaction performed, as shown in the table that follows.

CPF Configuration Field Values			CAF or CAFD ATC Field Updated		
ATC CHECK on Screen 3	**ATC CHECK TYPE on Screen 11**	**CAP ATC UPDATE on Screen 13**	**Contactless Magnetic Stripe Transactions**	**EMV Transactions**	**CAP Token Validation Transactions**
0	0/1	0	BASE	EMV	CAP
0	0/1	1	BASE	EMV	EMV
0/1	8/9	0	BASE	BASE	CAP
0/1	8/9	1	BASE	BASE	BASE
2	0/1	0	BASE	EMV	CAP
2	0/1	1	BASE	EMV	EMV
2/3	8/9	0	BASE	BASE	CAP
2/3	8/9	1	BASE	BASE	BASE

The ATC is updated whenever dynamic card verification (contactless magnetic stripe transactions), request cryptogram verification (EMV transactions) or CAP token verification (CAP token validation transactions) is successful.

ATC Checking

ATC checking is enabled through the ATC CHECK field on CPF screen 3 and the ATC CHECK TYPE field on CPF screen 11. ATC checking is not performed on CAP token validation transactions.

Two checks are performed on the ATC received in the authorization request to detect fraudulent cards used in contactless magnetic stripe transactions or EMV transactions:

- The value of the ATC in the message must be greater than the value in the appropriate ATC number field in the CAF.
- The value of the ATC in the message must not be greater than the sum of the appropriate ATC number field in the CAF and a configurable ATC limit.

The following fields are used to hold the ATC limit:

- ATC LIMIT on CPF screen 3
- ATC LIMIT on CAF screen 13

The values in the CPF fields mentioned above specify whether the BASE ATC or the EMV ATC checked, and whether the CAF ATC limit or the CPF ATC limit is used for each type of transaction performed, as shown in the table that follows.

CPF Configuration Field Values		**CAF or CAFD ATC Field Checked**		**CAF or CPF ATC Limit Field Checked**	
ATC CHECK on Screen 3	**ATC CHECK TYPE on Screen 11**	**Contactless Magnetic Stripe Transactions**	**EMV Transactions**	**Contactless Magnetic Stripe Transactions**	**EMV Transactions**
0/1	0	None	None	None	None
0/1	1	None	EMV	None	CAF
0	8	None	None	None	None
0	9	None	None	None	None
1	8	None	BASE	None	CPF
1	9	None	BASE	None	CAF
2/3	0	BASE	None	CPF	None
2/3	1	BASE	EMV	CPF	CAF
2	8	BASE	None	CPF	None
2	9	BASE	None	CPF	None
3	8	BASE	BASE	CPF	CPF
3	9	BASE	BASE	CPF	CAF

ATC limit checking is performed only if the appropriate ATC LIMIT field contains a non-zero value.

Status Field Verification

The EMV data elements sent in request messages include two status fields:

- Terminal verification results (TVR)
- Card verification results (CVR)

These fields contain information about processing performed by an EMV-capable terminal and the ICC at the acquiring point.

In case a host or issuer is unavailable and BASE24 is not using the Positive or Positive Balance Authorization method, status field checking enables BASE24 to authorize or decline a transaction. You can also configure status field checking as a prescreen check using the TVR/CVR field in the CPF.

Status field checking is performed using preconfigured edit files (CVRTBL and TVRTBL) that are compiled into the BASE24-pos Device Handler/Router/Authorization process. These files identify which field information is to be checked and the required actions. Options available for each table are listed later in this section.

These edit files support a number of sets of test conditions and actions. The value of the issuer application data value used for a specific transaction determines which set of action index options is used. Refer to the “Issuer Application Data” topic earlier in this section for an explanation of how the issuer application data is determined for a specific transaction.

Issuer Application Data Used	Action Index Used
0	0
EMV IAD FORMAT field on CPF screen 11	STATUS CHECK ACTION INDEX field on CPF screen 11
ALTERNATE IAD FORMAT field on CPF screen 11	ALTERNATE ACTION INDEX field on CPF screen 11
EMV IAD FORMAT field on CAF screen 13	STATUS CHECK ACTION INDEX field on CAF screen 13

You can edit the configuration files to support your specific requirements. For example, you can preconfigure alternative tests or actions. In addition, the bits are checked in the order in which they appear in the edit file. Therefore, further flexibility is available in that you can configure the order in which the bits are checked.

The following lists examples of changes you can make in the edit files:

- Adjust the actions to be performed for each test by modifying column four (described below).
- Revise the sequence of the checks by reordering the entries within a given action index.
- Add checks to a specific set.
- Add new test sets by allocating a new action index and adding entries for all required checks to the end of the table. All checks for a single action index must be contiguous entries in the table.

You can copy entries into the actual table to build additional test sets, adjusting the action index and authorization action fields accordingly.

Note that where the force online flag is set to N and the authorization action is set to C (i.e., the byte and bit settings have no effect), the entries can be left out of the table.

Edit File Layout

The edit files contain five columns with the following information.

Action Index

The first column, action index, identifies the test set to which the test and prescribed actions belong. This index is used to select the tests and actions to be applied for a particular card scheme by matching the value in the STATUS CHECK ACTION INDEX field configured on CPF screen 11, ALTERNATE ACTION INDEX field configured on CPF screen 11, or STATUS CHECK ACTION INDEX field configured on CAF screen 13.

Byte and Bit

The second and third columns, Byte and Bit, specify which byte and bit of the CVR or TVR data element are to be tested. Note that in the numbering of the bits, bit 8 is the high order bit.

Force Online Flag

The fourth column, Force Online Flag, specifies the action to be taken for transactions that are under the floor limit in the Authorization Selection Table (AST). This value is applicable to level 3 authorization. Valid values are as follows:

Y = Yes, force online to the over-limit authorizer
N = No, do not force online to the over-limit authorizer

A restriction to the usage of the Force Online Flag exists for a card prefix that has both of the following cards:

- Some cards that conform to the M/Chip 4 or CCD issuer application data format
- Some cards that conform to a different issuer application data format

When this condition applies to a card prefix, the Force Online Flag must be set to a value of N in all CVR Table entries for the action index defined in the relevant CPF record. BASE24-pos routing logic, of which force online processing is a part, does not have access to card-specific information such as the appropriate CAF record.

Authorization Action

The fifth column, Authorization Action, is applicable to all offline authorization methods and optionally as a prescreen check for online to host authorization methods. Valid values are as follows:

C = Continue processing (equivalent to no check)
D = Decline
K = Decline and retain (keep) card
R = Refer

Card Verification Results Table

The Card Verification Results Table (CVRTBL) specifies the particular bits from the CVR status bytes that are to be tested and the actions to be taken when a specific bit is set on.

The CVR is a proprietary EMV data element (included in the issuer application data). It is therefore possible for different card schemes to use alternative definitions for the bit settings.

The CVRTBL contains the following CVR record templates:

- MasterCard M/Chip 2.1 (EMV 1996-compliant) cards and Visa VIS (as used for the UKIS ICC implementation) cards
- MasterCard M/Chip 4 (EMV 2000-compliant) cards
- CCD cards

M/Chip 2.1 and VIS CVR Records

M/Chip 2.1 and VIS use the same interpretations, except for a few differences that are noted in the following tables. All settings are applicable to both VIS and M/Chip 2.1 unless followed by VIS or M/Chip 2.1 in parentheses.

The CVRTBL defines two preconfigured test sets plus a group of alternative tests for M/Chip 2.1 and VIS cards.

The default test set (action index 1) for VIS or M/Chip 2.1 cards identifies that no checks are required. (The single null test is included as a placeholder.)

Action Index	Byte	Bit	Force Online Flag	Authorization Action	Description
1	4	4	N	C	Issuer script processing failed on last transaction

The base test set (action index 2) provides default tests for VIS or M/Chip 2.1 cards.

Action Index	Byte	Bit	Force Online Flag	Authorization Action	Description
2	3	8	Y	C	Last online transaction not completed
2	3	6	Y	C	Exceeded velocity checking counters
2	3	5	Y	C	New card
2	3	4	Y	R	Issuer authentication failed on last online transaction
2	3	1	Y	C	Static data authentication failed on last transaction and transaction declined offline
2	4	4	Y	C	Issuer script processing failed on last transaction

The following table lists other alternative tests for VIS or M/Chip 2.1 cards.

Action Index	Byte	Bit	Force Online Flag	Authorization Action	Description
n	2	4	Y	R	Issuer authentication failed
n	2	3	Y	R	Offline PIN verification performed
n	2	2	Y	R	Offline PIN verification failed
n	2	1	Y	R	Unable to go online
n	3	8	Y	R	Last online transaction not completed
n	3	7	Y	R	PIN try limit exceeded
n	3	6	Y	R	Exceeded velocity checking counters
n	3	5	Y	R	New card
n	3	4	Y	R	Issuer authentication failed on last online transaction

Action Index	Byte	Bit	Force Online Flag	Authorization Action	Description
n	3	3	Y	R	Issuer authentication not performed after online authorization
n	3	2	Y	R	Application blocked by card because PIN try limit exceeded
n	3	1	Y	R	Static data authentication failed on last transaction and transaction declined offline
n	4	5	Y	R	DDA failed on last online transaction and transaction declined offline (M/Chip 2.1)
n	4	4	Y	R	Issuer script processing failed on last transaction.
n	4	3	Y	R	Lower consecutive offline limit or lower cumulative offline amount exceeded (M/Chip 2.1) DDA failed on last online transaction and transaction declined offline (VIS)
n	4	2	Y	R	Upper consecutive offline limit or upper cumulative offline amount exceeded (M/Chip 2.1) Offline DDA performed (VIS)
n	4	1	Y	R	Maximum offline transaction amount exceeded (M/Chip 2.1)

M/Chip 4 CVR Records

The CVRTBL defines two preconfigured test sets plus a group of alternative tests for M/Chip 4 cards.

The default test set (action index 1) identifies that no checks are required. (The single null test is included as a place holder.)

Action Index	Byte	Bit	Force Online Flag	Authorization Action	Description
1	4	4	N	C	PIN try limit exceeded

The base test set (action index 3) provides default tests for M/Chip 4 cards.

Action Index	Byte	Bit	Force Online Flag	Authorization Action	Description
3	4	7	Y	C	Unable to go online
3	5	4	Y	C	Go online on next transaction was set
3	5	3	Y	R	Issuer authentication failed
3	5	1	Y	C	Script failed

The following table lists the alternative tests available for M/Chip 4 cards.

Action Index	Byte	Bit	Force Online Flag	Authorization Action	Description
m	4	7	Y	C	Unable to go online
m	4	6	Y	C	Offline PIN verification not performed
m	4	5	Y	C	Offline PIN verification failed
m	4	4	Y	C	PIN try limit exceeded
m	4	3	Y	C	International transaction
m	4	2	Y	C	Domestic transaction
m	4	1	Y	C	Terminal erroneously considers offline PIN OK
m	5	8	Y	C	Lower consecutive offline limit exceeded

Action Index	Byte	Bit	Force Online Flag	Authorization Action	Description
m	5	7	Y	C	Upper consecutive offline limit exceeded
m	5	6	Y	C	Lower cumulative offline limit exceeded
m	5	5	Y	C	Upper cumulative offline limit exceeded
m	5	4	Y	C	Go online on next transaction was set
m	5	3	Y	C	Issuer authentication failed
m	5	2	Y	C	Script received
m	5	1	Y	C	Script failed
m	6	2	Y	C	Match found in additional check table
m	6	1	Y	C	No match found in additional check table

CCD (IAD Format A) CVR Records

The CVRTBL defines two preconfigured test sets plus a group of alternative tests for CCD (IAD format A) cards.

The default test set (action index 1) identifies that no checks are required. (The single null test is included as a place holder.)

Action Index	Byte	Bit	Force Online Flag	Authorization Action	Description
1	4	4	N	C	Issuer script processing failed on last transaction

The base test set (action index 4) provides default tests for CCD (IAD format A) cards.

Action Index	Byte	Bit	Force Online Flag	Authorization Action	Description
4	1	1	Y	R	Issuer authentication failed
4	4	4	Y	C	Issuer script processing failed
4	4	2	Y	C	Go online on next transaction was set
4	4	1	Y	C	Unable to go online

The following table lists other alternative tests for CCD (IAD format A) cards.

Action Index	Byte	Bit	Force Online Flag	Authorization Action	Description
c	1	4	Y	R	CDA performed
c	1	3	Y	R	Offline DDA performed
c	1	2	Y	R	Issuer authentication not performed
c	1	1	Y	R	Issuer authentication failed
c	2	4	Y	R	Offline PIN verification performed
c	2	3	Y	R	Offline PIN verification performed and PIN not successfully verified
c	2	2	Y	R	PIN try limit exceeded
c	2	1	Y	R	Last online transaction not completed
c	3	8	Y	R	Lower offline transaction count limit exceeded
c	3	7	Y	R	Upper offline transaction count limit exceeded
c	3	6	Y	R	Lower cumulative offline amount limit exceeded

Action Index	Byte	Bit	Force Online Flag	Authorization Action	Description
c	3	5	Y	R	Upper cumulative offline amount limit exceeded
c	4	4	Y	R	Issuer script processing failed
c	4	3	Y	R	Offline data authentication failed on previous transaction
c	4	2	Y	R	Go online on next transaction was set
c	4	1	Y	R	Unable to go online

Terminal Verification Results Table

The Terminal Verification Results Table (TVRTBL) is used to specify the particular bits from the TVR status bytes that are to be tested and the actions to be taken when a specific bit is set on.

The TVR is a standard EMV data element. The tables below document the TVR values as defined by EMV, and provide descriptions of each byte and bit that can be tested.

The TVRTBL defines two preconfigured test sets.

The default test set (action index 1) identifies that no checks are required. (The single null test is included as a place holder.)

Action Index	Byte	Bit	Force Online Flag	Authorization Action	Description
1	1	8	N	C	Offline data authentication was not performed

The base test set (action index 2) provides default tests.

Action Index	Byte	Bit	Force Online Flag	Authorization Action	Description
2	1	8	Y	R	Offline data authentication was not performed
2	1	7	Y	R	Offline static data authentication failed
2	1	6	Y	R	ICC data missing
2	1	5	Y	K	Card appears on terminal exception file
2	2	7	Y	D	Expired Application
2	2	6	Y	D	Application not yet effective
2	2	5	Y	D	Requested service not allowed for card product
2	2	4	Y	C	New card
2	3	8	Y	R	Cardholder verification was not successful
2	4	7	Y	C	Lower consecutive offline limit exceeded
2	4	5	Y	C	Transaction selected randomly for online processing
2	4	4	Y	C	Merchant forced transaction online

The following table lists other alternative tests.

Action Index	Byte	Bit	Force Online Flag	Authorization Action	Description
n	1	8	Y	R	Offline data authentication was not performed
n	1	7	Y	R	Offline static data authentication failed
n	1	6	Y	R	ICC data missing

Action Index	Byte	Bit	Force Online Flag	Authorization Action	Description
n	1	5	Y	R	Card appears on terminal exception file
n	1	4	Y	R	Offline DDA authentication failed
n	2	8	Y	R	ICC and terminal have different application versions
n	2	7	Y	R	Expired application
n	2	6	Y	R	Application not yet effective
n	2	5	Y	R	Requested service not allowed for card product
n	2	4	Y	R	New card
n	3	8	Y	R	Cardholder verification was not successful
n	3	7	Y	R	Unrecognized CVM
n	3	6	Y	R	PIN Try limit exceeded
n	3	5	Y	R	PIN entry required and PIN PAD not present or not working
n	3	4	Y	R	PIN entry required, PIN PAD present, but PIN was not entered
n	3	3	Y	R	Online PIN entered
n	4	8	Y	R	Transaction exceeds floor limit
n	4	7	Y	R	Lower consecutive offline limit exceeded
n	4	6	Y	R	Upper consecutive offline limit exceeded
n	4	5	Y	R	Transaction selected randomly for online processing
n	4	4	Y	R	Merchant forced transaction online
n	5	8	Y	R	Default TDOL used
n	5	7	Y	R	Issuer authentication was unsuccessful

Action Index	Byte	Bit	Force Online Flag	Authorization Action	Description
n	5	6	Y	R	Script processing failed before final GENERATE AC command
n	5	5	Y	R	Issuer script processing failed after final GENERATE AC command

Offline PIN Support

The result of the PIN verification performed at the POS device can affect the authorization processing performed by BASE24. In particular, the following capabilities are required by the UK Domestic Maestro card scheme:

- The acquirer must be able to forward transactions to the issuer for authorization, even if the transaction value is below the Online Limit configured in the Authorization Selection Table (AST), under the following circumstances:
  - When offline PIN verification fails
  - When the offline PIN try limit is exceeded
  - When the EMV application is blocked by the card because the offline PIN try limit is exceeded
  - When PIN entry is required, but a PIN pad is not present or is not working
  - When PIN entry is required and a PIN pad is available, but a PIN is not entered
- The acquirer must not approve transactions, if the transaction cannot be forwarded to the issuer for authorization under the following circumstances:
  - When offline PIN verification fails
  - When the offline PIN try limit is exceeded
  - When the EMV application is blocked by the card because the offline PIN try limit is exceeded
  - When PIN entry is required, but a PIN pad is not present or is not working
  - When PIN entry is required and a PIN pad is available, but a PIN is not entered

Support for these functions is provided using status field verification.

Note: The CPF can be configured to require a PIN to be entered for all POS transactions, requiring the transaction to be declined if an online PIN is not present and the CVR indicates that an offline PIN has not been entered. Support for this function is provided using status field verification.

Reason Online Code

The reason online code is a data element that indicates the reason the transaction was sent online from a terminal for authorization. Reason online code processing utilizes the prescreen and offline authorization checks in a similar way as the CVR/TVR status checks described above. The Reason Online Code Table (ROCTBL) defines the reason online code checks and actions that are to be tested during authorization of EMV transactions and the actions to be taken for a specific reason code.

The value of the issuer application data value used for a specific transaction determines which set of reason online code checks is used. Refer to the “Issuer Application Data” topic earlier in this section for an explanation of how the issuer application data is determined for a specific transaction.

Issuer Application Data Used	Action Index Used
0	0
EMV IAD FORMAT field on CPF screen 11	STATUS CHECK ACTION INDEX field on CPF screen 11
ALTERNATE IAD FORMAT field on CPF screen 11	ALTERNATE ACTION INDEX field on CPF screen 11
EMV IAD FORMAT field on CAF screen 13	STATUS CHECK ACTION INDEX field on CAF screen 13

In the ROCTBL edit file described below, the first column, action index, identifies the test set to which the test and prescribed actions belong. This index is used to select the tests and actions to be applied for a particular card scheme by matching the value in the appropriate action index shown in the table above.

The second column is the reason code against which the processing is to be tested.

The third column, force online flag, specifies the action to be taken for transactions that are under the floor limit in the Authorization Selection Table (AST). This value is applicable to level 3 authorization. Valid values are as follows:

Y = Yes, force online to the over-limit authorizer
N = No, do not force online to the over-limit authorizer

The fourth column, authorization action, is applicable to all offline authorization methods and optionally as a prescreen check for online to host authorization methods. Valid values are as follows:

C = Continue processing (equivalent to no check)
R = Refer

The *default test set* identifies that no checks are required. A default test set is not identified in the ROCTBL. The following base test set (action index 2) satisfies the rules for the UK Domestic Maestro card scheme. Note that the base test set contains the same entries as the template, except with an action index of 2. You can specify a subset of the template by using a different value for the action index and including only those entries that you require.

Action Index	Reason Code	Force Online Flag	Authorization Action	Description
2	1500	Y	R	ICC Application, Common data file unable to process
2	1501	Y	R	ICC Application, Application data file unable to process
2	1502	Y	C	ICC Random selection
2	1503	N	C	Terminal Random Selection
2	1504	Y	C	Terminal not able to process ICC
2	1505	Y	C	Online forced by ICC (CDF or IDF)
2	1506	Y	C	Online forced by Card Acceptor
2	1507	Y	C	Online forced by CAD
2	1508	N	C	Online forced by Terminal
2	1509	Y	C	Online forced by Issuer
2	1510	N	C	Over Floor Limit
2	1511	Y	C	Merchant suspicious

Fallback Checking

A fallback transaction is defined as a transaction performed on an ICC at an EMV-capable terminal but the microchip is not read. This can occur as a result of a faulty chip reader or a faulty chip, but could also be an indication of a fraudulent card. The fallback, or default, is to read the card information from the magnetic stripe. The FALLBACK field in the CPF allows you to configure whether to include fallback checking as a prescreen check. You can also specify whether the BASE24-pos Device Handler/Router/Authorization process should continue processing the transaction, refer the transaction, or decline the transaction and return the card.

If the BASE24-pos Device Handler/Router/Authorization process declines the transaction, a response code of 408 is sent in the response message. If the transaction is referred, a response code of 134 is sent in the response message.

Issuer Script Commands

Issuer script commands allow the issuer to alter the data or functionality of cards already in circulation. The BASE24-pos EMV product supports scripts originated by BASE24 as well as the issuer. These script commands are transmitted in the response message, and the card uses the script data to reconfigure itself. The supported script commands can disable the microchip or update card limits. Script data passed between the issuer and the card is secured with a message authentication code (MAC) generated by a security module.

The EMV product supports the following script commands:

- CARD BLOCK (an irreversible disabling of the microchip)
- PUT DATA (updates the LCOL field on the card)

The issuer script commands are configured by BASE24 in the LCONF params EMV-CARD-BLOCK-SCRIPT and EMV-PUT-DATA-SCRIPT. BASE24 can originate script commands based on an indicator in the CAF or when requested by the issuer using a flag in the EMV Response Data token. BASE24 supports the generation of only one script command in a response message.

Issuer script data is supported either in pass-through mode or as response scripts generated by BASE24. For pass-through mode, BASE24 passes any script data received from an external authorizer through to the acquirer without validation or modification. If the EMV Script Data token is received from an issuer in a response message, the token is passed on to the acquirer end-point.

A PUT DATA script command is originated by BASE24 only if the card is configured for a positive authorization method. When a PUT DATA script is generated by BASE24, the SEND PUT DATA flag on the appropriate CAF record is reset to N only if the transaction is approved. This means that PUT DATA scripts are resent to the card on every subsequent EMV transaction, until an EMV transaction performed on the card is approved.

Configuring Issuer Script Commands

The issuer script commands are configured by BASE24 at three levels—system, card prefix, and cardholder. The system level configuration uses the LCONF params EMV-CARD-BLOCK-SCRIPT and EMV-PUT-DATA-SCRIPT, respectively.

An issuer can use the SCRIPT TEMPLATE TAG and SCRIPT MAC LENGTH fields on CPF screen 11 to configure card prefix level overrides of the MAC length specified in the LCONF params for script commands. An issuer can also use the SCRIPT TEMPLATE TAG and SCRIPT MAC LENGTH fields on CAF screen 13 to configure card level overrides of the MAC length specified in the CPF for script commands.

The MAC lengths specified in the CPF cannot exceed those specified in the LCONF. The MAC lengths specified in the CAF can exceed those specified in the CPF, but they cannot exceed those specified in the LCONF. The MAC lengths configured in the CPF and CAF do not apply to Common Core Definition cards because those cards require a MAC length of 4 bytes.

Issuer Script Formats

The hexadecimal representations of binary data contained in the two LCONF params specific to the EMV product are EMV issuer scripts. EMV issuer scripts are executed on the card itself by the EMV terminal.

Each EMV issuer script consists of a script header followed by a sequence of one or more issuer script command application protocol data units (APDUs) to be delivered serially to the ICC. The script header and each script command are composed of a tag (T), length (L), and value (V). The value for the script header is composed from the script ID, which in itself, is composed of a tag, length, and value. This structure is depicted in the following table where two commands are included in the script. All lengths must be provided in hexadecimal format.

Script Header					**Command 1**			**Command 2**		
T	L	V (script ID)			T	L	V	T	L	V
T	L	T	L	V	T	L	V	T	L	V

Script Header

The tag for the script header is always set to 71 or 72. These tags indicate the sequence of execution for processing the script by the EMV terminal. EMV terminals process scripts with a tag of 71 before issuing the final GENERATE AC command. EMV terminals process scripts with a tag of 72 after issuing the final GENERATE AC command. The first length field in the script header represents the length, in bytes, of the optional script ID plus the length of the first length field itself. If the optional script ID is not included in the script header, the length in the second length field is set to 00. The tag field for the script ID is set to always set to the tag (9F18) for the Issuer Script Identifier data element. If the optional script ID is included in the script header, the second length field is set to the length of the script ID value plus the length of the second length field itself. The script ID is optional and is not interpreted by the EMV terminal. If provided, the script ID value consists of 4 bytes.

Script Command

Each script command provided in the script is composed of a tag set to 86 for the Issuer Script Command data element, the length of the command, and the command value. The length of the command represents the length, in bytes, of the command value plus the length of the command length field itself. The command value is provided in Application Protocol Data Unit (APDU) format, which consists of a mandatory header of four bytes followed by a conditional body of variable length, as shown in the following table.

CLA	INS	P1	P2	Lc	Data	Le
<-- Mandatory Header -->				<-- Conditional Body -->		

The mandatory header consists of the class byte of the command message (CLA), the instruction byte of the command message (INS), and two parameter bytes (P1 and P2). The conditional body contains the length of command data field (Lc), variable-length command data (Data), and the length of expected data field (Le).

Class Byte. The most significant nibble of the class byte indicates the type of command as shown in the following table.

Hexadecimal Value	Description
0	Inter-industry command
8	Proprietary to EMV specifications
Any other value	Outside the scope of the EMV specifications

A command proprietary to the EMV specification is introduced by the most significant nibble of the class byte set to 8; in other words, the structure of the command and response messages is according to the ISO/IEC 7816-4 standard, ***Information technology -- Identification cards -- Integrated circuit(s) cards with contacts -- Part 4: Interindustry commands for interchange***, and the coding of the messages is defined within the context of these specifications.

The least significant nibble of the class byte indicates secure messaging and logical channel mechanisms, according to ISO/IEC 7816-4. Valid values are as follows:

C = Format 1 secure messaging used with the CCD EMV scheme
4 = Format 2 secure messaging used with other EMV schemes

Instruction Byte. The INS byte of a command indicates the action to be performed on the ICC. The following table lists the valid INS byte values used by BASE24, their meanings, and the relationship of INS byte values to CLA byte values. For more details on other possible values for the INS byte, refer to section 9.4.1 of Book 1 and section 6 of Book 3 in the EMV specifications.

CLA	INS	Command Instruction
8x	16	CARD BLOCK
04	DA	PUT DATA

Parameter Bytes. The parameter bytes (P1 and P2) can have any value. If they are not used, a parameter byte has a value of 00.

Length of Command Data. The number of data bytes sent in the command APDU (C-APDU) is denoted by Lc (length of command data field). If no command data is sent, this field is set to 00.

Command Data. When present, a command APDU message data field consists of a string of data elements in Basic Encoding Rules - Tag Length Value (BER-TLV) format. Up to 256 bytes of data can be sent for a command.

Length of Expected Data. The maximum number of data bytes expected in the response APDU is denoted by Le (length of expected data). When Le is present and contains the value zero, the maximum number of available data bytes (up to 256) is expected. When required in a command message, Le must always be set to 00.

Issuer Script Examples

The following tables illustrate the script data actually sent by the issuer (i.e., the contents of the EMV Script Token) in Card Block and Put Data scripts.

Format 1 Issuer Scripts

Format 1 issuer scripts are used with the CCD card scheme. When supporting secure messaging according to Format 1, the script data actually sent by the issuer (i.e., the contents of the EMV Script Token), must contain the values specified in the table below. Note that the lower-order nibble of the CLA byte must be set to a value of C to indicate secure messaging according to Format 1. While MAC lengths of 04, 06, and 08 bytes are available, BASE24 supports only 4-byte MACs for Format 1 secure messaging.

Byte Description	CARD BLOCK	PUT DATA
Template Tag	71 or 72	71 or 72
Template Length	10 or 14	13 or 17
Script ID Tag (2 bytes)	9F18	9F18
Script ID Length	00 or 04	00 or 04
Script ID Value (4 bytes)[1]	XXXXXXXX[2]	XXXXXXXX[2]
Script Tag	86	86
Script Length	0B	0E
CLA	8C	0C
INS	16	DA
P1	00	Data Element Tag (first byte)
P2	00	Data Element Tag (second byte)
Data and MAC Length	06	09
Command Tag	Not present	81
Command Length	Not present	01
Command Data	Not present	XX
MAC Tag	8E	8E
MAC Length[3]	04	04
MAC Value[3] (4 bytes)	XXXX[2]	XXXX[2]

[1] The script ID value is optional. The script ID length identifies whether the script ID is present.

[2] The value X represents any hexadecimal digit.

Format 2 Issuer Scripts

Format 2 issuer scripts are used with card schemes other than CCD. When supporting secure messaging according to Format 2, the script data actually sent by the issuer (i.e., the contents of the EMV Script Token) must contain the values specified in the table below. Note that the lower-order nibble of the CLA byte must be set to a value of 4 to indicate secure messaging according to Format 2. MAC lengths of 04, 06, and 08 bytes are available for Format 2 secure messaging.

Byte Description	CARD BLOCK	PUT DATA
Template Tag	71 or 72	71 or 72
Template Length	0E or 12 10 or 14 12 or 16	0F or 13 11 or 15 13 or 17
Script ID Tag (2 bytes)	9F18	9F18
Script ID Length	00 or 04	00 or 04
Script ID Value (4 bytes)[1]	XXXXXXXX[2]	XXXXXXXX[2]
Script Tag	86	86
Script Length	09, 0B, or 0D	0A, 0C, or 0E
CLA	84	04
INS	16	DA
P1	00	Data Element Tag (first byte)
P2	00	Data Element Tag (second byte)

Byte Description	CARD BLOCK	PUT DATA
Data and MAC Length	04, 06, or 08	05, 07, or 09
Command Data	Not present	FF
MAC Value (4, 6, or 8 bytes)	XXXXXXXX[2]	XXXXXXXX[2]

[1] The script ID value is optional. The script ID length identifies whether the script ID is present.

[2] The value X represents any hexadecimal digit.

Prefix Routing for EMV Transactions

EMV processing allows routing of EMV transactions by prefix using a prefix route field in the EMV segment of the CPF. For EMV transactions, the prefix route code in the EMV segment of the CPF overrides the prefix route code in the base segment of the CPF. When an IDF record is used, this prefix route code allows EMV transactions to be routed and authorized separately from non-EMV transactions. When an IDF record is not used, the EMV Issuer code provides the same functionality. The EMV Issuer code overrides the value of the ISSUER field in the POS segment of the CPF.

When an IDF record is used, the EMV Issuer code provides additional functionality. This code allows EMV transactions to select a different entry in the Authorization Selection Table (AST) record to non-EMV transactions. This means that EMV transactions can be configured to use different online and offline limits than those used by non-EMV transactions.

Processing EMV Card Transactions as Standard Track 2 Card Transactions

Some card issuers start distributing integrated circuit cards before being able to support transactions initiated by users with those cards. The BASE24-pos Device Handler/Router/Authorization process can handle transactions initiated with these cards at EMV-enabled terminals by processing them as if they are initiated with standard Track 2 cards. This processing is also known as downgrading a transaction from EMV to magnetic stripe.

The CAM CHECK TYPE field in the CPF identifies how the BASE24-pos Device Handler/Router/Authorization process authenticates integrated circuit cards for each card prefix. You can set this field to instruct the BASE24-pos Device Handler/Router/Authorization process to process transactions initiated with integrated circuit cards at EMV-enabled terminals as if they are initiated with standard Track 2 cards instead of attempting to verify the authorization request cryptogram (ARQC).

The ARQC-VRFY field in the EMV Status token identifies integrated circuit card transactions processed as standard Track 2 card transactions without verifying the ARQC.

Authorization Request and Response Cryptograms

EMV integrated circuit cards are produced with both a magnetic stripe and a microchip. The chip also includes an electronic version of the magnetic stripe data as well as other data used for authentication and risk management. When used in an EMV device, the card and terminal perform offline risk management to determine whether the transaction should be approved or declined offline, or sent online by the acquirer for authorization. If sent online, an authorization request cryptogram, also known as an ARQC, is calculated using card and terminal data. The request cryptogram and component cryptogram data are sent to the issuer with the authorization request.

The issuer checks the request cryptogram, using the cryptogram data and a secure DES key, and calculates an authorization response cryptogram, also known as an ARPC, using response data. This response cryptogram is returned to the card by the acquirer, which can then confirm that the issuer is genuine. One call is made to the security module to both verify the request cryptogram and generate the response cryptogram. If the request cryptogram is not successfully verified, standard CVV/CVC checking is applied.

On successful completion of an online or offline transaction, a cryptogram called the transaction certificate is calculated for use in verifying the offline clearing message. For a transaction authorized offline, the transaction certificate is included in the advice message sent from the POS terminal. For a transaction authorized online, the SPDH message protocol allows the transaction certificate to be sent to BASE24-pos using a message subtype of E. BASE24 does not verify the transaction certificate.

A set of EMV-specific security utilities, called CAM utilities, must be used in conjunction with the Base security utilities to support this cryptogram verification. The CAM utilities support the Thales and Atalla security modules. For more information on these security modules, see the appropriate vendor documentation.

All EMV authorization processing is initiated by the presence of the EMV Request Data token and the appropriate configuration data in the CPF, CAF, and KEYI. Support for a prevalidated CAM (e.g., by Visa on the issuer’s behalf) is also provided.

CAM checking is configurable at a prefix level, and is performed only on known prefixes configured in the CPF. If a card has a CAF record, application transaction counter (ATC) checking can be performed, as described later in this section.

When processing as an acquirer, the EMV product modules operate in pass-through mode, routing the CAM data in the EMV Request Data token to the issuer. The CAM data is logged to the TLF in the appropriate tokens (i.e., EMV Request Data, EMV Response Data, and EMV Script Data tokens). The BASE24-pos Device Handler, Host Interface, or Interchange Interface process adds the EMV Request Data token.

In the event of an issuer being unavailable and BASE24 is not using the Positive or Positive Balance Authorization method, status field verification enables BASE24 to authorize or decline an EMV transaction. Status field verification is described in more detail later in this section.

For BASE24 processing as an issuer, the BASE24-pos Device Handler/Router/Authorization process optionally performs CAM authentication using either a Thales or Atalla security module.

Offline Update Counters

BASE24 includes configuration parameters at the card prefix level on CPF screen 11 that allow issuers to specify how the card should process its offline counters when an EMV response message is received from the issuer. There is one parameter for approved financial transactions, one parameter for approved non-financial transactions, and one parameter for non-approved transactions.

Card verification, balance inquiry, check verification, and check guarantee are non-financial transactions. All other BASE24-pos transactions are treated as financial transactions.

The values supported for each parameter are the same, and include do not update offline counters, set counters to upper offline limits, reset counters to zero, and add the transaction to the appropriate counter.

BASE24 uses the appropriate value when formatting the data used in generating the Authorization Response Cryptogram (ARPC) in the Issuer Authentication Data.

Offline counter processing applies only to M/Chip 4 and Common Core Definition (CCD) cards, as these are the only EMV applications supported by BASE24 that define these bit settings.

EMV Card Verification

BASE24-pos allows issuers to validate the card verification digits (CVD) during EMV authorization processing, where the CVD on the Track 2 equivalent data is different from the CVD on the magnetic stripe.

An EMV transaction typically contains an authorization request cryptogram (ARQC), which is used by the issuer to authenticate the chip card. The ARQC verification is based on more powerful cryptography than the CVD validation, and is therefore more likely to be able to distinguish between genuine and counterfeit EMV cards.

It is possible for EMV issuers to receive chip-read transactions that do not contain an ARQC; e.g., MasterCard *partial grade* transactions. It is also possible for issuers to configure BASE24 to continue processing EMV transactions when the ARQC has not been verified successfully. Under these circumstances, BASE24-pos allows issuers optionally to validate the CVD during EMV authorization processing.

As described above, validating the CVD in an EMV transaction where ARQC verification has been performed successfully serves little purpose, because the validation authenticates only the Track 2 equivalent data. However, to provide complete flexibility, BASE24-pos allows issuers optionally to validate the CVD during EMV authorization processing, even when the ARQC has been verified successfully.

The EMV extension modules to POS Authorization use four fields in the EMV segment of the Card Prefix File (CPF) to determine whether to perform card verification on EMV transactions.

- An EMV card verification check type field specifies the circumstances (if any) under which card verification is to be performed on EMV transactions.
- An EMV card verification method field specifies the algorithm and data used when card verification is performed on EMV transactions.
- An EMV card verification effective date field specifies the earliest card expiration date for which card verification is to be performed on EMV transactions.
- An EMV card verification data field contains additional data that is required when card verification is performed on EMV transactions.

Cryptographic Support

EMV cryptographic support is provided for the following security functions:

- Request and response cryptogram processing
- Secure messaging with integrity for issuer script data

BASE24-pos provides cryptographic support for VIS, M/Chip, and CCD card schemes. All of the required cryptographic support can be provided by both the BASE24-pos Device Handler/Router/Authorization process (using the CAM Utilities module) and the BASE24 Transaction Security Services process, except that only the Transaction Security Services process can provide cryptographic support for the EMV 2000 method of session key derivation and CCD (IAD format A) cards.

The value in the KEYI GROUP field on CPF screen 11 is used to access the security data (e.g., master keys) required for cryptographic processing. The value in the KEYI GROUP field is used to access either the KEYI (when the Router/Authorization process provides cryptographic support) or the TSS database (when the Transaction Security Services process is used). The expiration date of the card is also used to access the data, to allow different EMV security profiles to be configured as cards are reissued.

Both the KEYI and the TSS database support up to 24 authentication (cryptogram) master keys and one secure messaging (integrity) key for each EMV security profile. When requesting a specific security function, the Router/Authorization process retrieves the derivation key index passed in requests, and passes it on to the appropriate CAM utilities procedure or TSS procedure to locate a specific cryptogram key or secure messaging key.

To support different security implementations for individual payment schemes, the EMV SCHEME field in the KEYI or the TSS database identifies the specific implementation associated with an EMV profile (i.e., authentication data). This field, in conjunction with the cryptogram version number passed in requests, identifies the specific implementation for the card, and is used by the Router/Authorization process or the Transaction Security Services process to determine the specific security module calls and security functions to be performed.

The following subtopics discuss BASE24 cryptographic support without the BASE24 Transaction Security Services process. For information on cryptographic support provided with the Transaction Security Services process, including EMV 2000 cryptographic support, refer to the ***BASE24 Transaction Security Services Processing Guide.***

EMV Scheme and Cryptogram Version Number

To support different security implementations for individual payment schemes, the EMV SCHEME field on KEYI screen 1 identifies the specific EMV implementation associated with an EMV profile (i.e., authentication data). This field, in conjunction with the cryptogram version number passed in requests, identifies the specific implementation associated with a KEYI record and is used by the BASE24-pos Device Handler/Router/Authorization process to determine the specific security module calls and security functions to be performed. Each of the valid EMV schemes are explained below.

- VIS—Defined by Visa in the ***Visa Integrated Circuit Card Specification***. (Note this format is used for UKIS cards).
- M/Chip—Defined for MasterCard MCPA (M/Chip 2.1) cards.
- CCD—Mandated for CCD (IAD Format A) cards.

  Note: The CCD scheme does not use the KEYI because it requires the TSS.

Cryptogram Version Numbers in EMV

The cryptogram version number is used (in conjunction with the EMV scheme) to identify the security requirements in EMV online transaction processing. In particular, the cryptogram version number defines the required processing in the following areas:

- Card key derivation method
- Session key derivation method for ARQC calculation
- Input data in ARQC/TC calculation
- Method used to calculate ARQC
- Session key derivation method for ARPC calculation
- Method used to calculate ARPC
- Session key derivation method for secure messaging
- Data format in secure messaging MAC calculation
- Padding of encrypted data for secure messaging
- Method used to encrypt data in secure messaging

The tables below indicate the security processing options, data formats, and methods used for each supported combination of EMV scheme and cryptogram version number. You can obtain further details on each of these security processing options, data formats, and methods in the EMV and card scheme specifications.

Throughout these tables, *x* represents the characters 0–9 and A–F in cryptogram version number %h0*x*, and *y* represents the characters A–F in cryptogram version number %h*y*5.

The processing performed within security devices for these functions is depicted in several diagrams at the end of this topic.

Card Key Derivation Method

Security Processing Options	EMV Application	
	EMV Scheme	**CVN**
Option A – Format a 16-byte data block by concatenating the PAN and application PAN sequence number (APSN), using padding or truncation if required. The left half of the key is computed by 3DES encrypting the PAN/APSN with the master key. The right half of the key is computed by 3DES encrypting the inverted (bitwise) PAN/APSN with the master key.	VIS	%h0A
	VIS	%h11[1]
	VIS	%h0E
	M/Chip	%h0*x*
	M/Chip	%h10
	M/Chip	%h11
	M/Chip	%h12
	M/Chip	%h13
	M/Chip	%h14
	M/Chip	%h15

Security Processing Options	EMV Application	
	EMV Scheme	**CVN**
Option B – Use the SHA-1 hashing algorithm on the PAN and APSN, and then create a 16-byte data block by selecting decimal digits from the output of the hash result. The left half of the key is computed by 3DES encrypting the PAN/APSN with the master key. The right half of the key is computed by 3DES encrypting the inverted (bitwise) PAN/APSN with the master key.	CCD	%h*y*5

[1] Cryptogram version number 17 for Visa cards is used to support quick Visa Smart Debit/Credit card (qVSDC) transactions. These transactions use a minimum of data to ensure quick transactions over a contactless interface, such as one employed for a low value payment toll booth operation.

Session Key Derivation Method For ARQC Calculation

Security Processing Options	EMV Application	
	EMV Scheme	**CVN**
None – Use card key.	VIS	%h0A
	VIS	%h11
EMV 2000 – Use a specified function that contains a branch factor, a *parent* key, and a *grandparent* key. Use the function to take bits from the ATC to derive a session key that is unique for each value of the ATC.	VIS	%h0E
	M/Chip	%h12
	M/Chip	%h13
MasterCard-proprietary – Format an 8-byte data block, and use it to generate the left and right halves of the session key. The left half is computed by replacing the third byte of the data block with %hF0, and 3DES encrypting the result with the card key. The right half is computed by replacing the third byte of the data block with %h0F, and 3DES encrypting the result with the card key.	M/Chip	%h0*x*
	M/Chip	%h10
	M/Chip	%h11

Security Processing Options	EMV Application	
	EMV Scheme	CVN
Common – Format an 8-byte *diversification value*, which is the 2-byte ATC concatenated with 6 bytes of %h00. Use the diversification value to generate the left and right halves of the session key. The left half is computed by replacing the third byte of the data block with %hF0, and 3DES encrypting the result with the card key. The right half is computed by replacing the third byte of the data block with %h0F, and 3DES encrypting the result with the card key.	M/Chip	%h14
	M/Chip	%h15
	CCD	%h*y*5

Input Data In ARQC/TC Calculation

Input Data Format	EMV Application	
	EMV Scheme	CVN
Amount, Auth Amount, Other Terminal Country Code Terminal Verification Results (TVR) Transaction Currency Code Transaction Date Transaction Type Unpredictable Number Application Interchange Profile (AIP) Application Transaction Counter (ATC) Card Verification Results (4 bytes)	VIS	%h0A
	VIS	%h0E
	M/Chip	%h0*x*
Amount, Auth Transaction Currency Code Unpredictable Number	VIS	%h11

<table>
<tr><th rowspan="2">Input Data Format</th><th colspan="2">EMV Application</th></tr>
<tr><th>EMV Scheme</th><th>CVN</th></tr>
<tr><td rowspan="3">Amount, Auth
Amount, Other
Terminal Country Code
Terminal Verification Results (TVR)
Transaction Currency Code
Transaction Date
Transaction Type
Unpredictable Number
Application Interchange Profile (AIP)
Application Transaction Counter (ATC)
Card Verification Results (6 bytes)</td><td>M/Chip</td><td>%h10</td></tr>
<tr><td>M/Chip</td><td>%h12</td></tr>
<tr><td>M/Chip</td><td>%h14</td></tr>
<tr><td rowspan="3">Amount, Auth
Amount, Other
Terminal Country Code
Terminal Verification Results (TVR)
Transaction Currency Code
Transaction Date
Transaction Type
Unpredictable Number
Application Interchange Profile (AIP)
Application Transaction Counter (ATC)
Card Verification Results (6 bytes)
Counters (8 bytes)</td><td>M/Chip</td><td>%h11</td></tr>
<tr><td>M/Chip</td><td>%h13</td></tr>
<tr><td>M/Chip</td><td>%h15</td></tr>
<tr><td>Amount, Auth
Amount, Other
Terminal Country Code
Terminal Verification Results (TVR)
Transaction Currency Code
Transaction Date
Transaction Type
Unpredictable Number
Application Interchange Profile (AIP)
Application Transaction Counter (ATC)
Issuer Application Data</td><td>CCD</td><td>%hy5</td></tr>
</table>

Method Used to Calculate ARQC

<table>
<tr><th rowspan="2">Security Processing Options</th><th colspan="2">EMV Application</th></tr>
<tr><th>EMV Scheme</th><th>CVN</th></tr>
<tr><td rowspan="2">Visa-proprietary – Format a data block, comprising the input data padded to the right with bytes of %h00 until the data block is a multiple of 8 bytes in length. Calculate an 8-byte ARQC by using the key to generate a MAC over the data block.</td><td>VIS</td><td>%h0A</td></tr>
<tr><td>VIS</td><td>%h11</td></tr>
<tr><td rowspan="9">EMV – Format a data block, comprising the ARQC concatenated with the input data and one byte containing %h80. If necessary, append bytes of %h00 until the data block is a multiple of 8 bytes in length. Calculate an 8-byte ARQC by using the key to generate a MAC over the data block.</td><td>VIS</td><td>%h0E</td></tr>
<tr><td>M/Chip</td><td>%h0x</td></tr>
<tr><td>M/Chip</td><td>%h10</td></tr>
<tr><td>M/Chip</td><td>%h11</td></tr>
<tr><td>M/Chip</td><td>%h12</td></tr>
<tr><td>M/Chip</td><td>%h13</td></tr>
<tr><td>M/Chip</td><td>%h14</td></tr>
<tr><td>M/Chip</td><td>%h15</td></tr>
<tr><td>CCD</td><td>%hy5</td></tr>
</table>

Session Key Derivation Method For ARPC Calculation

Derivation Method	EMV Application	
	EMV Scheme	**CVN**
None – Use card key.	VIS	%h0A
	M/Chip	%h0*x*
	M/Chip	%h10
	M/Chip	%h11
EMV 2000 – Use a specified function that contains a branch factor, a *parent* key, and a *grandparent* key. Use the function to take bits from the ATC to derive a session key that is unique for each value of the ATC.	VIS	%h0E
	M/Chip	%h12
	M/Chip	%h13
Common – Format an 8-byte *diversification value*, which is the 2-byte ATC concatenated with 6 bytes of %h00. Use the diversification value to generate the left and right halves of the session key. The left half is computed by replacing the third byte of the data block with %hF0, and 3DES encrypting the result with the card key. The right half is computed by replacing the third byte of the data block with %h0F, and 3DES encrypting the result with the card key.	M/Chip	%h14
	M/Chip	%h15
	CCD	%h*y*5

Input Data and Data Length in ARPC Calculation

Input Data and Data Length	EMV Application	
	EMV Scheme	**CVN**
ISO 8583 (1987) Response Code (2 bytes). ARPC is 8 bytes.	VIS	%h0A
	VIS	%h0E
	M/Chip	%h0*x*

Input Data and Data Length	EMV Application	
	EMV Scheme	**CVN**
16-bit ARPC Response Code (2 bytes). ARPC is 8 bytes.	M/Chip	%h10
	M/Chip	%h11
	M/Chip	%h12
	M/Chip	%h13
	M/Chip	%h14
	M/Chip	%h15
Card Status Updates (4 bytes) plus Proprietary Authentication Data (up to 8 bytes). ARPC is 4 bytes (truncated from 8 bytes).	CCD	%h*y*5

Method Used To Calculate ARPC

Calculation Method	**EMV Application**	
	EMV Scheme	**CVN**
Method 1 - Format an 8-byte data block, comprising the input data padded to the right with bytes of %h00. Calculate an 8-byte ARPC by XORing the data block with the ARQC, and 3DES encrypting the result with the session key. **Note:** This method cannot support more than 8 bytes of input data.	VIS	%h0A
	VIS	%h0E
	M/Chip	%h0*x*
	M/Chip	%h10
	M/Chip	%h11
	M/Chip	%h12
	M/Chip	%h13
	M/Chip	%h14
	M/Chip	%h15
Method 2 – Format a data block, comprising the ARQC concatenated with the input data and one byte containing %h80. If necessary, append bytes of %h00 until the data block is a multiple of 8 bytes in length. Calculate an 8-byte ARPC by using the key to generate a MAC over the data block. **Note:** This method is also used for all schemes and CVNs to calculate the MAC required for secure messaging with integrity.	CCD	%h*y*5

Session Key Derivation Method For Secure Messaging

<table>
<tr><th rowspan="2">Derivation Method</th><th colspan="2">EMV Application</th></tr>
<tr><th>EMV Scheme</th><th>CVN</th></tr>
<tr><td>Visa-proprietary. Format an 8-byte diversification value, which is the 2-byte ATC right justified and padded to the left with 6 bytes of %h00. Use the diversification value to generate the left and right halves of the session key. The left half is computed by XORing the data block with the left half of the card key. The right half is computed by performing a bitwise inversion and XORing the result with the right half of the card key.</td><td>VIS</td><td>%h0A</td></tr>
<tr><td rowspan="3">EMV 2000 – Use a specified function that contains a branch factor, a parent key, and a grandparent key. Use the function to take bits from the ATC to derive a session key that is unique for each value of the ATC.</td><td>VIS</td><td>%h0E</td></tr>
<tr><td>M/Chip</td><td>%h12</td></tr>
<tr><td>M/Chip</td><td>%h13</td></tr>
<tr><td rowspan="6">Common – Format an 8-byte diversification value, which is the ARQC. Use the diversification value to generate the left and right halves of the session key. The left half is computed by replacing the third byte of the data block with %hF0, and 3DES encrypting the result with the card key. The right half is computed by replacing the third byte of the data block with %h0F, and 3DES encrypting the result with the card key.</td><td>M/Chip</td><td>%h0x</td></tr>
<tr><td>M/Chip</td><td>%h10</td></tr>
<tr><td>M/Chip</td><td>%h11</td></tr>
<tr><td>M/Chip</td><td>%h14</td></tr>
<tr><td>M/Chip</td><td>%h15</td></tr>
<tr><td>CCD</td><td>%hy5</td></tr>
</table>

Data Format In Secure Messaging MAC Calculation

Data Format	EMV Application	
	EMV Scheme	CVN
Format 2 - The data elements (including the MAC) contained in the data field of the secured command are not composed of TLV data objects. Transaction-specific data, (e.g., the ATC), is inserted into the data field (immediately before any command data) to ensure that the MAC value generated is unique per transaction.	VIS	%h0A
	VIS	%h0E
	M/Chip	%h0*x*
	M/Chip	%h10
	M/Chip	%h11
	M/Chip	%h12
	M/Chip	%h13
	M/Chip	%h14
	M/Chip	%h15
Format 1 - The data field of the secured command is composed of TLV data objects. If the command to be secured has command data, then the command data is carried in the first data object, and is itself encapsulated under an EMV tag. The second data object is the MAC, which is encapsulated under tag 8E.	CCD	%h*y*5

Padding Of Encrypted Data For Secure Messaging

Padding Method	EMV Application	
	EMV Scheme	CVN
Visa-proprietary - Prepend encrypted data with one byte containing the length of the encrypted data. Append encrypted data with one byte containing %h80. If necessary, append bytes of %h00 until data block is a multiple of 8 bytes in length.	VIS	%h0A
	VIS	%h0E

Padding Method	EMV Application	
	EMV Scheme	**CVN**
MasterCard-proprietary - If the enciphered data is not a multiple of 8 bytes in length, append one byte containing %h80. If necessary, append bytes of %h00 until data block is a multiple of 8 bytes in length.	M/Chip	%h0*x*
	M/Chip	%h10
	M/Chip	%h11
	M/Chip	%h12
	M/Chip	%h13
	M/Chip	%h14
	M/Chip	%h15
EMV - Append enciphered data with one byte containing %h80. If necessary, append bytes of %h00 until data block is a multiple of 8 bytes in length.	CCD	%h*y*5

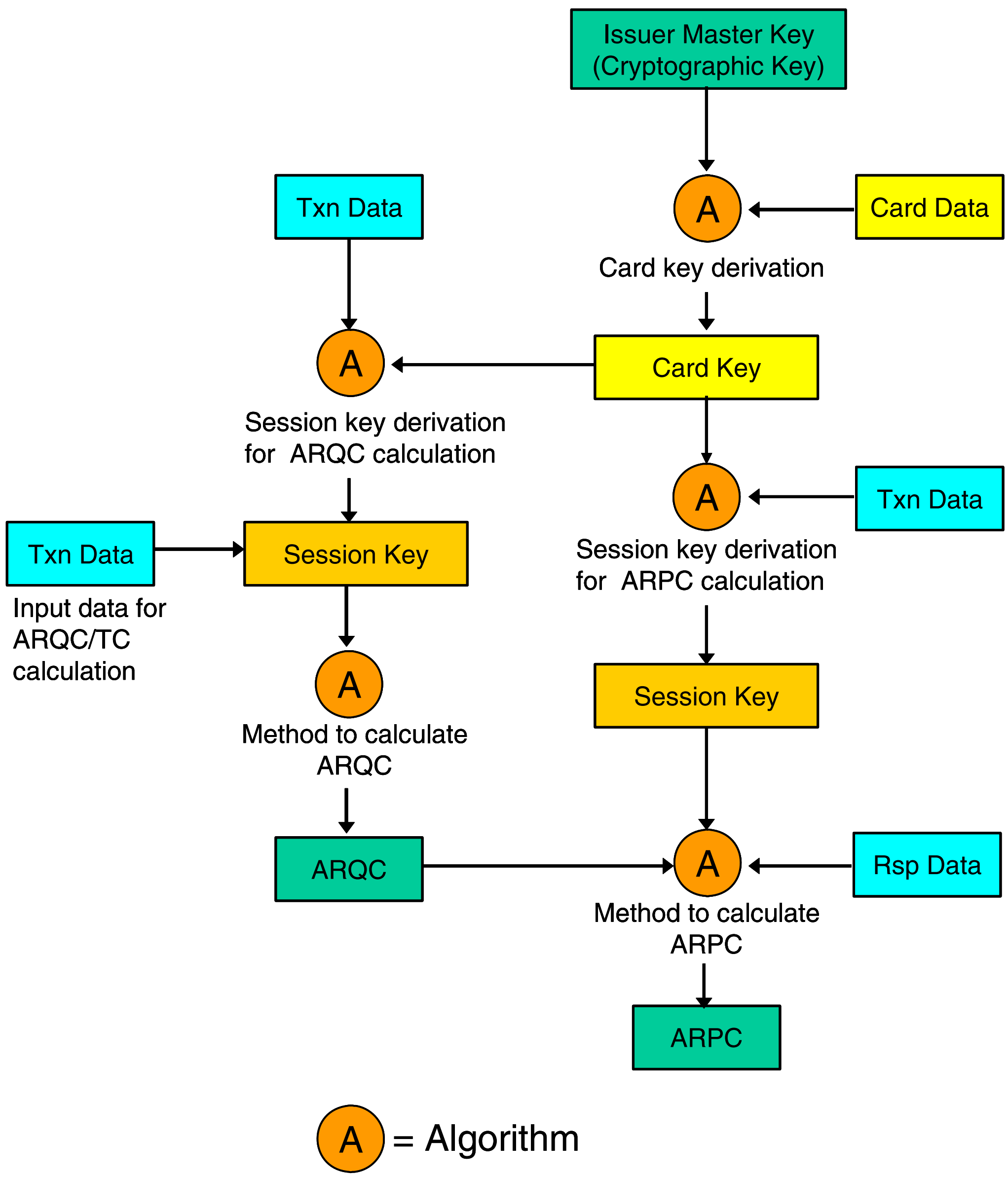
Method Used To Encrypt Data In Secure Messaging

Encryption Method	EMV Application	
	EMV Scheme	**CVN**
Electronic Code Book (ECB).	VIS	%h0A
	VIS	%h0E
Cipher Block Chaining (CBC).	M/Chip	%h0x
	M/Chip	%h10
	M/Chip	%h11
	M/Chip	%h12
	M/Chip	%h13
	M/Chip	%h14
	M/Chip	%h15
	CCD	%hy5

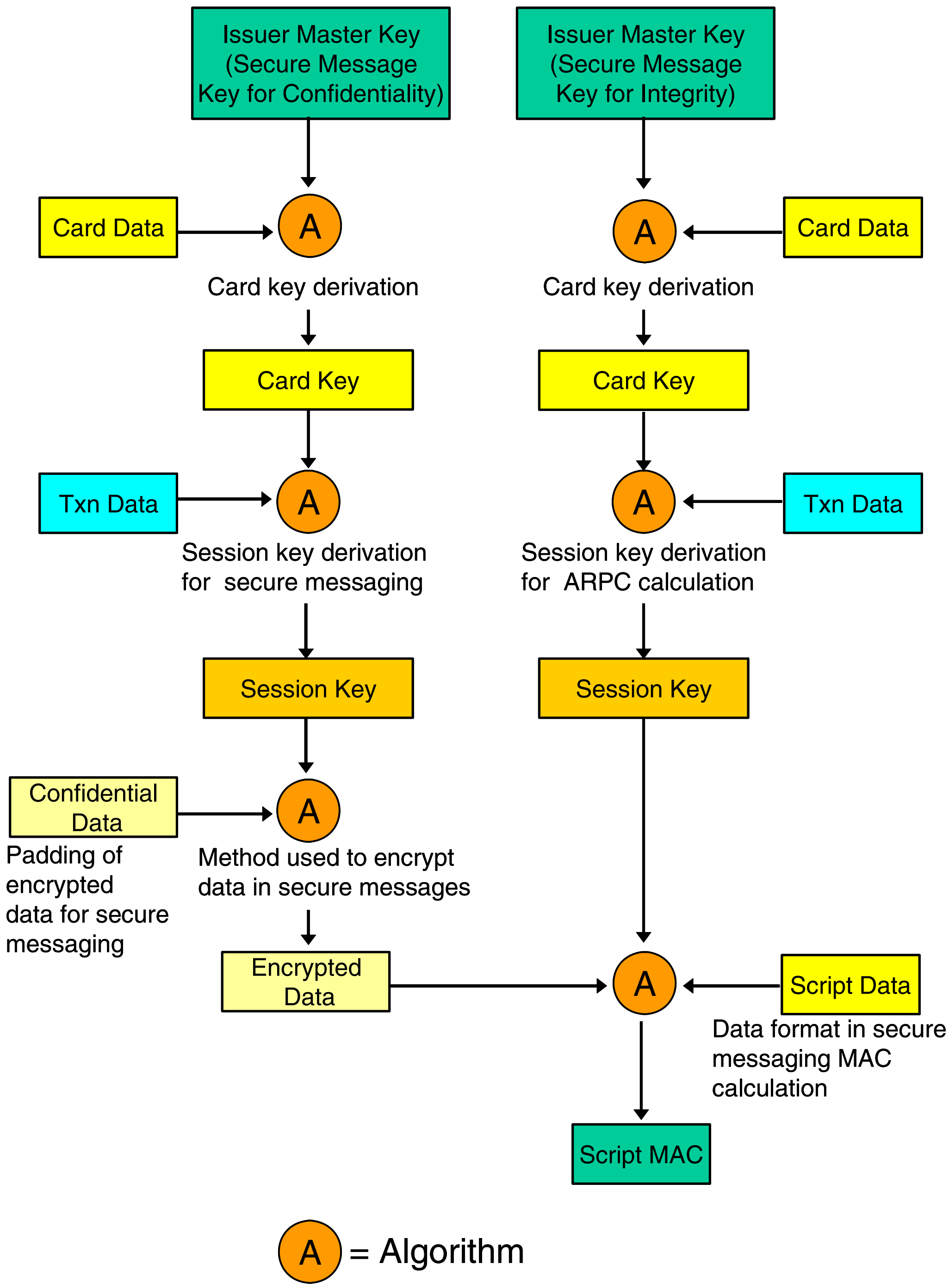
Cryptographic Processing Flows

The following diagrams depict the flow of cryptographic functions for authorization request and response cryptogram processing and for secure message processing. All of these functions are performed within a hardware security device.

Authorization Request and Response Processing



Secure Message Processing



ACI Worldwide, Inc.

5: CAP Token Validation

Through the use of the Transaction Security Services process, BASE24 supports chip authentication program (CAP) token validation for MasterCard's OneSMART Authentication solution and Visa's Dynamic Passcode Authentication (DPA) solution. CAP token validation provides two factor authentication (2FA) in card-not-present point-of-sale (POS) environments. Because two factor authentication uses two security items—the chip on the card and a PIN—it is more secure than using only chip or PIN verification. The Transaction Security Services process supports CAP token validation in Thales e-Security host security modules (HSMs).

For detailed information on the configuration requirements for CAP token validation, refer to section 2 of this manual and the ***BASE24 Transaction Security Services Processing Guide.***

CAP token validation can be used with contact and contactless EMV cards or on chip cards used only for CAP token validation, and is performed by an extension module of the BASE24-pos Router/Authorization module. With or without the CAP token validation extension module, the BASE24-pos Router/Authorization module can perform standard authorization processing (e.g., card status checking, expiration date checking) when it receives a card verification request that includes CAP token data in the Authentication Data token, if standard processing is configured for the card and the necessary data is available. However, no CAP token validation occurs without the CAP token validation extension module.

Although CAP functionality is based on the EMV infrastructure, there is little similarity in the authorization processing performed for an EMV payment transaction and a CAP token validation transaction. Because of this, a CAP token validation transaction does not use the EMV tokens, and does not invoke logic in EMV extension modules. However, a BASE24 customer has to implement the BASE24 EMV extension modules in order to retrieve the CAP data added to the EMV segments in the CPF and CAF. It is assumed that BASE24 customers who implement CAP token validation will also support EMV payment processing because the CAP solution uses the EMV technology, and a similar assumption applies to the TSS implementation.

Processing

For Level 2 (offline) and Level 3 (online/offline) authorization where the CAF is used in authorization processing, the CAP token validation extension module performs the following processing when the request contains CAP token data in the Authentication Data token.

1. Checks the values of the HOST CAP TOKEN VALIDATION OPTION and CAP GROUP fields on CPF screen 13 to determine whether BASE24 should perform CAP token validation.
2. Checks whether the value of the BAD CAP TOKEN VALIDATION ACCUMULATOR field for the primary or secondary card on CAF screen 13 has reached the value of the BAD CAP TOKEN VALIDATION LIMIT field on CPF screen 13. If the limit has been reached, the extension module checks the value of the BAD CAP TOKEN OVERRIDE FLAG on CAF screen 13 to determine whether the limit should be overridden, and either declines the transaction and returns control to the Router/Authorization module or continues CAP token validation processing with the next step.
3. Formats and passes the message to the Transaction Security Services process for validation and a response. The message is formatted using transaction data from the Authentication Data token and configuration data from the CPF and CAF. If the CAP token cannot be validated (e.g., due to an HSM error), the extension module declines the transaction and returns control to the Router/Authorization module. Otherwise, CAP token validation processing continues with the next step.
4. Updates the CAF based on the results of the CAP token validation.
   - If the validation succeeds, the extension module updates the value in the appropriate ATC field (refer to the "Application Transaction Counter Checking" topic in section 4) and resets the value in the appropriate BAD CAP TOKEN VALIDATION ACCUMULATOR field for the primary or secondary card on CAF screen 13.
   - If the validation fails, the extension module adds 1 to the value of the BAD CAP TOKEN VALIDATION ACCUMULATOR field for the primary or secondary card on CAF screen 13.

   Note: If an Issuer host system performs the CAP processing, the Router/Authorization extension module still updates the CAF based on the results.
5. Sets an appropriate response code in the PSTM and returns control to the Router/Authorization module.

6: Device Handler Processing

BASE24-pos Device Handler modules are bound with the BASE24-pos Device Handler/Router/Authorization process to provide an interface between POS devices and the BASE24-pos Router and Authorization modules.

This section introduces the functionality provided by device-specific BASE24-pos Device Handler EMV modules. The EMV modules are additional licensable modules that, when bound with standard BASE24-pos device-specific Device Handler modules, provide the ability to process EMV transactions.

Device Handler EMV Functionality

EMV Device Handler modules provide the following functionality:

- Identify EMV request messages
- Validate and store EMV-specific data sent in request messages
- Accept and process card details read from a chip
- Determine whether a MAC is present in an EMV request message and, if present, verify the MAC
- Set EMV-specific values in the PSTM (e.g., a point-of-service entry mode of 05)
- Determine which EMV-specific tokens are required and add them to the PSTM
- Identify EMV response messages and retrieve the appropriate tokens from the PSTM
- Include EMV-specific data in response messages
- Determine whether a MAC is required in an EMV response message and, if required, generate the MAC

EMV data is passed between processing modules in a BASE24 system in data tokens. For requests, the Device Handler module maps EMV data from the request message to tokens and appends these tokens to the PSTM. The module formats the EMV Request Data token, the EMV Status token, and the EMV Discretionary Data token, as needed.

For responses, the module maps EMV data from the EMV Response Data token and the EMV Script Data token to the response message sent back to the terminal.

EMV Processing

The Device Handler EMV extension module removes and saves the EMV data from an incoming request message and reformats the message as a device-specific message. The device-specific Device Handler module of the BASE24-pos Device Handler/Router/Authorization process performs normal processing on the message and then the EMV extension module builds the EMV tokens that are appended to the PSTM before passing the message to the Authorization module. The EMV extension module performs the following:

- If the terminal indicates that it is EMV-capable, EMV processing is initiated.
- If the EMV data is successfully retrieved, this data is formatted and loaded into the appropriate tokens and passed to the Router and Authorization modules of the BASE24-pos Device Handler/Router/Authorization process for further processing.
- If EMV terminal capabilities are not provided in the request message, the module checks the EMV terminal capability settings for the terminal on BASE24-pos Terminal Data files (PTD) screen 17 to determine which EMV tokens to include.

Response messages from the Authorization module are reformatted into the device-specific format using data passed from the Router and Authorization modules. If the EMV Response Data token and, optionally, the EMV Script Data token are present, the following processing occurs:

- EMV response data is added to the message and returned to the terminal. The PSTM can contain the EMV Response Data token, which is used in formatting the response to the terminal.
- The content of the EMV Script Data token, if present, is passed to the device in the appropriate field.
- If the request contained EMV tokens but the response does not contain the EMV Response Data token, no EMV data is included in the response to the terminal.

If the Device Handler module declines the transaction, no EMV data is included in the response to the terminal. Similarly, if the issuer does not return EMV data, none is returned in the response to the device.

Device-Specific Processing

Refer to the following appendixes for device-specific processing and message formats.

- Appendix A contains information for the BASE24-pos Standard POS Device Handler (SPDH) module.
- Appendix B contains information for the APACS 70 Device Handler module.
- Appendix C contains information for the Hypercom Device Handler module.

In addition, the Visa II Device Handler supports EMV data in contactless magnetic stripe transactions (MSD with CVN 17).

7: Interchange Interface Processing

This section describes how the following interchange interface processes handle EMV transactions.

- American Express Global Network (AMEX GNS)
- MasterCard (BankNet and MDS Maestro)
- Visa

EMV Support in the BASE24 External Message

The BIC ISO Interchange Interface and BASE24 ISO Host Interface processes handle EMV transactions using tokens to carry EMV information internally and placing these tokens in data element P-63 of the BASE24 external message.

The BIC ISO Interchange Interface and BASE24 ISO Host Interface processes use the standard technique for configuring the EMV-specific tokens to be included in external messages.

Enhanced EMV Response Code Mapping

BASE24 maps response codes carried internally with ISO response codes used in the BASE24 external message. Two types of mapping are supported: standard mapping and enhanced EMV mapping.

The difference between standard mapping and enhanced EMV mapping is how EMV-related response codes are handled. Using standard mapping, EMV-related response codes are mapped to more general response codes which can obscure the reason an EMV transaction is declined. Using enhanced EMV mapping, however, unique ISO response codes are used that allow EMV-related response codes to retain their EMV characteristics. For example, an outbound BASE24-pos EMV response code of 404 would map to 05 under standard mapping and U4 under enhanced EMV mapping as shown in the table below. Where the former does not retain the specific reason for the response (due to an ATC check failure), the latter does.

BASE24-pos	Standard Mapping	Enhanced EMV Mapping
404 ATC check failure	05 Do not honor	U4 ATC check failure

For a complete list of BASE24-pos response code mappings, refer to the ***BASE24 External Message Manual***.

The type of mapping used for BASE24-atm and BASE24-pos EMV messages is controlled by the ENHANCED-EMV-RC-MAPPING param in the LCONF. Param values control the type of mapping as follows:

Param Value	**Response Code Mapping**	
	BASE24-atm	**BASE24-pos**
0	Standard Mapping	Standard Mapping
1	Enhanced EMV Mapping	Standard Mapping
2	Standard Mapping	Enhanced EMV Mapping
3	Enhanced EMV Mapping	Enhanced EMV Mapping

Token Impacts

The following table describes how token processing is performed when an interchange is involved.

EMV Request Data token	The interchange EMV module adds the EMV Request Data token on receiving inbound EMV requests, and uses the token in formatting outbound EMV-format requests.
EMV Discretionary Data token	The interchange EMV module adds the EMV Discretionary Data token at the same time that it adds the EMV Request Data token. The EMV Discretionary Data token provides information about the capabilities of the terminal. None of this data is required for authorization.
EMV Status token	The interchange EMV module adds the EMV Status token on receiving EMV messages, or magnetic stripe transactions from an EMV-capable terminal.
EMV Response Data token	The interchange EMV module adds the EMV Response Data token on receiving an inbound response with EMV information, and uses the token in formatting outbound EMV-format responses.
EMV Script Data token	The interchange EMV module adds the EMV Script Data token on receiving inbound responses with EMV information, and with the Issuer Script bit set in the EMV Response token. It uses the token in formatting in outbound EMV-format responses.

Refer to the ***BASE24 Tokens Manual*** for configuring the tokens used by interface processes.

AMEX GNS Interchange Interface

The American Express Global Network (AMEX GNS) interchange uses data element P-55, ICC System Related Data, to pass chip card information from the acquirer to the issuer. This information is passed in the POS Authorization Request (1100) and POS Authorization Response (1110) messages. The data includes encrypted elements that only the issuer or issuer agent is able to use. The passing of this data provides for Online Mutual Authentication (OMA) as follows:

- The chip is able to authenticate itself to the issuer in the POS Authorization Request (1100) message.
- The issuer is able to authenticate itself to the chip in the POS Authorization Response (1110) message. American Express Global Network does not edit the contents of data element P-55.

The American Express Global Network interchange performs the following message edits:

- If data element P-55 is present in the 1100 POS Authorization Request message, then positions 1 (Card Data Input Capability) and 7 (Card Data Input Mode Code) of data element 22 (Point of Service Data Code) must be set to the value 5 (Integrated Circuit Card (ICC)).
- If data element P-55 is present in the 1110 POS Authorization Response message, then it must be present in the corresponding 1100 POS Authorization Request. Data element P-55 is mandatory in the 1110 POS Authorization Response message if it is present in the 1100 POS Authorization Request message.
- The American Express Global Network interchange mandates that in the 1110 POS Authorization Response messages, data element P-55 must contain the issuer authentication data and issuer script data only if the transaction is approved. Therefore, if the transaction is denied, the issuer authentication data and issuer script data is not included in the 1110 POS Authorization Response messages. Regardless of whether the transaction is approved or disapproved, the version header and version number must be echoed in the 1110 POS Authorization Response message.

BASE24 as Acquirer

For BASE24 as an acquirer, the American Express Global Network (AEGN) ISO Interface receives BASE24-pos Standard Internal Messages (PSTMs) with one or more EMV tokens attached. The AEGN EMV extension is called to complete the formatting of the SEM. The extent of this formatting is dependent upon the level of chip support determined by the contents of the transaction message.

For transactions performed at non-chip-capable terminals, no further formatting is required.

For transactions performed at chip-capable terminals where the PAN is not read from the chip, and no chip data is available, the EMV Status token is used to indicate that it is not a chip read transaction in the message sent to the American Express Global Network.

For transactions performed at chip-capable terminals where the PAN is read from the chip, but no chip data is available, the EMV Status token is used to indicate that it is a chip-read transaction in the message sent to American Express Global Network.

For transactions performed at chip-capable terminals where the PAN is read from the chip and chip data is available, both the EMV Status token and the EMV Request Data token are present in the internal message.

When receiving a response from the American Express Global Network interchange containing data element P-55, the PSTM to be returned is formatted as a response to the BASE24-pos Device Handler/Router/Authorization process. The AEGN EMV extension is called to complete the formatting of the PSTM. If the original request is contained in the EMV Request Data token, then this formatting can include adding the EMV Response token and the EMV Script Data token prior to sending the formatted PSTM.

BASE24 as Issuer

For BASE24 as issuer, the AEGN ISO Interface receives external request messages. The AEGN EMV extension is called to complete the formatting. The extent of this formatting is dependent upon the level of chip support determined by the contents of the transaction message.

If the terminal is EMV capable, but the chip data elements are not available, the AEGN EMV extension appends only the EMV Status token to the PSTM to be forwarded to the BASE24-pos Device Handler/Router/Authorization process. This token differentiates between transactions where the PAN is read from a chip and those where it is not.

If the terminal is EMV capable, the card data is read from the chip, and the chip data elements are available, the AEGN EMV extension appends both the EMV Status token and the EMV Request Data token to the PSTM to be forwarded to the BASE24-pos Device Handler/Router/Authorization process. The PSTM, with the tokens, is sent to the BASE24-pos Device Handler/Router/Authorization process. The BASE24-pos Device Handler/Router/Authorization process validates the request cryptogram, generates the response cryptogram, and verifies the Application Transaction Counter.

When a response is sent to American Express Global Network interchange, the AEGN EMV extension is called to complete the formatting. If data element P-55 is present in the original request from American Express Global Network interchange, this formatting includes extracting data from the EMV Response token and (if present) the EMV Script Data token. If the card authentication is successful, (issuer authentication data is present in the EMV Response token), this data is mapped to the corresponding field in data element P-55 in the response message sent to American Express Global Network interchange. If the card authentication is not successful, data element 55 is sent in the response message to American Express Global Network interchange with only the version header and version number subfields populated.

American Express Global Network EMV External Message

The American Express Global Network interchange uses data element P-55, Integrated Circuit Card System Related Data, to pass chip information from the acquirer to the issuer. This data element contains many subelements and is supported only for transactions where chip data is successfully read from the chip at a chip-capable terminal.

Most of the information in data element P-55 is also contained in the EMV-specific tokens. Although data element P-55 may be present in the external message, it is not logged to the ILF.

The format of EMV-specific tokens is described in the ***BASE24 Tokens Manual.***

EMV Request Data Token

The following table shows the mapping between fields in the EMV Request Data token and the EMV tag values in data element P-55 of the request message. All data elements in this token are required.

Token Field	EMV Tag Values in Data Element P-55 of the Request Message
CRYPTO-INFO-DATA	9F27
TVR	95
ARQC	9F26
AMT-AUTH	9F02
AMT-OTHER	9F03
AIP	82
ATC	9F36
TERM-CNTRY-CDE	9F1A
TRAN-CRNCY-CDE	5F2A
TRAN-DAT	9A
TRAN-TYPE	9C
UNPREDICT-NUM	9F37
ISS-APPL-DATA	9F10

EMV Discretionary Data Token

The following table shows the mapping between fields in the EMV Discretionary Data token and the EMV tag values in data element P-55 of the request message.

Token Field	EMV Tag Values in Data Element P-55 of the Request Message
TERM-SERL-NUM	9F1E

Token Field	EMV Tag Values in Data Element P-55 of the Request Message
EMV-TERM-CAP	9F33
EMV-TERM-TYPE	9F35
APPL-VER-NUM	9F09
CVM-RSLTS	9F34
DF-NAME	84

EMV Status Token

The following table shows the mapping between fields in the EMV Status token and the interchange/interface message.

Token Field	Value
PT-SRV-ENTRY-MDE	Set to the value 0*yx,* where *y* is the value of field 22 subfield 7 (Card Data Input Mode) and *x* is mapped from field 22 subfield 12 (PIN Capture Capability).
TERM-ENTRY-CAP	Set to the value 5 (EMV-capable terminal) if field 22 subfield 1 (Card Data Input Capability) equals the value 5; otherwise set to the value 0.
LAST-EMV-STAT	Set to the value 1 if either of the following exists: • Field 22 subfield 7 (Card Data Input Mode) equals 5. • Field 22 subfield 7 (Card Data Input Mode) equals 2, field 35 (Track 2 Data) is present, and the first digit of the Service Code (position 21 in field 35, excluding the length subfield) equals 2. Otherwise, set to the value 0.
DATA-SUSPECT	Set to the value 0.
APPL-PAN-SEQ-NUM	Set to the value of field 55 subfield 15 (Application PAN Sequence Number), if present; otherwise set to spaces.

Token Field	Value
DEV-INFO	Set to spaces.
RSN-ONL-CDE	Set to spaces.
ARQC-VRFY	Set to the value 0.

EMV Response Data Token

The following table shows the mapping between fields in the EMV Response Data token and the EMV tag values in data element P-55 of the response message.

Token Field	EMV Tag Values in Data Element P-55 of the Request Message
ISS-AUTH-DATA	91

EMV Script Data Token

The following table shows the mapping between fields in the EMV Script Data token and the EMV tag values in data element P-55 of the response message.

Token Field	EMV Tag Values in Data Element P-55 of the Request Message
ISS-SCRIPT-DATA	71 or 72

MasterCard Interchange Interfaces (BankNet and MDS Maestro)

The BankNet EMV module supports BASE24-pos EMV transactions when bound with the BankNet Interchange Interface process. The MDS Maestro EMV module supports BASE24-pos EMV transactions when bound with the BASE24 MDS Interchange Interface process. Both of these modules are additional licensable modules and comply with the message formats and rules developed by MasterCard.

The main function of the BASE24 MasterCard Debit Switch (MDS) Maestro EMV Interchange Interface module and the BankNet EMV Interchange Interface module is the formatting of internal message token data to and from EMV-specific external message fields and subfields. They also handle the initialization, compression, and expansion of the EMV-related fields in the external message.

In this section, the BankNet interchange and the MDS Maestro interchange are collectively referred to as the MasterCard interchanges.

BASE24 as Acquirer

For BASE24 as an acquirer, the BankNet or MDS Maestro Interchange Interface process receives BASE24-pos Standard Internal Messages (PSTMs) with one or more EMV tokens attached. Standard BASE24 external message formatting is performed and then the EMV module completes the formatting. The extent of this formatting is dependent on the level of chip support determined by the contents of the transaction message:

- For transactions performed at terminals that are not chip-capable, no further formatting is required.
- For transactions performed at chip-capable terminals where the PAN is not read from the chip and no chip data is available, the EMV Status token is used to set data element P-22 (Point of Service Entry Mode) in the message sent to the MasterCard interchanges to indicate that it is not a chip-read transaction.
- For transactions performed at chip-capable terminals where the PAN is read from the chip, but no chip data is available, the EMV Status token is used to set data element P-22 in the message sent to the MasterCard interchanges to indicate that it is a chip-read transaction.

- For transactions performed at chip-capable terminals where the PAN is read from the chip and chip data is available, both the EMV Status token and the EMV Request Data token are present in the internal message. The EMV Discretionary Data token may also be present. In addition to the processing described above, the data in the EMV Request Data token (and optionally from the EMV Discretionary Data token) are moved to data element P-55 (ICC System-Related Data) in the message sent to the MasterCard interchanges.

When receiving a response from the MasterCard interchanges containing EMV data, existing logic handles the initial formatting of the PSTM. The EMV module then completes the formatting of the PSTM. If the original request contained the EMV Request Data token, this formatting may include adding the EMV Response Data token and possibly the EMV Script Data token prior to sending the formatted PSTM.

BASE24 as Issuer

For BASE24 as an issuer, the BankNet or MDS Maestro Interchange Interface process receives external request messages from the MasterCard interchanges. Standard BASE24 internal message formatting is performed, and then the EMV module completes the formatting, adding the appropriate tokens. The extent of this formatting is dependent on the level of chip support determined by the contents of the transaction message:

- For transactions not containing data element P-55, no specific EMV processing is performed.
- If the terminal was EMV-capable, but the chip data elements were not available (i.e., data element P-55 is not present), the EMV module appends only the EMV Status token to the PSTM. The information in this token differentiates between transactions where the PAN is read from a chip and those where it is not.
- If the terminal was EMV-capable, the card data was read from the chip, and the chip data elements are available, the EMV module appends both the EMV Status token and the EMV Request Data token (and optionally the EMV Discretionary Data token) to the PSTM before sending it to the BASE24-pos Device Handler/Router/Authorization process.

When the BASE24-pos Device Handler/Router/Authorization process sends a response to the BankNet or MDS Maestro Interchange Interface process, existing logic formats the external response and then the EMV module completes the formatting. If data element P-55 was present in the original request from the

MasterCard interchanges, the EMV module extracts data from the EMV Response Data token and (if present) the EMV Script Data token to include in data element P-55 of the response message.

BankNet and MDS Maestro EMV External Message

The MasterCard interchanges use data element P-55, Integrated Circuit Card System Related Data, to pass chip information from the acquirer to the issuer. This data element contains many subelements and is supported only for transactions where chip data is successfully read from the chip at a chip-capable terminal.

The MDS Maestro interchange passes this information only in Financial Transaction Request (0200), Financial Transaction Request Response (0210), Financial Transaction Advice (0220), and Financial Transaction Advice Response (0230) messages. The BankNet interchange passes this information only in Authorization Request (0100), Authorization Request Response (0110), Authorization Advice (0120), and Authorization Advice Response (0130) messages. Internally, BASE24 handles messages sent from the BankNet interchange as 0200, 0210, 0220, and 0230 messages.

Most of the information in data element P-55 is also contained in the EMV-specific tokens. Although data element P-55 may be present in the external message, it is not logged to the Interchange Log File (ILF).

The format of EMV-specific tokens is described in the ***BASE24 Tokens Manual.***

EMV Request Data Token

The following table shows the mapping between fields in the EMV Request Data token and the EMV tag values in data element P-55 of the request message. All data elements in this token are required.

Token Field	EMV Tag Values in Data Element P-55 of the Request Message
CRYPTO-INFO-DATA	9F27
TVR	95
ARQC	9F26
AMT-AUTH	9F02

Token Field	EMV Tag Values in Data Element P-55 of the Request Message
AMT-OTHER	9F03
AIP	82
ATC	9F36
TERM-CNTRY-CDE	9F1A
TRAN-CRNCY-CDE	5F2A
TRAN-DAT	9A
TRAN-TYPE	9C
UNPREDICT-NUM	9F37
ISS-APPL-DATA	9F10

EMV Discretionary Data Token

The following table shows the mapping between fields in the EMV Discretionary Data token and the EMV tag values in data element P-55 of the request message.

Token Field	EMV Tag Values in Data Element P-55 of the Request Message
TERM-SERL-NUM	9F1E
EMV-TERM-CAP	9F33
EMV-TERM-TYPE	9F35
APPL-VER-NUM	9F09
CVM-RSLTS	9F34
DF-NAME	84

EMV Status Token

The following table shows the mapping between fields in the EMV Status token and the interchange/interface message.

Token Field	Value
PT-SRV-ENTRY-MDE	Set to the value 01*x* if field 22 (Point-of-Service Entry Mode) equals the value 79*x*, set to the value 02*x* if field 22 equals the value 80*x* or 90*x*; otherwise set to the value of field 22.
TERM-ENTRY-CAP	Set to the value 5 (EMV-capable terminal) if field 61 subfield 11 (POS Card Data Terminal Input Capability Indicator) equals the value 5, 8, or 9.
LAST-EMV-STAT	Set to the value 1 if field 22 (Point-of-Service Entry Mode) equals the value 05*x*, 79*x*, or 80*x*; otherwise set to the value 0.
DATA-SUSPECT	Set to the value 0.
APPL-PAN-SEQ-NUM	Set to the value of field 23 (Card Sequence Number), if present; otherwise set to spaces.
DEV-INFO	Set to spaces.
RSN-ONL-CDE	Set to spaces.
ARQC-VRFY	Map from field 48 subfield 71 (On-Behalf Services), if present; otherwise set to the value 0.

EMV Response Data Token

The following table shows the mapping between fields in the EMV Response Data token and the EMV tag values in data element P-55 of the response message.

Token Field	EMV Tag Values in Data Element P-55 of the Request Message
ISS-AUTH-DATA	91

EMV Script Data Token

The following table shows the mapping between fields in the EMV Script Data token and the EMV tag values in data element P-55 of the response message.

Token Field	EMV Tag Values in Data Element P-55 of the Request Message
ISS-SCRIPT-DATA	71 or 72

Visa Interchange Interface

The BASE24 Visa EMV product module is an additional licensable module which, when bound with the BASE24 VisaNet ISO Interchange Interface module (or VisaNet Interface), complies with the message formats and rules developed by Visa for the Visa Smart Debit/Credit (VSDC) Service. It offers support for early and full option issuer processing, or general EMV acquirer processing, as described by Visa, and supports BASE24-pos EMV processing.

The main function of the BASE24 Visa EMV module is the formatting of internal message token data to and from EMV-specific external message fields and subfields. It also handles the initialization, compression, and expansion of the EMV-related fields in the external message.

BASE24 as Acquirer

For EMV acquirers, the following levels of support are provided:

Full Support	A number of EMV data elements are included in the message to Visa. The BASE24 Visa EMV module assumes this mode if the EMV Request Data token is appended to the PSTM.
Easy Entry	The EMV data elements do not need to be included in the message to Visa. The BASE24 Visa EMV module assumes this mode if the EMV Request Data token is not appended to the PSTM, but the card details were read from the chip.

For BASE24 as an acquirer, the BASE24 VisaNet Interchange Interface process receives standard internal request messages with EMV Request Data, EMV Discretionary Data, and EMV Status tokens attached. Standard external message formatting is performed and then the BASE24 Visa EMV module completes the formatting. If only the EMV Status token is present, the transaction is deemed to be either an Easy Entry transaction or a magnetic stripe transaction from an EMV-capable terminal, and the internal message is formatted without the EMV data elements.

When receiving a response from Visa containing EMV data, the Interchange Interface process formats the PSTM to return to the BASE24-pos Device Handler/Router/Authorization process. If the response contains EMV data, the Visa EMV module formats the EMV Response Data token, and, if applicable, the EMV Script Data token prior to sending the formatted PSTM.

BASE24 as Issuer

For issuer processing, the two levels of support offered by Visa provide varying degrees of impact on the current message structure:

Early Option	If issuers request this option, Visa removes all EMV-specific fields from inbound request messages. Issuers receive new values in existing fields. The BASE24 Visa EMV module uses these fields to format the EMV Status token appended to the PSTM for forwarding to the BASE24-pos Device Handler/Router/Authorization process. The BASE24-pos Device Handler/Router/Authorization process does not take any specific action on this token data. This form of processing is known as pass-through.
Full Option	Issuers receive a number of new EMV data elements. The new data items are added to the standard EMV Status and EMV Request Data tokens, which contain all the information needed for the BASE24-pos Device Handler/Router/Authorization process or back-end host to perform card authentication processing. The EMV Response Data and EMV Script Data tokens returned from the BASE24-pos Device Handler/Router/Authorization process are used in formatting the EMV data elements in the response to Visa. Full Option issuers may use the VisaNet Card and Issuer Authentication Services.

For card authentication by Visa, the BASE24 Visa EMV module sets a flag in the EMV Status token to indicate a prevalidated request cryptogram. This causes the BASE24-pos Device Handler/Router/Authorization process to suppress the card and issuer authentication processing. The processing assumes that the VisaNet Issuer Authentication Service is also used, and that this service returns issuer authentication data in the response message to the acquirer.

Where BASE24 is the issuer, the BASE24 VisaNet Interchange Interface process receives external request messages with formats dependent on the level of issuer support:

Early Option	External request message processing moves the EMV-specific data into an EMV Status Token.

Full Option	The PSTM request is formatted and the Visa EMV module completes the formatting with EMV changes, adding the EMV Status and EMV Request Data tokens. The PSTM, with tokens appended, is sent to the BASE24-pos Device Handler/Router/Authorization process. When the BASE24-pos Device Handler/Router/ Authorization process responds, the response to Visa is formatted. The Visa EMV module completes the formatting, extracting the data required to format the EMV data elements from the EMV Response and (if present) EMV Script Data tokens.

For fallback transactions the EMV Status Token is added to the PSTM.

On transactions acquired from VisaNet, a check is made on how the card details were entered for the transaction. If the card details are read from the chip, the Point of Service Entry Mode field in the PSTM must be set to 05*x,* where *x* indicates the PIN entry capability of the terminal.

VisaNet EMV External Message

The EMV external message can use data element P-55 or an extension to the standard VIP (BASE I/SMS) messages that incorporates a tertiary (or third) bit map and associated fields. Both formats are provided in this discussion.

The USE-FIELD-55-FOR-EMV-DATA param in the Logical Network Configuration File (LCONF) controls the format of outbound messages. The VisaNet Interchange Interface process uses the tertiary bit map format if this param is missing or invalid. For inbound messages, the VisaNet Interchange Interface process can accept messages containing either format or a combination of formats.

The following tables show the mapping between fields in the EMV data tokens and fields in the VisaNet external message. Note that token fields not listed have no comparable field in the VisaNet message.

The format of EMV-specific tokens is described in the ***BASE24 Tokens Manual.***

EMV Request Data Token (Data Element P-55)

The following table shows the mapping between fields in the EMV Request Data token and the EMV tag values in data element P-55. All data elements in this token are required.

Token Field	EMV Tag Values in Data Element P-55
CRYPTO-INFO-DATA	9F27
TVR	95
ARQC	9F26
AMT-AUTH	9F02
AMT-OTHER	9F03
AIP	82
ATC	9F36
TERM-CNTRY-CDE	9F1A
TRAN-CRNCY-CDE	5F2A
TRAN-DAT	9A
TRAN-TYPE	9C
UNPREDICT-NUM	9F37
ISS-APPL-DATA	9F10

EMV Discretionary Data Token (Data Element P-55)

The following table shows the mapping between fields in the EMV Discretionary Data token and the EMV tag values in data element P-55.

Token Field	EMV Tag Values in Data Element P-55
TERM-SERL-NUM	9F1E
EMV-TERM-CAP	9F33

Token Field	EMV Tag Values in Data Element P-55
EMV-TERM-TYPE	9F35
APPL-VER-NUM	9F09
CVM-RSLTS	9F34
DF-NAME	84

EMV Status Token (Data Element P-55)

The following table shows the mapping between fields in the EMV Status token and the interchange/interface message using data element P-55.

Token Field	External Message Field	Visa Field Name
PT-SRV-ENTRY-MDE	Set to the value 05*x* if field P-22 is equal to the value 95*x*, set to the value 02*x* if field P-22 is equal to the value 90*x*; otherwise set to the value of field P-22.	Point of Service Entry Mode (field P-22)
TERM-ENTRY-CAP	P-60, subfield 2	Terminal Entry Capability
LAST-EMV-STAT	P-60, subfield 3	Chip Condition Code
DATA-SUSPECT	P-60, subfield 7	Chip Authentication Reliability Indicator
APPL-PAN-SEQ-NUM	P-23	Card Sequence Number
ARQC-VRFY	P-44, subfield 8	Card Authentication Results Code

EMV Response Data Token (Data Element P-55)

The following table shows the mapping between fields in the EMV Response Data token and the EMV tag values in data element P-55.

Token Field	EMV Tag Values in Data Element P-55
ISS-AUTH-DATA	91

EMV Script Data Token (Data Element P-55)

The following table shows the mapping between fields in the EMV Script Data token and the EMV tag values in data element P-55.

Token Field	EMV Tag Values in Data Element P-55
ISS-SCRIPT-DATA	71 or 72

EMV Supplementary Data Token (Data Element P-55)

The following table shows the mapping between fields in the EMV Supplementary Data token and the EMV tag values in data element P-55. This mapping is performed when the DATASET-ID field in the token is set to the value 01.

Token Field	EMV Tag Values in Data Element P-55
SUPPL-DATA	9F6E, 9F7C, 9F66

EMV Request Data Token (Tertiary Bit Map)

The following table shows the mapping between fields in the EMV Request Data token and the tertiary bit map of the VisaNet message.

Token Field	Visa Field	Visa Field Name
TVR	T-131	Terminal Verification Results
ARQC	T-136	Cryptogram
AMT-AUTH	T-147	Cryptogram Amount
AMT-OTHER	T-149	Cryptogram Cashback Amount
AIP	T-138	Application Interchange Profile
ATC	T-137	Application Transaction Counter
TERM-CNTRY-CDE	T-145	Terminal Country Code
TRAN-CRNCY-CDE	T-148	Cryptogram Currency Code
TRAN-DAT	T-146	Terminal Transaction Date
TRAN-TYPE	T-144	Cryptogram Transaction Type
UNPREDICT-NUM	T-132	Unpredictable Number
LGTH	T-134	Length of field T-134
ISS-APPL-DATA	T-134	Issuer Application Data

EMV Discretionary Data Token (Tertiary Bit Map)

The following table shows the mapping between fields in the EMV Discretionary Data token and the tertiary bit map of the VisaNet message.

Token Field	Visa Field	Visa Field Name
TERM-SERL-NUM	T-133	Terminal Serial Number
EMV-TERM-CAP	T-130	Terminal Capability Profile

EMV Status Token (Tertiary Bit Map)

The following table shows the mapping between fields in the EMV Status token and the tertiary bit map of the VisaNet message.

Token Field	External Message Field	Visa Field Name
PT-SRV-ENTRY-MDE	Set to the value 05*x* if field P-22 is equal to the value 95*x*, set to the value 02*x* if field P-22 is equal to the value 90*x*; otherwise set to the value of field P-22.	Point of Service Entry Mode (field P-22)
TERM-ENTRY-CAP	P-60, subfield 2	Terminal Entry Capability
LAST-EMV-STAT	P-60, subfield 3	Chip Condition Code
DATA-SUSPECT	P-60, subfield 7	Chip Authentication Reliability Indicator
APPL-PAN-SEQ-NUM	P-23	Card Sequence Number
ARQC-VRFY	P-44, subfield 8	Card Authentication Results Code

EMV Response Data Token (Tertiary Bit Map)

The following table shows the mapping between fields in the EMV Response Data token and the tertiary bit map of the VisaNet message. The tertiary bit used for this data depends on whether BASE24 is the issuer or the acquirer.

Token Field	Visa Field		Visa Field Name
	BASE24 as Issuer	**BASE24 as Acquirer**	
ARPC	T-139, subfield 1	T-140, subfield 1	Authorization Response Cryptogram (ARPC)
ISS-RESP-CDE	T-139, subfield 2	T-140, subfield 2	ARPC Response Code

EMV Script Data Token (Tertiary Bit Map)

The following table shows the mapping between fields in the EMV Request Data token and the tertiary bit map of the VisaNet message.

Token Field	Visa Field	Visa Field Name
ISS-SCRIPT-DATA	T-142	Issuer Script

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A: SPDH Device Support

The BASE24 Standard POS Device Handler (SPDH) module of the BASE24-pos Device Handler/Router/Authorization process provides an interface between terminals that use the SPDH message format and the Router and Authorization modules of the BASE24-pos Device Handler/Router/Authorization process.

Section 6 summarizes the functionality provided by all device handler EMV modules.

This appendix describes the functionality that is unique to the BASE24-pos SPDH Device Handler EMV module. This module is an additional licensable module that, when bound with the standard BASE24-pos SPDH Device Handler module, provides the ability to process EMV transactions.

For information about the functionality of the standard SPDH Device Handler module, see the ***BASE24-pos Standard POS Device Configuration Manual***.

Field Mapping

The format of the EMV-specific tokens is described in the ***BASE24 Tokens Manual.*** The following tables show the mapping between fields in the data tokens and fields in the SPDH request and response messages.

EMV Request Data Token

The following table shows the mapping between fields in the EMV Request Data token and the SPDH request message. This mapping occurs regardless of EMV card scheme, which is defined in the Smart Card Scheme field in FID 6, subFID O of the ACI Standard POS Message. Valid values for the Smart Card Scheme are as follows:

00 = EMV version 1
01 = EMV version 2

Token Field	Source Field in SPDH Request Message	Default Value
CRYPTO-INFO-DATA	FID 6, subFID O	None (always present in request)
TVR	FID 6, subFID O	None (always present in request)
ARQC	FID 6, subFID O	None (always present in request)
AMT-OTHER	FID C	000000000000
AIP	FID 6, subFID O	None (always present in request)
ATC	FID 6, subFID O	None (always present in request)
TERM-CNTRY-CDE	FID 6, subFID O	None (always present in request)
TRAN-DAT	FID 6, subFID O	None (always present in request)
TRAN-TYPE	FID 6, subFID O	None (always present in request)
UNPREDICT-NUM	FID 6, subFID O	None (always present in request)
ISS-APPL-DATA	FID 6, subFID O	None (always present in request)

The following table shows the additional mapping between fields in the EMV Request Data token and the SPDH request message. This mapping occurs only when the EMV card scheme is set to a value of 00 (EMV version 1).

Token Field	Source Field in SPDH Request Message	Default Value
AMT-AUTH	FID B	000000000000
TRAN-CRNCY-CDE	FID 6, subFID I	None (always present in request)

The following table shows the additional mapping between fields in the EMV Request Data token and the SPDH request message. This mapping occurs only when the EMV card scheme is set to a value of 01 (EMV version 2).

Token Field	Source Field in SPDH Request Message	Default Value
AMT-AUTH	FID 6, subFID O	None
TRAN-CRNCY-CDE	FID 6, subFID O	None

EMV Discretionary Data Token

The following table shows the mapping between fields in the EMV Discretionary Data token and the fields in the SPDH request message. This mapping occurs regardless of EMV card scheme.

Token Field	Source Field in SPDH Request Message	Default Value
EMV-TERM-TYPE	FID 6, subFID P	None (BIT MAP is set to indicate presence or absence of this field)

The following table shows the additional mapping between fields in the EMV Discretionary Data token and the fields in the SPDH request message. This mapping occurs only when the EMV card scheme is set to a value of 01 (EMV version 2).

Token Field	Source Field in SPDH Request Message	Default Value
CVM-RSLTS	FID 6, subFID P	None
APPL-VER-NUM	FID 6, subFID P	None
DF-NAME	FID 6, subFID P	None

EMV Status Token

The following table shows the mapping between fields in the EMV Status token and the fields in the SPDH request message. This mapping occurs regardless of EMV card scheme.

Token Field	Source Field in SPDH Request Message	Default Value
PT-SRV-ENTRY-MDE	FID 6, subFID E	None (always present in request)
TERM-ENTRY-CAP	Processing Flag 2	None
LAST-EMV-STAT	None	Set to a value of 1 if the first byte of the Track 2 service code contains a value of 2 or 6, otherwise set to a value of 0
DATA-SUSPECT	None	Set to a value of 0
APPL-PAN-SEQ-NUM	FID 6, subFID P	Spaces
RSN-ONL-CDE	FID 6, subFID N	Spaces
ARQC-VRFY	None	Set to a value of 0

The following table shows the additional mapping between fields in the EMV Status token and the fields in the SPDH request message. This mapping occurs only when the EMV card scheme is set to a value of 01 (EMV version 2).

Token Field	Source Field in SPDH Request Message	Default Value
CVM-RSLTS	FID 6, subFID P	None

EMV Response Data Token

The following table shows the mapping between fields in the EMV Response Data token and the fields in the SPDH response message. This mapping occurs regardless of EMV card scheme.

Token Field	Destination Field in SPDH Response Message	Default Value
ISS-AUTH-DATA	FID 6, subFID Q	None

EMV Script Data Token

The following table shows the mapping between fields in the EMV Script Data token and the fields in the SPDH response message. This mapping occurs regardless of EMV card scheme.

Token Field	Destination Field in SPDH Response Message	Default Value
ISS-SCRIPT-DATA	FID 6, subFID R	None

EMV Supplementary Data Token

The following table shows the mapping between fields in the EMV Supplementary Data token and the fields in the SPDH request message. This mapping occurs regardless of EMV card scheme.

Token Field	Source Field in SPDH Request Message	Default Value
DATASET ID	First two characters of FID 6, subFID q	None
LGTH	None	Set to the length of data in the SUPPL-DATA field
SUPPL-DATA	The remainder of FID 6, subFID q	None

B: APACS 70 Device Support

The APACS 70 Device Handler module of the BASE24-pos Device Handler/Router/Authorization process provides an interface between terminals that conform to the APACS 70 standards and the Router and Authorization modules of the BASE24-pos Device Handler/Router/Authorization process. This Device Handler module supports the functions described in Book 2 of the APACS 70 Standard, up to and including the October 2004 version.

Section 6 summarizes the functionality provided by all device handler EMV modules.

This appendix describes the functionality that is unique to the BASE24-pos APACS 70 Device Handler EMV module. This module is an additional licensable module that, when bound with the standard BASE24-pos APACS 70 Device Handler module, provides the ability to process EMV transactions.

For information about the functionality of the standard APACS 70 Device Handler module, see the ***BASE24-pos APACS 70 Device Handler Manual***.

Field Mapping

The format of the EMV-specific tokens is described in the ***BASE24 Tokens Manual.*** The following tables show the mapping between fields in the data tokens and fields in the APACS 70 request and response messages.

EMV Request Data Token

The following table shows the mapping between fields in the EMV Request Data token and the APACS request message:

Token Field Name	Source Field in APACS Request Message	Default Value
CRYPTO-INFO-DATA	Field 29m, if present	If not present, the token field is set to 40 for post-event advice message types, and 80 for all other message types
TVR	Field 29e	None (always present in request)
ARQC	Field 29a	None (always present in request)
AMT-AUTH	Field 9 (Transaction Amount)	None (always present in request)
AMT-OTHER	Field 17, if present (Cash Amount)	000000000000
AIP	Field 29b	None (always present in request)
ATC	Field 29c	None (always present in request)
TERM-CNTRY-CDE	Field 23	None (always present in request)
TRAN-CRNCY-CDE	Field 25	None (always present in request)
TRAN-DAT	Field 19	None (always present in request)
TRAN-TYPE	Field 29f	None (always present in request)
UNPREDICT-NUM	Field 29d	None (always present in request)

Token Field Name	Source Field in APACS Request Message	Default Value
ISS-APPL-DATA (Length is specified by ISS-APPL-DATA-LGTH field)	Field 29g	None (always present in request)

EMV Discretionary Data Token

The following table shows the mapping between fields in the EMV Discretionary Data token and the fields in the APACS request message:

Token Field Name	Source Field in APACS Request Message	Default Value
EMV-TERM-TYPE	Field 21	None (BIT MAP set to indicate this field is absent)
APPL-VER-NUM	Field 29k	None (BIT MAP set to indicate this field is absent)
CVM-RSLTS	Field 29o, if present	None (BIT MAP set to indicate this field is absent)
DF-NAME (Length is specified by DF-NAME-LGTH field)	Field 29i	None (BIT MAP set to indicate this field is absent)

EMV Status Token

The following table shows the mapping between fields in the EMV Status token and the fields in the APACS request message:

Token Field Name	Source Field in APACS Request Message	Default Value
PT-SRV-ENTRY-MDE	None	The first two digits are set to a value of 05 if the first digit of field 7 is numeric

Token Field Name	Source Field in APACS Request Message	Default Value
TERM-ENTRY-CAP	Field 3 (Terminal Type)	Set to a value of 5 if the first digit of field 3 contains a value of 1 or 3
LAST-EMV-STAT	None	Set to a value of 1 if the first byte of the Track 2 service code contains a value of 2 or 6, otherwise set to a value of 0
DATA-SUSPECT	None	Set to a value of 0
APPL-PAN-SEQ-NUM	Field 7c or 7e, if present	Spaces
DEV-INFO	Field 29o, if present	Spaces
RSN-ONL-CDE	Field 27	None (always present in request)
ARQC-VRFY	None	Set to a value of 0

EMV Response Data Token

The following table shows the mapping between fields in the EMV Response Data token and the fields in the APACS response message:

Token Field Name	Destination Field in APACS Response Message	Default Value
ISS-AUTH-DATA (Length is specified by ISS-AUTH-DATA-LGTH field)	Field 18a	None

EMV Script Data Token

The following table shows the mapping between fields in the EMV Script Data token and the fields in the APACS response message:

Token Field Name	Destination Field in APACS Response Message	Default Value
ISS-SCRIPT-DATA (Length is specified by ISS-SCRIPT-DATA-LGTH field)	Field 18c	None

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C: Hypercom Device Support

The BASE24-pos Hypercom Device Handler (HPDH) module of the BASE24-pos Device Handler/Router/Authorization process provides an interface between Hypercom terminals and the Router and Authorization modules of the BASE24-pos Device Handler/Router/Authorization process.

Section 6 summarizes the functionality provided by all device handler EMV modules.

This appendix describes the functionality that is unique to the BASE24-pos Hypercom Device Handler EMV module. This module is an additional licensable module that, when bound with the standard BASE24-pos Hypercom Device Handler module, provides the ability to process EMV transactions.

For information about the functionality of the standard HPDH Device Handler module, see the ***BASE24-pos Hypercom Device Support Manual***.

Field Mapping

The format of the EMV-specific tokens is described in the ***BASE24 Tokens Manual.*** The following tables show the mapping between fields in the data tokens and fields in the HPDH request and response messages. Note that token fields not listed have no comparable field in the HPDH message.

EMV Request Data Token

The following table shows the mapping between fields in the EMV Request Data token and the tag values in data element P-55 of the HPDH request message. All fields in this token are required.

Token Field	EMV Tag Values in Data Element P-55 of the HPDH Request Message
CRYPTO-INFO-DATA	9F27
TVR	95
ARQC	9F26
AMT-AUTH	9F02
AMT-OTHER	9F03
AIP	82
ATC	9F36
TERM-CNTRY-CDE	9F1A
TRAN-CRNCY-CDE	5F2A
TRAN-DAT	9A
TRAN-TYPE	9C
UNPREDICT-NUM	9F37
ISS-APPL-DATA	9F10

EMV Discretionary Data Token

The following table shows the mapping between fields in the EMV Discretionary Data token and the tag values in data element P-55 of the HPDH request message:

Token Field	EMV Tag Values in Data Element P-55 of the HPDH Request Message	Default Value
TERM-SERL-NUM	9F1E	Set to 0
EMV-TERM-CAP	9F33	PTD.EMV-TERM-CAP
EMV-TERM-TYPE	9F35	Set to 0
APPL-VER-NUM	9F09	Set to 0
CVM-RSLTS	9F34	Set to 0
DF-NAME	84	Set to spaces

EMV Status Token

The following table shows the mapping between fields in the EMV Status token and the token values set from the HPDH request message:

Token Field	Value
PT-SRV-ENTRY-MDE	Set to the value in field 22 (Point-of-Service Entry Mode)
TERM-ENTRY-CAP	Set to 5 (EMV-capable terminal)
LAST-EMV-STAT	Set to 1 if field 22 is equal to 05*x* or 80*x*; otherwise set to 0
DATA-SUSPECT	Set to 0
APPL-PAN-SEQ-NUM	Set to the contents of EMV tag 5F34 in data element P-55 of the HPDH request message, if present
DEV-INFO	Set to spaces

Token Field	Value
RSN-ONL-CDE	Set to spaces
ARQC-VRFY	Set to 0

EMV Response Data Token

The following table shows the mapping between fields in the EMV Response Data token and the tag values in data element P-55 of the HPDH response message:

Token Field	EMV Tag Values in Data Element P-55 of the HPDH Response Message
ISS-AUTH-DATA	91

EMV Script Data Token

The following table shows the mapping between fields in the EMV Script Data token and the tag values in data element P-55 of the HPDH response message:

Token Field	EMV Tag Values in Data Element P-55 of the HPDH Response Message
ISS-SCRIPT-DATA	71 or 72

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